

Contributions to Graph Compression and Preconditioning

by

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Abstract

From social networks to global trade systems, much of today’s data is naturally modeled as graphs. These graphs are often enormous, making them difficult to store and analyze efficiently. As a result, many classical graph algorithms and data structures become computationally infeasible at scale.

This thesis studies two core strategies for addressing this challenge: graph compression and graph preconditioning. Graph compression seeks compact graph representations that approximately preserve key structural information, such as connectivity and distances. Graph preconditioning seeks to transform graphs, for example through augmentations or decompositions, to enable faster algorithms.

We establish new upper and lower bounds for several fundamental graph-theoretic objects arising in these areas, including distance preservers, reachability preservers, strong connectivity preservers, hopsets, shortcut sets, and DAG covers.

Chapter 1

Introduction

Many large-scale systems in computing, science, and society are naturally modeled as graphs. For example, the internet can be viewed as a network of interconnected computers and servers. Similarly, social platforms can be represented as networks of users. These graphs are often massive, and their scale poses new engineering challenges.

We often want to extract information from such graphs or answer queries about them. A navigation system, for example, may use a road network to answer shortest-path queries and provide directions. On graphs of modern scale, however, even algorithms with satisfactory worst-case guarantees may be too slow to be practical. Space can be just as serious an obstacle: for massive networks like the internet, storing the full graph explicitly may be prohibitively expensive. These considerations motivate methods that reduce the size of the input graph, or transform it into a more structured form, while preserving the information needed for computation.

Graph compression asks how to replace a graph by a much smaller representation that still retains, at least approximately, the information needed for a given task. The relevant information depends on the application. For example, one may want to approximate distances between pairs of vertices in the graph, or approximate cut values in the graph up to small error, or maintain spectral information about the graph. The study of graph compression has spawned a broad family of graph-theoretic objects, such as spanners [PS89; Alt+93; BS07], cut sparsifiers [Ben95; BK15; Fun+19; FV22], spectral sparsifiers [ST11; SS11; BSS12; LS15], distance oracles and labeling schemes [TZ05; Gav+04], vertex and flow sparsifiers [Moi09; LM10; Cha+10], distance and reachability preservers [CE06; AB24a], and many other related notions.

Graph compression is attractive for two reasons. First, a smaller representation reduces storage costs, which matters for very large networks. For example, many graph data structures and oracles rely on compressed representations of the input graph in order to answer information queries efficiently [TZ05; Gav+04; vS19; PSY22; LS22a]. Second, compression can also speed up

downstream computation by shrinking the effective input size. Many fast graph algorithms rely on compression primitives such as emulators, spanners, and cut, flow, or spectral sparsifiers as subroutines [BR11; ST04; KMP11; Kel+14; Cen+21; Che+22]. For many settings where graph compression is useful, it is often acceptable to preserve only a “lossy” representation of the input graph in exchange for substantial savings in space, provided the loss in accuracy is well understood. A central objective in graph compression is to understand this tradeoff between the size of the compressed representation and the quality of the information it preserves.

Graph preconditioning [Li21] takes a complementary viewpoint. Rather than primarily shrinking the graph, it seeks to transform the input into a form on which algorithms run more efficiently, through carefully chosen augmentations, decompositions, or reductions [ST04; KMP11; She17]. The point is to expose useful structure in the graph that can be exploited by algorithms. For instance, *expander decomposition* lets one treat a graph as a collection of highly connected pieces [KVV04; ST04; ACL06; AP09; SW19; CS20], while *tree embeddings* reduce certain metric questions to the setting of trees [Bar96b; Bar98a; FRT04a; Răc08a; ABN07; AN12]. Techniques of this kind play an important role in fast graph algorithms and data structures, including nearly-linear-time Laplacian solvers and flow algorithms [ST04; KMP11; Chr+11; Mad10; Pen16; She17] and dynamic graph algorithms based on expander decompositions and hierarchies [NSW17; Gor+21a].

This thesis proves new upper and lower bounds for several objects studied in these two areas, including distance preservers [CE06; Bod17], reachability preservers [AB24a], strong connectivity preservers [CC20], hopsets [Coh00; EN19a], shortcut sets [Tho93], and DAG covers [AHW25a]. The remainder of this introduction provides background for the main results and places them in context. Broadly speaking, the results are organized into contributions to graph compression and contributions to graph preconditioning.

1.1 Graph Compression

This section surveys the graph-compression results proved in this thesis.

1.1.1 Distance Preservers

Distance preservers are a basic graph-compression object for shortest-path information. Given a graph G and a set of demand pairs $P \subseteq V(G) \times V(G)$, one seeks a sparse subgraph $H \subseteq G$ that preserves the distances for the pairs in P , either exactly or approximately. This problem lies within a broad category of graph compression problems concerned with preserving distance information the graph. In particular, distance preservers are closely connected to spanners and distance oracles,

and they have found applications in data structures and fast graph algorithms [CE06; EP16; ES23; BCE05; BV15; AB16; AB17; ABP17; EFN17; GK17; BV21].¹

Formally, given an n -vertex graph G and a set of demand pairs $P \subseteq V(G) \times V(G)$ of size $|P| = p$, a subgraph $H \subseteq G$ is an α -approximate distance preserver if

$$\text{dist}_H(s, t) \leq \alpha \text{dist}_G(s, t) \quad \text{for every } (s, t) \in P \text{ with } \text{dist}_G(s, t) < \infty.$$

When $\alpha = 1$, we obtain *exact distance preservers*. When $\alpha = 1 + \varepsilon$, we obtain approximate distance preservers. If we drop any finite stretch requirement and ask only that the demand pairs remain reachable, we obtain *reachability preservers*.

The study of exact distance preservers was initiated by Coppersmith and Elkin [CE06], whose work also identified a benchmark that shaped the area for many years: the size of a *consistent tiebreaking scheme*. Informally, such a scheme chooses one shortest path for each demand pair so that no two chosen paths meet, diverge, and later meet again. This consistency condition is strong enough to imply exact preserver upper bounds, and for many years the best known bounds for exact preservers were precisely the bounds obtained through consistent tiebreaking. In particular, one obtains bounds of $O(\min\{n^{1/2}p + n, np^{1/2}\})$ edges in undirected graphs and $O(\min\{n^{2/3}p + n, np^{1/2}\})$ edges in directed graphs [CE06; BV21; Bod17; BHT23].

More recent work shows that this benchmark is not always tight. In undirected unweighted graphs, exact preservers with $O(n^{2/3}p^{2/3} + np^{1/3})$ edges are known [BV21; Bod17]. For $(1 + \varepsilon)$ -approximate preservers, almost optimal constructions are known in undirected graphs, and $\tilde{O}_\varepsilon(n^{2/3}p^{2/3} + np^{1/3})$ -edge constructions are known in directed graphs [KP22b; BW23a; ES23]. Reachability preservers are known to be sparser still, with $O(n^{3/4}p^{1/2} + n^{0.59}p^{0.71} + n)$ edge bounds [BHT23; BL25]. These results raise two natural questions. Can one beat consistent-tiebreaking bounds for exact preservers in natural graph classes? And how different are the finite-stretch and infinite-stretch settings?

The first contribution of this section makes progress on both questions.

Theorem 1.1 (Main Result 1, [HXX25]). *For n -vertex directed graphs with p demand pairs:*

1. *exact distance preservers in the unweighted setting can be constructed with $\tilde{O}(n^{5/6}p^{2/3} + n)$ edges; and*
2. *for every fixed finite stretch α , there are instances for which every α -approximate distance preserver requires $\Omega(n^{2/3}p^{2/3})$ edges.*

¹The new results in this subsection first appeared in joint work with Yinzhan Xu and Zixuan Xu at SODA 2025 [HXX25].

The first item yields, in particular, a separation between exact distance preservers and consistent tiebreaking in unweighted directed graphs. Compared with the classical directed bound $O(\min\{n^{2/3}p + n, np^{1/2}\})$, it matches the classical bound at the endpoints $p = n^{1/2}$ and $p = n$ and is better in the regime $n^{1/2} \leq p \leq n$. The second item partially separates finite-stretch preservers from reachability preservers, since the lower bound above exceeds the best known reachability-preserver upper bounds in an appropriate range of parameters. Thus allowing infinite stretch can fundamentally change the sparsity landscape. The lower bound also requires some care in interpretation: it does not contradict the nearly matching $(1 + \varepsilon)$ -preserver upper bounds of [KP22b; BW23a], which apply only to graphs of $2^{\tilde{O}(1)}$ aspect ratio, whereas the lower-bound construction here has $\exp(n)$ aspect ratio.

The second contribution establishes a new reduction from directed exact distance preservers to the undirected case. Let $\text{UDP}(n, p)$ denote the best upper bound for exact distance preservers in undirected graphs with n vertices and p demand pairs, and let $\text{DDP}(n, p)$ denote the corresponding quantity for directed graphs.

Theorem 1.2 (Main Result 2, [HXX25]). *If $\text{UDP}(n, p) = O(n^\lambda p^\mu + n)$ for constants $\lambda, \mu > 0$, then*

$$\text{DDP}(n, p) = O\left(n^{\frac{1}{2-\lambda}} p^{\frac{1+\mu-\lambda}{2-\lambda}} + n^{1/2} p + n\right).$$

This reduction is strong enough to recover the $O(n^{2/3}p + n)$ arm of the current best directed upper bound from the corresponding $O(n^{1/2}p + n)$ undirected bound. More importantly, the theorem shows that any future undirected improvement of the form $O(n^\lambda p^\mu + n)$ would automatically yield a corresponding directed improvement. In particular, the consistent-tiebreaking barrier in directed graphs cannot remain optimal on the full regime $p \leq n^{2/3}$ unless the analogous barrier is already optimal in undirected graphs up to $p \leq n$.

Corollary 1.3 ([HXX25]). *Consistent tiebreaking is optimal for directed distance preservers when $p \leq n^{2/3}$ only if consistent tiebreaking is optimal for undirected distance preservers when $p \leq n$.*

1.1.2 Strong Connectivity Preservers

Connectivity is a basic structural feature of graphs that one may wish to preserve under compression. Directed graphs, however, pose unique challenges for sparsifying while preserving connectivity. General reachability is inherently asymmetric, and in the worst case the full reachability relation cannot be represented in $o(n^2)$ space. Sparse subgraphs therefore cannot, in general, preserve all directed connectivity information. Strong connectivity is a more tractable target. It captures mutual

reachability while avoiding the full complexity of arbitrary reachability queries. The results in this section show that this symmetric notion admits sparse certificates even in fault-tolerant settings.²

For a directed graph $G = (V, E)$, a *strong connectivity preserver* is a subgraph $H \subseteq G$ whose strongly connected components are exactly those of G . More generally, a *k -fault-tolerant strong connectivity preserver* is a subgraph H such that for every set $F \subseteq E$ with $|F| \leq k$, the strongly connected components of $G - F$ and $H - F$ coincide. In other words, H preserves the decomposition of the graph into strongly connected components even after any k edge failures. In undirected graphs the same definition reduces to preserving connected components under up to k edge deletions.

This question is well understood in the undirected setting. Established by the work of Nagamochi and Ibaraki, every undirected graph admits a k -fault-tolerant connectivity preserver with $O(nk)$ edges, and this bound is optimal up to constant factors [NI92; BK96; Cho21]. These sparse certificates have played an important role in graph algorithms, beginning with applications to minimum cuts and cut sparsification [BK96; BK15] and later extending to dynamic, distributed, parallel, and streaming models [Epp+97; Tho07; Abr+16; Thu95; Dag+19; Par19; AGM12; GMT15; AS23]. A parallel line of work studied stronger failure models, including simultaneous edge and vertex failures, bounded-degree failures, and color failures, while still achieving near-optimal $\tilde{O}(nk)$ bounds [CKT93; FIN93; EIR95; Fül05; BJ05b; Nag06; PT25; PST24; Par+25]. More broadly, fault-tolerant compression has been studied beyond connectivity itself, with analogous questions for approximate distances, exact distances, diameter, and spectral structure [Che+10; DK11; Bod+18; BDR22; Par22; BDN22; BHP24; PST24; PT25; Par+25; Bod+25; PP16; Bod+17; GK17; Bil+23; Zhu+19].

By contrast, directed graphs remain less well understood. Baswana, Choudhary, and Roditty gave an $O(2^k n)$ -edge preserver for the *single-source* version of a k -fault-tolerant strong connectivity preserver, which preserves only the strongly connected component containing a fixed source vertex, and this bound is tight [BCR18]. For the general all-pairs problem, Chakraborty and Choudhary obtained $\tilde{O}(k2^k n^{2-1/k})$ edges [CC20]. In particular, even for $k = 2$ their construction can be superlinear by a polynomial factor. They explicitly asked whether the large gap between upper and lower bounds could be closed.

Here we present a theorem which closes this gap up to a logarithmic factor.

Theorem 1.4 (Near-optimal fault-tolerant strong connectivity preservers, [HSW25]). *For every positive integer k , every n -vertex directed graph admits a k -fault-tolerant strong connectivity preserver with $O(k4^k n \log n)$ edges.*

²The new results in this subsection first appeared in joint work with Thatchaphol Saranurak and Benyu Wang at FOCS 2025 [HSW25]. Please note the contribution statement at the end of this subsection.

For every constant k , Theorem 1.4 yields an $O(n \log n)$ -edge preserver, which is near-optimal up to the logarithmic factor. Quantitatively, it improves the previous general upper bound of $\tilde{O}(k2^k n^{2-1/k})$ to $O(k4^k n \log n)$. It also indicates that the all-pairs directed problem is much closer to the undirected and single-source settings than earlier bounds suggested. The result yields compact query structures for reporting the strongly connected components of $G - F$ after up to k edge failures, and by the standard vertex-splitting reduction it extends to vertex failures as well.

Fault tolerance is only one notion of robustness. A weaker requirement is to preserve the higher-order edge-connectivity structure of the graph. For vertices $s, t \in V$, let $\lambda_G(s, t)$ denote the maximum number of edge-disjoint directed paths from s to t . We say that s and t are *strongly i -connected* if $\lambda_G(s, t) \geq i$ and $\lambda_G(t, s) \geq i$. For each i , this relation induces a partition of the vertex set into strongly i -connected components. A *k -strong-connectivity preserver*, is a subgraph $H \subseteq G$ such that for every $i \leq k$, the strongly i -connected components of H are exactly those of G . Every $(k - 1)$ -fault-tolerant strong connectivity preserver is automatically a k -connectivity preserver, but the converse is false. In directed graphs, preserving higher connectivity is strictly easier than preserving strong connectivity under faults.

Here too the results in the undirected setting are classical. The same $O(nk)$ bound of Nagamochi and Ibaraki yields optimal k -connectivity preservers in undirected graphs [NI92]. In directed graphs, however, tight bounds were previously known only in special cases. Classical results imply that if the whole graph is already k -edge-connected, then it contains a k -edge-connected spanning subgraph with $O(nk)$ edges [Dal77; Mad85; BJ05a]. More recently, linear-size preservers were obtained for the case $k = 2$ [Geo+16], and optimal $\Theta(nk)$ bounds were shown for the single-source variant [Ban+24]. For general directed graphs, the best previous bounds again followed indirectly from fault-tolerant constructions and were far from linear.

As an immediate consequence of Theorem 1.4, one obtains near-optimal higher-connectivity preservers when k is constant.

Corollary 1.5 ([HSW25]). *For every positive integer k , every n -vertex directed graph admits a k -connectivity preserver with $O(k4^k n \log n)$ edges.*

We present a different upper bound that remains nontrivial even when k grows polynomially with n .

Theorem 1.6 ([HSW25]). *For every positive integer k , every n -vertex directed graph admits a k -connectivity preserver with $O(n\sqrt{nk})$ edges.*

Theorem 1.6 gives the first $o(n^2)$ -size k -connectivity preserver for every regime $k = o(n)$. This stands in sharp contrast with the fault-tolerant setting. Because even the single-source fault-tolerant

problem requires $\Omega(n^k)$ edges [BCR18], no $o(n^2)$ upper bound is possible for $k \geq \log_2 n$ in that model. Higher-order connectivity, by contrast, continues to admit subquadratic compression well beyond the constant- k regime. Thus directed graphs display a genuine separation between preserving robustness against failures and preserving higher connectivity levels. In the undirected world these two notions coincide up to constant factors, but in directed graphs they lead to fundamentally different compression behavior.

Together, these results clarify the compression landscape for strong connectivity in directed graphs. Strong connectivity is much more compressible than general reachability, yet it already exhibits phenomena that do not arise in undirected graphs, including an exponential dependence on fault tolerance and a sharp gap between fault tolerance and higher-order connectivity.

Contribution Statement. The results in this section are based on joint work with Thatchaphol Saranurak and Benyu Wang [HSW25]. Benyu Wang and I were the main contributors to this paper, and we both made roughly equal contributions.

1.2 Graph Preconditioning

This section surveys the graph preconditioning results proved in this thesis.

1.2.1 Hopsets and Shortcut Sets

In the parallel, distributed, and streaming settings, algorithm complexity often scales with the *hop-diameter* of the graph, i.e., the maximum number of edges on any shortest path in the graph. Therefore, a popular strategy to optimize these algorithms is *diameter reduction* – adding a few edges to the input graph with the goal to reduce hop-diameter while leaving the relevant distance or reachability information unchanged. This set of additional edges is called a hopset, and was first introduced in the 90s [Tho93; UY91]. Since then, many algorithmic applications for hopsets and their relatives have been discovered [Ber12; Coh00; Fin19].³

Shortcut sets capture the reachability version of this idea. They were introduced explicitly by Thorup [Tho93], after appearing implicitly in earlier work, and they have since become a useful primitive in a variety of algorithmic settings [UY91; KS97; HKN14b; HKN14a; HKN15; FN18; JLS19; Fin19; GW20; BGW20; CFR20; KS21; ASZ20]. Informally, a shortcut set adds only edges that were already implied by reachability in the original graph, but it does so in a way that makes every reachable pair connectable by a path with far fewer hops.

³The new results in this subsection first appeared in joint work with Greg Bodwin at FOCS 2023 [BH23b].

Definition 1.7 (Shortcut sets). Let G be a directed graph. A set of additional edges H is a D -diameter-reducing shortcut set for G if the following hold:

- Every edge $(u, v) \in H$ lies in the transitive closure of G , that is, there is a directed path from u to v in G .
- For every pair of vertices s, t with t reachable from s in G , there is a path from s to t in $G \cup H$ that uses at most D edges.

For many years, the only general construction was essentially the folklore sampling argument: sample about $n^{1/2}$ vertices and add a shortcut between each reachable sampled pair. Any sufficiently long directed path is likely to contain sampled vertices near its beginning and end, so one new shortcut can bypass most of the path. This yields a shortcut set of size $|H| = O(n)$ with a diameter bound of $\tilde{O}(n^{1/2})$, and later work removed the logarithmic factors [UY91; BRR10a]. Despite its simplicity, this construction resisted polynomial improvement for roughly three decades. In a breakthrough result, Kogan and Parter [KP22d] later showed that every n -node graph admits an $O(n)$ -size shortcut set with diameter $\tilde{O}(n^{1/3})$. Their construction samples not only vertices but also carefully chosen paths, showing for the first time that the folklore $n^{1/2}$ exponent is polynomially suboptimal for shortcut sets.

Hopsets are the distance-preserving analogue of shortcut sets. Instead of merely asking that reachability be realizable in few hops, one asks for near-shortest, or exact shortest, paths to be realizable in few hops after adding auxiliary edges. Hopsets have become an important tool in the design of distance algorithms and data structures, especially in the parallel, distributed or streaming settings where the number of rounds of computation or communication steps must be bounded.

Definition 1.8 (Hopsets). Let G be a weighted graph, possibly directed, and let $\varepsilon \geq 0$. A set of additional edges H is a D -diameter-reducing $(1 + \varepsilon)$ -hopset for G if the following hold:

- Every edge $(u, v) \in H$ is assigned weight $w(u, v) = \text{dist}_G(u, v)$.
- For every pair of vertices s, t with $\text{dist}_G(s, t) < \infty$, there is a path π from s to t in $G \cup H$ using at most D edges such that $w(\pi) \leq (1 + \varepsilon) \text{dist}_G(s, t)$.

When $\varepsilon = 0$, we call H an *exact hopset*.

The same folklore sampling idea extends immediately to hopsets, giving exact and approximate hopsets of size $O(n)$ with hopbound $\tilde{O}(n^{1/2})$. More recently, Kogan and Parter [KP22d] improved the hopbound for $(1 + \varepsilon)$ -hopsets to $\tilde{O}(n^{2/5})$, and Bernstein and Wein [BW23a] refined their approach

to obtain a bound of $\tilde{O}(n^{1/3})$, matching the best known upper bound for shortcut sets. Thus, by the time of this work, the folklore barrier had been beaten for shortcut sets and for $(1 + \varepsilon)$ -hopsets, but not for exact hopsets. The case of exact hopsets remained unresolved: no construction was known with hopbound $n^{1/2-c}$, and the best lower bound was only $\Omega(n^{1/3})$ [KP22d; BHT23]. This raised a basic structural question: is exactness a genuine obstacle, or should exact hopsets ultimately behave like approximate hopsets and shortcut sets?

We resolve this question, up to polylogarithmic factors, for linear-size hopsets.

Theorem 1.9 (Exact hopset lower bound, [BH23b]). *There are n -node weighted graphs such that every exact hopset on $O(n)$ edges reduces the diameter only to $\tilde{\Omega}(n^{1/2})$.*

This theorem shows that the folklore sampling algorithm is essentially optimal for exact hopsets. In particular, it yields a clean separation between exact hopsets and $(1 + \varepsilon)$ -hopsets: allowing even arbitrarily small positive stretch changes the best known hopbound from essentially $n^{1/2}$ to $\tilde{O}(n^{1/3})$, whereas preserving exact distances provably does not. The lower bound holds for both directed and undirected input graphs, but it relies crucially on edge weights, so it does not directly settle the unweighted setting. The stronger theorem proved later in the thesis also treats general size budgets $O(p)$ for $p \in [1, n^2]$, showing that the folklore size-versus-hopbound tradeoff is near-optimal throughout that full parameter range.

The second main result of this subsection concerns shortcut sets. Before this work, the best lower bound for $O(n)$ -size shortcut sets was the $\Omega(n^{1/6})$ construction of Huang and Pettie [HP21]. Since Kogan and Parter’s upper bound is $\tilde{O}(n^{1/3})$, there remained a substantial polynomial gap in our understanding of the true optimum. We strengthen the lower-bound side as follows.

Theorem 1.10 (Shortcut set lower bound, [BH23b]). *There are n -node directed graphs such that every shortcut set on $O(n)$ edges reduces the diameter only to $\tilde{\Omega}(n^{1/4})$.*

This improves the previous exponent from $1/6$ to $1/4$. A polynomial gap to the best known upper bound of $\tilde{O}(n^{1/3})$ still remains, so determining the correct shortcut-set exponent is an important open problem. Since every $(1 + \varepsilon)$ -hopset is also a shortcut set, the same lower bound immediately applies to directed $(1 + \varepsilon)$ -hopsets.⁴ As with exact hopsets, the full theorem proved later in the thesis is more general and gives improved lower bounds for $O(p)$ -size shortcut sets over a nontrivial range of edge budgets.

Taken together, these results sharpen the emerging picture of sparse diameter-reducing augmentations. Exact hopsets are fundamentally more rigid than approximate hopsets, and the

⁴Because the shortcut-set lower bound is proved for directed graphs, this implication is immediate only in the directed setting. For undirected weighted graphs, much smaller hopbound is known for $(1 + \varepsilon)$ -hopsets [EGN19].

folklore $n^{1/2}$ construction is already near-optimal in that regime. Shortcut sets admit strictly better upper bounds, but our lower bound indicates that the room for subsequent improvements is limited. These results therefore complement the recent upper-bound progress by clarifying which improvements are possible and which are not.

The discussion here concerns existential size-diameter tradeoffs rather than the computational cost of constructing the augmenting edges. Efficient algorithmic constructions have also been studied extensively, most notably by Fineman [Fin19], Jambulapati, Liu, and Sidford [JLS19], and Kogan and Parter [KP22a]. There is likewise a distinct literature on $(1 + \varepsilon)$ -hopsets in undirected unweighted graphs, where dramatically smaller hopbounds, even $n^{o(1)}$, are known [KS97; SS99; Coh00; HP19; EN19b; ABP17; NS22]. These results underscore that the weighted exact regime studied here is qualitatively different.

Subsequent Work. The subsequent works of [VXX24] and [HXX24] directly simplified and improved some of the lower bounds presented in this section. Additionally, the works of [Ber+25] and [HRZ26] introduced the notions of Steiner shortcut sets and certified shortcut sets, respectively, and established new lower bounds for these objects.

1.2.2 DAG Covers

Another preconditioning strategy is to reduce a problem on a complicated graph to the same problem on a simpler class of graphs. In undirected graphs, a standard example is probabilistic tree embedding. Introduced explicitly by Alon, Karp, Peleg, and West [Alo+95], a probabilistic tree embedding maps a graph to a distribution over trees such that tree distances dominate the original distances and the expected distortion is small. Building on a sequence of earlier works [Kar89; Alo+95; Bar96a; Bar98b], Fakcharoenphol, Rao, and Talwar [FRT04b] obtained the asymptotically optimal $O(\log n)$ distortion bound. Fast constructions and extensions have since been studied in dynamic, online, distributed, parallel, hop-constrained, and derandomized settings [BGS17; MS09; FGH21; Bar96a; BFU20; Kha+08; GL14; BGT12; FL18; Fil22; HHZ21; Cha+99]. These embeddings have found applications in a broad range of problems, including routing, distance oracles, oblivious routing, buy-at-bulk network design, metric labeling, group Steiner tree, minimum bisection, and linear system solvers [BGT12; Kha+08; BGS17; R  c08b; HHR03; Rac02; KT02; GKR00; KLS13; Coh+14].

A related primitive is a *tree cover*. Instead of requiring a single sampled tree to preserve every pair in expectation, a tree cover asks for a small collection of trees such that for every pair of vertices, some tree in the collection approximates their distance well. Sampling from a probabilistic

tree embedding yields a tree cover with the same distortion and $O(\log n)$ trees, and there are also direct tradeoffs between distortion and the number of trees [AP92; AKP94; TZ05; BFN19; BFN22]. Tree covers have been particularly useful in routing and in metrics with additional geometry, such as Euclidean, doubling, ultrametric, and planar metrics [AKP94; AP92; AKP91; AP95; Cha+23; Ary+95; BFN22; Cha+16; Bar+05; MN07; GKR05; Kah+22; FL22].

The success of these ideas is largely confined to undirected graphs. Directed distances still lack a comparably rich collection of structural primitives. Some familiar objects either do not exist or behave much more poorly in directed graphs, while others remain much less understood. This relative scarcity of tools is reflected in the state of several basic algorithmic problems for directed graphs, including diameter approximation, disjoint shortest paths, and not-shortest path problems [Ahm+20; CGR16; Loc21; KN04]. One goal of the results presented in this section is to begin filling this gap by developing directed analogs of tree covers and probabilistic tree embeddings.⁵

The natural target class is that of directed acyclic graphs. DAGs are among the simplest directed graph families, and many distance problems become substantially easier on them than on general directed graphs. For example, shortest paths with negative weights can be computed in linear time on a DAG, whereas the analogous problem on arbitrary directed graphs remains a major challenge despite recent breakthroughs [BNW22; BCF23; Fin24; HJQ24]. Likewise, several distance-preserving primitives admit better bounds on DAGs than on general directed graphs [Bod17; KP22c; BW23b]. This makes DAGs a plausible preconditioning target in precisely the way trees are in the undirected setting. By contrast, arborescences are too restrictive for our purposes: even preserving reachability in simple directed graphs can require linearly many of them.

A first obstacle is that a subgraph-only notion is too rigid. Even a directed cycle already shows that one cannot hope for a nontrivial DAG cover if every DAG must be a subgraph of the input. At the opposite extreme, if arbitrarily many new edges are allowed, the problem becomes trivial. Fix a total order on V . Let one DAG contain, for every reachable pair $u \rightsquigarrow_G v$ with u preceding v in the order, an edge of length $\text{dist}_G(u, v)$, and let a second DAG contain the corresponding edges for pairs ordered the other way. Each DAG is acyclic, and every reachable pair appears as a single exact edge in one of the two DAGs. The right formulation therefore allows a controlled number of *additional edges*. This leads to the following definition.

Definition 1.11 (DAG cover). Let $G = (V, E)$ be a weighted directed graph. A *DAG cover* of G with f additional edges and g DAGs is a collection $C = \{D_1, \dots, D_g\}$ of weighted DAGs on vertex

⁵The new results in this subsection first appeared in joint work with Sepehr Assadi and Nicole Wein at STOC 2025 [AHW25a].

set V such that every edge of $E \cap E(D_i)$ keeps its weight from G and

$$\left| \left(\bigcup_{i=1}^g E(D_i) \right) \setminus E \right| \leq f.$$

The cover is *reachability-preserving* if for all $s, t \in V$,

$$s \rightsquigarrow_G t \iff \exists i \in [g] \text{ such that } s \rightsquigarrow_{D_i} t.$$

It is α -*distance-preserving* if it is reachability-preserving and, for all $s, t \in V$ with $s \rightsquigarrow_G t$,

$$\text{dist}_G(s, t) \leq \min_{i \in [g]} \text{dist}_{D_i}(s, t) \leq \alpha \cdot \text{dist}_G(s, t).$$

A literal directed analogue of probabilistic tree embedding is impossible, because in any DAG at least one of $\text{dist}(s, t)$ and $\text{dist}(t, s)$ must be infinite. We therefore use a one-sided definition that conditions on the event that reachability from s to t is preserved.

Definition 1.12 (DAG embedding). Let $TC(G) = \{(s, t) \in V(G) \times V(G) : s \rightsquigarrow_G t\}$ denote the transitive closure of G . An α -*distortion DAG embedding* of G with f additional edges is a distribution $\mathcal{D}$ over weighted DAGs on vertex set $V(G)$ such that

$$\left| \left(\bigcup_{D \in \text{supp}(\mathcal{D})} E(D) \right) \setminus E(G) \right| \leq f,$$

and for every $(s, t) \in TC(G)$, if $D \sim \mathcal{D}$ then

$$\Pr[s \rightsquigarrow_D t] \geq \frac{1}{2} \quad \text{and} \quad \mathbb{E}[\text{dist}_D(s, t) \mid s \rightsquigarrow_D t] \leq \alpha \cdot \text{dist}_G(s, t).$$

An important theme in our results is that the difficulty of the problem depends strongly on whether the additional-edge budget is expressed in terms of n or in terms of m . In the n -parameterized regime, even allowing almost quadratically many new edges does not accomplish much.

Theorem 1.13 ([AHW25a]). *There exists a family of n -vertex directed graphs such that any reachability-preserving DAG cover with at most $n^{2-o(1)}$ additional edges requires at least $n^{1-o(1)}$ DAGs.*

This is essentially the strongest possible impossibility result in this regime, since n DAGs are always enough by taking one shortest-path DAG from each source. The picture changes completely

when the number of added edges is measured relative to the original sparsity m . Reachability alone is already easy in this setting: every directed graph admits a reachability-preserving DAG cover with $O(m)$ additional edges and only two DAGs (this follows straightforwardly from the observation that every directed graph is a DAG of its strongly connected components). Thus the interesting question in the near-linear-in- m regime is not reachability but approximately preserving shortest-path distances.

The main upper-bound result we prove for DAG covers is that, in the $\tilde{O}(m)$ regime, one can recover a directed analogue of the classical tree-cover guarantee.

Theorem 1.14 ([AHW25a]). *Let G be an n -vertex weighted directed graph with polynomially bounded positive integer edge weights. Then G admits an $O(\log^3 n \cdot \log \log n)$ -distance-preserving DAG cover with $O((m + n) \log^3 n)$ additional edges and $O(\log n)$ DAGs. Moreover, such a cover can be constructed in $\tilde{O}(m)$ time with high probability.*

This theorem gives polylogarithmic bounds on both the distortion and the number of DAGs while keeping the total size of the cover nearly linear. From the perspective of preconditioning, this is the key point: the simplified objects are sparse enough to remain faithful to the input size, but structured enough to support algorithms that exploit acyclicity.

A randomized version of the same construction yields a directed analogue of probabilistic tree embedding.

Theorem 1.15 ([AHW25a]). *Let G be an n -vertex weighted directed graph with positive integer edge weights and polynomial aspect ratio. Then there exists a distribution $\mathcal{D}$ over weighted DAGs on vertex set $V(G)$ satisfying the following properties:*

1. *The union of the edge sets of DAGs in the support of $\mathcal{D}$ uses at most $O((m + n) \log^3 n)$ additional edges.*
2. *For every $(s, t) \in TC(G)$, if $D \sim \mathcal{D}$ then s can reach t in D with probability at least $1/2$, and*

$$\mathbb{E}[\text{dist}_D(s, t) \mid s \rightsquigarrow_D t] = O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

3. *A sample from $\mathcal{D}$ can be generated in $\tilde{O}(m)$ time.*

Taken together, these results show that the near-linear regime behaves much more like the undirected world than the nearly quadratic regime does. With $\tilde{O}(m)$ additional edges, DAG covers and DAG embeddings achieve polylogarithmic distortion and admit near-linear-time construction,

whereas with $n^{2-o(1)}$ additional edges one still cannot hope for a subpolynomial number of DAGs in general.

The full version of the paper underlying this subsection also proves a complementary lower bound for exact distance preservation in the near-linear regime [AHW25a]. More precisely, it constructs n -vertex, m -edge graphs for which any exact distance-preserving DAG cover with $m^{1+\varepsilon}$ additional edges requires $\Omega(n^{1/6})$ DAGs, for a sufficiently small constant $\varepsilon > 0$. That result is omitted from this thesis for space reasons. Interested readers can consult the full version of the paper for details.

Subsequent Work. The subsequent work [Fil26] greatly simplifies the proof of our upper bound result for DAG covers (Theorem 1.14) and obtains an improved approximation factor of $\tilde{O}(\log n)$. Additionally, the work of [HJS26] introduces the related notion of DAG projections, which allow duplicate vertices and Steiner vertices in their DAG constructions. Remarkably, DAG projections can preserve distances up to a $(1 + \varepsilon)$ -factor.

1.3 Organization

The remainder of this thesis is organized as follows. Chapters 2 and 3 concern results in graph compression, while Chapters 4 and 5 concern results in graph preconditioning.

- Chapter 2 develops the distance-preserver results, including new upper and lower bounds for exact and finite-stretch preservers and a reduction from directed exact distance preservers to the undirected setting. The contents of this chapter originally appeared in [HXX24] and are based on joint work with Yinzhan Xu and Zixuan Xu that appeared in SODA 2025.
- Chapter 3 develops the strong-connectivity-preserver results, including fault-tolerant strong connectivity preservers and higher-order connectivity preservers in directed graphs. The contents of this chapter originally appeared in [HSW25] and are based on joint work with Thatchaphol Saranurak and Benyu Wang that appeared in FOCS 2025.
- Chapter 4 develops the results on hopsets and shortcut sets, including the lower bounds proved for exact hopsets and for shortcut sets. The contents in this section originally appeared in [BH23a] and are based on joint work with Greg Bodwin that appeared in FOCS 2023.
- Chapter 5 develops the results on DAG covers and DAG embeddings, including the near-linear-size upper bounds and the accompanying lower bounds. The contents in this section originally appeared in [AHW25b] and are based on joint work with Sepehr Assadi and Nicole Wein.

- Chapter 6 concludes with a summary of the main contributions and a discussion of open problems.

Chapter 2

Distance Preservers

This chapter develops several results about distance preservers, including new upper and lower bounds for exact and finite-stretch preservers and a reduction from directed exact distance preservers to the undirected setting.¹

2.1 Preliminaries

In this section, we present the notations and definitions used in this section.

Notation

- For integers $n \geq 0$, we use $[n]$ to denote $\{1, \dots, n\}$ and $\llbracket n \rrbracket$ to denote $\{0, \dots, n-1\}$. For integers $i, j \geq 0$, where $i < j$, we use $[i, j]$ to denote $\{i, i+1, \dots, j-1, j\}$.
- Let $G = (V, E, w)$ be a weighted graph with weight function $w : E \rightarrow \mathbb{R}$. For vertex $v \in V$, we use $\deg_G(v)$ to denote the number of edges incident to v in G . For $s, t \in V$, we use $w(s, t)$ to denote the edge weight of the edge from s to t , $\text{dist}_G(s, t)$ to denote the distance from s to t in G and $\pi_G(s, t)$ to denote a shortest path from s to t . The subscripts G might be dropped if it is clear from context.
- For $n, p \geq 1$, we use $\text{UDP}(n, p)$ (resp. $\text{DDP}(n, p)$, $\text{DADP}(n, p)$) to denote the best bound on the size of exact distance preservers for n -node undirected (resp. directed, directed acyclic) weighted graphs with p demand pairs.
- For a vertex $v \in V(G)$, we use $\text{outdeg}_G(v)$ to denote the number of outgoing edges incident

¹The contents of this chapter originally appeared in [HXX24] and are based on joint work with Yinzhan Xu and Zixuan Xu that appeared in SODA 2025.

to v in G , and we use $\text{indeg}_G(v)$ to denote the number of incoming edges incident to v in G . We let $\text{deg}_G(v) = \text{indeg}_G(v) + \text{outdeg}_G(v)$.

- We will frequently use symbol π to denote paths. For a simple path π , $|\pi|$ denotes its number of edges and $w(\pi)$ denotes the total edge weights of edges on π . For two vertices x, y on π where x appears before y , $\pi[x, y]$ or $\pi[x \rightsquigarrow y]$ denotes the subpath of π from x to y .
- Given a path π and a set of nodes S , we use $\pi \cap S$ to denote the subpath of π induced on the nodes of S . Formally, $\pi \cap S$ is the path obtained by listing the nodes in $V(\pi) \cap S$ in the order they appear on path π .
- Given a graph G , a set of demand pairs $P \subseteq V(G) \times V(G)$, and a distance preserver H of G, P , we say that H is a minimal distance preserver if no edge can be deleted from H while ensuring that H remains a distance preserver of G, P . If shortest paths in G are unique shortest paths, then there is a unique minimal distance preserver H of G .

Definitions

Definition 2.1 (Tiebreaking Scheme). Given a graph G and a set of demand pairs $P \subseteq V(G) \times V(G)$, we define a tiebreaking function $\pi(\cdot, \cdot)$ with respect to G, P as follows. For each demand pair $(s, t) \in P$, fix a shortest path $\pi_{s,t}$ in G . Then we define the output of tiebreaking function $\pi(\cdot, \cdot)$ on input (s, t) to be

$$\pi(s, t) = \pi_{s,t}$$

for all $(s, t) \in P$. Additionally, we let $\pi(P)$ denote the image of P under $\pi(\cdot, \cdot)$, i.e., $\pi(P)$ denotes the collection of paths $\{\pi(s, t) \mid (s, t) \in P\}$. Likewise, we let

$$\cup_{\pi \in \pi(P)} \pi$$

denote the graph obtained by taking the (non-disjoint) union of the paths in set $\pi(P)$.

Definition 2.2 (Consistent Tiebreaking Scheme). Let $\pi(\cdot, \cdot)$ be a tiebreaking scheme with respect to G, P . Then $\pi(\cdot, \cdot)$ is consistent if for all paths $\pi_1, \pi_2 \in \pi(P)$ and nodes $x, y \in V(G)$ such that x precedes y in both π_1 and π_2 , then we have that $\pi_1[x \rightsquigarrow y] = \pi_2[x \rightsquigarrow y]$.

Definition 2.3 (Branching Events [CE06]). Let G be a directed graph, let P be a set of demand pairs, and let $\pi(\cdot, \cdot)$ be a tiebreaking scheme. Let $H = \cup_{(s,t) \in P} \pi(s, t)$ be the exact distance preserver implied by $\pi(\cdot, \cdot)$.

A branching event is a pair of directed edges $(x, y), (x, z) \in E(H)$ in H , for any $x, y, z \in V(G)$. For each branching event $(x, y), (x, z)$, we associate a pair of paths $\pi_1, \pi_2 \in \pi(P)$ such that $(x, y) \in \pi_1$ and $(x, z) \in \pi_2$. If there is more than one choice of paths π_1, π_2 , then we choose arbitrarily.

We say that a branching event $(x, y), (x, z)$ is *between* paths $\pi_1, \pi_2 \in \pi(P)$ if paths π_1, π_2 are associated with branching event $(x, y), (x, z)$ as in Definition 2.3.

2.2 Directed to Undirected Preserver Reduction

The goal of this section will be to prove Theorem 1.2 and Corollary 1.3. We restate Theorem 1.2 below.

Theorem. *If $\text{UDP}(n, p) = O(n^\lambda p^\mu + n)$ for constants $\lambda, \mu > 0$, then*

$$\text{DDP}(n, p) = O\left(n^{\frac{1}{2-\lambda}} p^{\frac{1+\mu-\lambda}{2-\lambda}} + n^{1/2} p + n\right).$$

We will need several intermediate claims and lemmas before we can prove Theorem 1.2. Let G, P be an n -node directed, weighted graph and a set of $|P| = p$ demand pairs, such that the minimal distance preserver H of G, P has

$$|E(H)| = \Theta(\text{DDP}(n, p))$$

edges. Let $\pi(\cdot, \cdot)$ be a tiebreaking scheme associated with G, P . We will make the following assumptions on G, P, H , and $\pi(\cdot, \cdot)$, which we can assume without loss of generality:

1. Path $\pi(s, t)$ is a unique shortest $s \rightsquigarrow t$ path in G for all $s, t \in V(G)$. In particular, this implies that $\pi(\cdot, \cdot)$ is consistent.
2. Paths in $\pi(P)$ are pairwise edge-disjoint.
3. Every node $v \in V(G)$ has degree at least $\deg_H(v) \geq \frac{|E(H)|}{4n}$ in H .
4. Every path $\pi \in \pi(P)$ has at least $|\pi| \geq \frac{|E(H)|}{4p}$ edges.

Assumptions 1 and 2 hold without loss of generality due to the Independence Lemma for weighted distance preservers (Lemma 33 of [BHT23]). Note that Assumption 1 implies that tiebreaking scheme $\pi(\cdot, \cdot)$ is consistent. Assumptions 3 and 4 follow from standard arguments in this area (complete proofs of these assumptions can be found in Appendix A of [HXX24]).

Our extremal reduction from $\text{DDP}(n, p)$ to $\text{UDP}(n, p)$ will happen in two steps. First, in Section 2.2, we will reduce DDP to DADP , culminating in the proof of Lemma 2.8. Second, in Section 2.2 we will show that $\text{DADP}(n, p) \leq \text{UDP}(n, p)$. We combine these two reductions in Section 2.2 to complete the proofs of Theorem 1.2 and Corollary 1.3.

Directed to DAG Reduction

We begin our reduction from DDP to DADP with the following observation about the structure of paths in $\pi(P)$.

Claim 2.4 (Path Intersection Claim). *Fix two distinct shortest paths $\pi_1, \pi_2 \in \pi(P)$. Let*

$$v_1, \dots, v_k \in V(\pi_1) \cap V(\pi_2)$$

be the nodes in $V(\pi_1) \cap V(\pi_2)$, listed in the order they appear in path π_1 . Then nodes in $V(\pi_1) \cap V(\pi_2)$ appear on path π_2 in the reverse order $v_k, \dots, v_1$.

Proof. Suppose for the sake of contradiction that there are two vertices $x, y \in V(\pi_1) \cap V(\pi_2)$, where $x \neq y$, such that node x comes before node y in path π_1 and in path π_2 . Then by the consistency of tiebreaking scheme $\pi(\cdot, \cdot)$, it follows that $\pi_1[x, y] = \pi_2[x, y]$. However, this implies that there exists an edge $e \in E(\pi_1) \cap E(\pi_2)$, contradicting Assumption 2 about the collection of paths $\pi(P)$. $\square$

Now, for any path $\pi^* \in \pi(P)$, we will define an associated collection of acyclic paths Π_{DAG} .

Definition 2.5 (DAG Path Collection). Fix a path $\pi^* \in \pi(P)$. For every other path $\pi \in \pi(P) \setminus \{\pi^*\}$ we will define a subpath π_{DAG} of π induced on the nodes of π^* . This will yield an acyclic collection of paths $\Pi_{\text{DAG}} = \{\pi_{\text{DAG}} \mid \pi \in \pi(P) \setminus \{\pi^*\}\}$. Given a path $\pi \in \pi(P) \setminus \{\pi^*\}$, we construct π_{DAG} as follows.

- Let $v_1, \dots, v_k \in V(\pi^*) \cap V(\pi)$ be the nodes in $V(\pi^*) \cap V(\pi)$, listed in the order they appear in path π .
- We define π_{DAG} to be

$$\pi_{\text{DAG}} = (v_1, \dots, v_k).$$

- We assign edge $(v_i, v_{i+1}) \in \pi_{\text{DAG}}$ weight $w(v_i, v_{i+1}) = \text{dist}_G(v_i, v_{i+1})$.

We define directed graph D be the graph obtained by taking the (non-disjoint) union of paths in Π_{DAG} , i.e.,

$$D = \bigcup_{\pi_{\text{DAG}} \in \Pi_{\text{DAG}}} \pi_{\text{DAG}}.$$

We now verify that Π_{DAG} is an acyclic collection of shortest paths in graph D .

Claim 2.6. *Fix a path $\pi^* \in \pi(P)$, and let Π_{DAG} and D be as defined in Definition 2.5. Then graph D is a DAG, and π_{DAG} is a unique shortest path in D for all $\pi_{\text{DAG}} \in \Pi_{\text{DAG}}$.*

Proof. Let $<_D$ be the total ordering on the vertices of $V(D)$ that is the reverse of the ordering of the vertices in path π^* . That is, if node x comes before node y in π^* , then $y <_D x$. It is immediate that $<_D$ is a valid topological ordering for directed graph D because every path $\pi_{\text{DAG}} \in \Pi_{\text{DAG}}$ respects $<_D$ by Claim 2.4.

To see that all paths in Π_{DAG} are unique shortest paths in D , observe the following properties of graph D :

- For all $s, t \in V(D)$,

$$\text{dist}_D(s, t) \geq \text{dist}_G(s, t).$$

- For all vertices $s, t \in V(D)$, if $s, t \in V(\pi_{\text{DAG}})$ for some $\pi \in \Pi_{\text{DAG}}$, then

$$\text{dist}_D(s, t) = \text{dist}_G(s, t).$$

These two properties imply that every path $\pi_{\text{DAG}} \in \Pi_{\text{DAG}}$ is a shortest path in D . To see that π_{DAG} is a *unique* shortest path in D , observe that if π_{DAG} is an $s \rightsquigarrow t$ path in D , then path π_{DAG} is the unique (possibly non-contiguous) subpath of path $\pi(s, t) \subseteq G$ in D . Then since $\pi(s, t)$ is a unique shortest path in G , we conclude that π_{DAG} is a unique shortest path in D . $\square$

Our end goal is to upper bound the size $|E(H)|$ of graph H by upper bounding the size $|E(D)|$ of DAG D . As such, we need to prove that $|E(D)|$ grows with $|E(H)|$. We achieve this in a roundabout way with the following claim.

Claim 2.7. *Fix a shortest path $\pi^* \in \pi(P)$, and let Π_{DAG} and D be as defined in Definition 2.5. Let $Q \subseteq P$ be the set of demand pairs (s, t) in P such that $\pi(s, t)$ intersects path π^* at exactly 1 node. Then*

$$\sum_{v \in V(\pi^*)} \deg_H(v) = \Theta(|\pi^*| + |Q| + |E(D)|).$$

Proof. In order to prove Claim 2.7, it will suffice to tightly bound the number of distinct edges incident to nodes in π^* , since

$$\sum_{v \in V(\pi^*)} \deg_H(v) = \Theta(|\{e \in E(H) \mid e = (u, v), \text{ where } u \in \pi^* \text{ or } v \in \pi^*\}|).$$

Let $\Pi \subseteq \pi(P)$ be the set of all paths in $\pi(P)$ in graph H that have a nonempty intersection with path π^* . Observe that every edge incident to a node in π^* in H is contained in a path in Π . We partition the set of paths Π into sets Π_1, Π_2 , and Π_3 , where

$$\Pi_1 = \{\pi^*\}, \quad \Pi_2 = \pi(Q), \quad \text{and} \quad \Pi_3 = \Pi \setminus (\Pi_1 \cup \Pi_2).$$

We let E_1, E_2 , and E_3 denote the set of edges in paths in Π_1, Π_2 , and Π_3 , respectively, that are incident to nodes in π^* . Then since sets Π_1, Π_2 , and Π_3 partition Π , and paths in Π are pairwise edge-disjoint by Assumption 2, we conclude that

$$\sum_{v \in V(\pi^*)} \deg_H(v) = \Theta(|E_1| + |E_2| + |E_3|).$$

We will bound the sizes of E_1, E_2 , and E_3 separately.

- (Size of $|E_1|$.) Set E_1 has size $|E_1| = \Theta(|\pi^*|)$.
- (Size of $|E_2|$.) Every path $\pi \in \pi(Q)$ has either 1 or 2 edges incident to a node in π^* . Then since paths in $\pi(P)$ are pairwise edge-disjoint by Assumption 2, the edges of the paths in $\pi(Q)$ contribute $\Theta(|Q|)$ edges to the sum of the degrees of the nodes in π^* in H .
- (Size of $|E_3|$.) Fix a path $\pi \in \Pi_3$, and let π_{DAG} denote the corresponding path in Π_{DAG} . We observe that since $|V(\pi) \cap V(\pi^*)| > 1$,

$$|\{(u, v) \in E(\pi) \mid u \in V(\pi^*) \text{ or } v \in V(\pi^*)\}| = \Theta(|V(\pi) \cap V(\pi^*)|) = \Theta(|\pi_{\text{DAG}}|).$$

Then we conclude that

$$|E_3| = \sum_{\pi \in \Pi_3} |\{(u, v) \in E(\pi) \mid u \in V(\pi^*) \text{ or } v \in V(\pi^*)\}| = \Theta\left(\sum_{\pi \in \Pi_3} |\pi_{\text{DAG}}|\right) = \Theta(|E(D)|),$$

where the first and last equality follow from the fact that paths in Π are pairwise edge-disjoint and that paths in Π_{DAG} are pairwise edge-disjoint, respectively.

Putting it all together, we conclude that

$$\sum_{v \in V(\pi^*)} \deg_H(v) = \Theta(|E_1| + |E_2| + |E_3|) = \Theta(|\pi^*| + |Q| + |E(D)|),$$

as desired. □

We now have all the ingredients we need to prove our reduction from DDP to DADP.

Lemma 2.8 (Directed to DAG Reduction).

$$\text{DDP}(n, p) \leq O \left(\text{DADP} \left(\frac{\text{DDP}(n, p)}{p}, p \right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right).$$

Proof. Let G, P be an n -node directed, weighted graph and a set of $|P| = p$ demand pairs, such that the minimal distance preserver H of G, P has

$$|E(H)| = \Theta(\text{DDP}(n, p))$$

edges. Let $\pi(\cdot, \cdot)$ be a tiebreaking scheme associated with G, P . We can assume that Assumptions 1-4 discussed at the beginning of Section 2.2 hold without loss of generality. In particular, these assumptions imply that there exists a path $\pi^* \in \pi(P)$ such that $|\pi^*| = \Theta \left(\frac{|E(H)|}{p} \right)$, since the average length of paths in $\pi(P)$ is $\Theta \left(\frac{|E(H)|}{p} \right)$, and every path in $\pi(P)$ has length at least $\frac{|E(H)|}{4p}$.

Let the collection of paths Π_{DAG} and DAG D be as defined in Definition 2.5, with respect to path π^* . By Assumption 3, we have that

$$\sum_{v \in V(\pi^*)} \deg_H(v) = \Omega \left(\frac{|E(H)|}{p} \cdot \frac{|E(H)|}{n} \right) = \Omega \left(\frac{\text{DDP}(n, p)^2}{np} \right).$$

Applying Claim 2.7, it follows that

$$\Omega \left(\frac{\text{DDP}(n, p)^2}{np} \right) \leq \Theta(|\pi^*| + |Q| + |E(D)|),$$

where $Q \subseteq P$ is the set of demand pairs $(s, t) \in P$ such that $\pi(s, t)$ intersects path π^* at exactly 1 node. We make the following observations:

- $|\pi^*| = \Theta \left(\frac{\text{DDP}(n, p)}{p} \right)$ by our choice of π^* ,
- $|Q| \leq p$ since $Q \subseteq P$, and
- $|E(D)| \leq \text{DADP}(|V(D)|, |\Pi_{\text{DAG}}|)$.

The final observation follows from the fact that each path $\pi_{\text{DAG}} \in \Pi_{\text{DAG}}$ is a unique shortest path

between its endpoints in D by Claim 2.6. Consequently, if we define demand pairs

$$P_D = \{(s, t) \in V(D) \times V(D) \mid \exists \pi_{\text{DAG}} \in \Pi_{\text{DAG}} \text{ s.t. } \pi_{\text{DAG}} \text{ is an } s \rightsquigarrow t \text{ path in } D\},$$

then any distance preserver of D, P_D will contain every path in Π_{DAG} and thus $E(D)$. Using our observations and the fact that $|\Pi_{\text{DAG}}| \leq p$ and $|V(D)| \leq |V(\pi^*)| = \Theta(\frac{\text{DDP}(n, p)}{p})$, we obtain the following inequality:

$$\Omega\left(\frac{\text{DDP}(n, p)^2}{np}\right) \leq \Theta(|\pi^*| + |Q| + |E(D)|) \leq \Theta\left(\frac{\text{DDP}(n, p)}{p} + p + \text{DADP}\left(\frac{\text{DDP}(n, p)}{p}, p\right)\right)$$

We now split our analysis into three cases, based on which of the three terms on the right hand side is the largest.

- (Term $\frac{\text{DDP}(n, p)}{p}$ dominates.) In this case, we find that

$$\frac{\text{DDP}(n, p)^2}{np} \leq O\left(\frac{\text{DDP}(n, p)}{p}\right).$$

Rearranging, we conclude that

$$\text{DDP}(n, p) = O(n).$$

- (Term p dominates.) In this case, we find that

$$\frac{\text{DDP}(n, p)^2}{np} \leq O(p).$$

Then multiplying both sides of the inequality by np and taking the square root of both sides, we get

$$\text{DDP}(n, p) = O(n^{1/2}p).$$

- (Term $\text{DADP}\left(\frac{\text{DDP}(n, p)}{p}, p\right)$ dominates.) In this case, our inequality is

$$\frac{\text{DDP}(n, p)^2}{np} \leq O\left(\text{DADP}\left(\frac{\text{DDP}(n, p)}{p}, p\right)\right).$$

Multiplying both sides by np , and taking the square root of both sides, we obtain

$$\text{DDP}(n, p) = O\left(\text{DADP}\left(\frac{\text{DDP}(n, p)}{p}, p\right)^{1/2} \cdot n^{1/2}p^{1/2}\right)$$

Combining our three bounds from our three different cases, we conclude that

$$\text{DDP}(n, p) \leq O \left(\text{DADP} \left(\frac{\text{DDP}(n, p)}{p}, p \right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right),$$

completing the proof. $\square$

DAG to Undirected Reduction

We now present our reduction to convert shortest paths in DAGs to shortest paths in undirected graphs.

Lemma 2.9 (DAG to Undirected Reduction).

$$\text{DADP}(n, p) \leq \text{UDP}(n, p)$$

Proof. Let $G = (V, E, w)$ be an n -node weighted directed acyclic graph, and let $P \subseteq V(G) \times V(G)$ be a set of $|P| = p$ demand pairs. Let $<_G$ be a topological ordering of DAG G , and let $v_1, \dots, v_n$ be the nodes of G , ordered according to $<_G$. Let W be a sufficiently large number (e.g., chose W to be larger than the sum of all edge weights in G). We may assume without loss of generality that for all $(s, t) \in P$, there exists a unique shortest $s \rightsquigarrow t$ path $\pi(s, t)$ in G , by the Independence Lemma for weighted distance preservers (Lemma 33 of [BHT23]).

We define an undirected graph $G' = (V, E', w')$ as follows. For every directed edge $(v_i, v_j) \in E(G)$, add the undirected edge (v_i, v_j) to $E(G')$, and assign it weight

$$w'(v_i, v_j) := w(v_i, v_j) + W \cdot (j - i).$$

This completes the construction of graph G' .

Fix a pair of nodes $v_i, v_j \in V$ such that v_i can reach v_j in G . Let π denote the unique shortest $v_i \rightsquigarrow v_j$ path in G , and let π' denote a shortest $v_i \rightsquigarrow v_j$ path in G' . We claim that paths π and π' are identical, up to a reweighting of their edges. Formally, we claim if path π has the corresponding node sequence $(x_1, x_2, \dots, x_k)$, then π' has the corresponding node sequence $(x_1, x_2, \dots, x_k)$, as well.

Suppose that path π has the corresponding node sequence $(x_1, x_2, \dots, x_k)$, where $x_1 = v_i$ and $x_k = v_j$. For all $\ell \in [1, k]$, let $q_\ell \in [1, n]$ be the index such that node $x_\ell = v_{q_\ell}$. Notice that since π

is a path in DAG G , we must have that $q_\ell < q_{\ell+1}$ for all $\ell \in [1, k-1]$. Then the distance between v_i and v_j in G' is at most

$$\begin{aligned}
\text{dist}_{G'}(v_i, v_j) &\leq w'(\pi) = \sum_{i=1}^{k-1} w'(x_i, x_{i+1}) = \sum_{i=1}^{k-1} (w(x_i, x_{i+1}) + W \cdot |q_{i+1} - q_i|) \\
&= \sum_{i=1}^{k-1} (w(x_i, x_{i+1}) + W \cdot (q_{i+1} - q_i)) \\
&= w(\pi) + W \cdot (q_k - q_1) \\
&= \text{dist}_G(s, t) + W \cdot (j - i).
\end{aligned}$$

In the above sequence of inequalities, we made use of the fact that since $q_\ell < q_{\ell+1}$ for all $\ell \in [1, k-1]$, it follows that

$$\sum_{i=1}^{k-1} |q_{i+1} - q_i| = \sum_{i=1}^{k-1} (q_{i+1} - q_i) = q_k - q_1 = j - i.$$

Let path π' have the corresponding node sequence $(y_1, y_2, \dots, y_{k'})$, where $y_1 = v_i$ and $y_{k'} = v_j$. For all $\ell \in [1, k']$, let $r_\ell \in [1, n]$ be the index such that node $y_\ell = v_{r_\ell}$. We claim that since π' is a shortest $v_i \rightsquigarrow v_j$ path in G' , we have that $r_\ell < r_{\ell+1}$ for all $\ell \in [1, k'-1]$. Informally speaking, this is equivalent to claiming that path π' respects the total order $<_G$ in undirected graph G' .

Suppose towards contradiction that there exists an $\ell \in [1, k]$ such that $r_\ell > r_{\ell+1}$. Then

$$\sum_{i=1}^{k'-1} |r_{i+1} - r_i| > \sum_{i=1}^{k'-1} (q_{i+1} - q_i) = q_{k'} - q_1 = j - i.$$

This immediately implies that

$$\begin{aligned}
w'(\pi') &= \sum_{i=1}^{k'-1} (w(y_i, y_{i+1}) + W \cdot |r_{i+1} - r_i|) \\
&\geq w(\pi') + W \cdot (j - i + 1) \\
&= (w(\pi') + W) + W \cdot (j - i) \\
&> w(\pi) + W \cdot (j - i) \\
&= \text{dist}_{G'}(s, t),
\end{aligned}$$

where the final inequality follows the fact that if W is sufficiently large (e.g., larger than the sum of all edge weights in G), then $W > w(\pi)$. This contradicts our assumption that π' is a shortest

$v_i \rightsquigarrow v_j$ path in G' , so we conclude that $r_\ell < r_{\ell+1}$ for all $\ell \in [1, k']$.

Then by identical calculations as before, we have that

$$\begin{aligned} w'(\pi') &= \sum_{i=1}^{k'-1} (w(y_i, y_{i+1}) + W \cdot (q_{i+1} - q_i)) \\ &= w(\pi') + W \cdot (q_k - q_1) \\ &= w(\pi') + W \cdot (j - i) \end{aligned}$$

Since π' is a shortest path in G' , then

$$w'(\pi') = w(\pi') + W \cdot (j - i) \leq \text{dist}_{G'}(v_i, v_j) \leq \text{dist}_G(v_i, v_j) + W \cdot (j - i),$$

so $w(\pi') \leq \text{dist}_G(v_i, v_j)$. Since path π is the unique shortest $v_i \rightsquigarrow v_j$ path in G , we conclude that paths π and π' are identical, e.g., $(x_1, \dots, x_k) = (y_1, \dots, y_{k'})$.

Then every path $\pi \in \pi(P)$ is a unique shortest path between its endpoints in undirected graph G' . Informally, this means that the unique shortest paths between demand pairs P are identical in G and G' . As a consequence, any distance preserver of G, P is a distance preserver of G', P , and vice versa. We conclude that $\text{DADP}(n, p) \leq \text{UDP}(n, p)$. $\square$

Finishing the Reduction

We can chain our reductions in Lemmas 2.8 and 2.9 to prove Theorem 1.2.

Proof of Theorem 1.2. Suppose that $\text{UDP}(n, p) = O(n^\lambda p^\mu + n)$. Then we calculate

$$\begin{aligned} \text{DDP}(n, p) &\leq O \left(\text{DADP} \left(\frac{\text{DDP}(n, p)}{p}, p \right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right) \\ &\leq O \left(\text{UDP} \left(\frac{\text{DDP}(n, p)}{p}, p \right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right) \\ &\leq O \left(\left(\left(\frac{\text{DDP}(n, p)}{p} \right)^\lambda p^\mu + \frac{\text{DDP}(n, p)}{p} \right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right) \\ &\leq O \left(\left(\frac{\text{DDP}(n, p)^{\lambda/2}}{p^{\lambda/2}} \cdot p^{\mu/2} + \frac{\text{DDP}(n, p)^{1/2}}{p^{1/2}} \right) \cdot n^{1/2} p^{1/2} + n^{1/2} p + n \right) \\ &\leq O \left(\text{DDP}(n, p)^{\lambda/2} \cdot n^{1/2} p^{\frac{1+\mu-\lambda}{2}} + \text{DDP}(n, p)^{1/2} \cdot n^{1/2} + n^{1/2} p + n \right) \end{aligned}$$

This upper bound has four terms. Suppose the first term $\text{DDP}(n, p)^{\lambda/2} \cdot n^{1/2} p^{\frac{1+\mu-\lambda}{2}}$ dominates. Then

$$\text{DDP}(n, p)^{1-\lambda/2} \leq O\left(n^{1/2} p^{\frac{1+\mu-\lambda}{2}}\right),$$

so in this case we conclude

$$\text{DDP}(n, p) \leq O\left(n^{\frac{1}{2-\lambda}} p^{\frac{1+\mu-\lambda}{2-\lambda}}\right).$$

Likewise, when the second term $\text{DDP}(n, p)^{1/2} \cdot n^{1/2}$ dominates, we get that $\text{DDP}(n, p) = O(n)$.

Putting it all together, we conclude that

$$\text{DDP}(n, p) = O\left(n^{\frac{1}{2-\lambda}} p^{\frac{1+\mu-\lambda}{2-\lambda}} + n^{1/2} p + n\right).$$

□

We can use a similar analysis to prove Corollary 1.3. We restate Corollary 1.3 in greater detail below for convenience.

Corollary 2.10 (cf. Corollary 1.3). *Suppose that consistent tiebreaking schemes are optimal for directed distance preservers when $p \leq n^{2/3}$, or equivalently,*

$$\text{DDP}(n, p) = \Theta(\min(n^{2/3} p + n, np^{1/2})) = \Theta(n^{2/3} p + n),$$

when $p \leq n^{2/3}$. Then consistent tiebreaking schemes are optimal for undirected distance preservers when $p \leq n$, or equivalently,

$$\text{UDP}(n, p) = \Theta(\min(n^{1/2} p + n, np^{1/2})) = \Theta(n^{1/2} p + n),$$

when $p \leq n$.

Proof. Suppose that

$$\text{DDP}(n, p) = \Theta(\min(n^{2/3} p + n, np^{1/2})) = \Theta(n^{2/3} p + n) = \Theta(n^{2/3} p),$$

when $n^{1/3} \leq p \leq n^{2/3}$. Now by combining Lemmas 2.8 and 2.9, we get the inequality

$$\text{DDP}(n, p) \leq O\left(\text{UDP}\left(\frac{\text{DDP}(n, p)}{p}, p\right)^{1/2} \cdot n^{1/2} p^{1/2} + n^{1/2} p + n\right).$$

Plugging in $\text{DDP}(n, p) = \Theta(n^{2/3}p)$, we compute that

$$n^{2/3}p \leq O\left(\text{UDP}\left(n^{2/3}, p\right)^{1/2} \cdot n^{1/2}p^{1/2} + n^{1/2}p + n\right).$$

Let $c > 0$ be a universal constant such that

$$n^{2/3}p \leq c \cdot \left(\text{UDP}\left(n^{2/3}, p\right)^{1/2} \cdot n^{1/2}p^{1/2} + n^{1/2}p + n\right),$$

for sufficiently large n . Then for all $10cn^{1/3} \leq p \leq n^{2/3}$,

$$n^{2/3}p \leq 10c \cdot \text{UDP}\left(n^{2/3}, p\right)^{1/2} \cdot n^{1/2}p^{1/2}.$$

Dividing both sides by $10cn^{1/2}p^{1/2}$ and taking the square root, we obtain

$$\text{UDP}(n^{2/3}, p) = \Omega(n^{1/3}p).$$

If we let $N = n^{2/3}$, this is equivalent to

$$\text{UDP}(N, p) = \Omega(N^{1/2}p),$$

for $10c \cdot N^{1/2} \leq p \leq N$. Moreover, it is true unconditionally that $\text{UDP}(N, p) = \Theta(N)$ when $p \leq N^{1/2}$. We conclude that

$$\text{UDP}(N, p) = \Omega(N^{1/2}p + N),$$

for $p \leq N$. Consistent tiebreaking schemes imply that

$$\text{UDP}(N, p) = O(\min(N^{1/2}p + N, Np^{1/2})) = O(N^{1/2}p + N),$$

when $p \leq N$, so we conclude that

$$\text{UDP}(N, p) = \Theta(N^{1/2}p + N) = \Theta(\min(N^{1/2}p + N, Np^{1/2})),$$

when $p \leq N$, as desired. □

2.3 Exact Distance Preserver Upper Bound

The goal of this section will be to prove the following theorem.

Theorem 2.11. *Every n -node directed unweighted graph G with demand pairs P of size $p = |P|$ admits a distance preserver with*

$$\tilde{O}\left(n^{5/6}p^{2/3} + n\right)$$

edges.

We will need to introduce some preliminary definitions first.

Preliminary Definitions

Throughout Section 2.3, we will use G to denote an n -node directed unweighted graph, and we will use $P \subseteq V(G) \times V(G)$ to denote an associated set of demand pairs of size $|P| = p$. Additionally, we will associate with G and P a *consistent* tiebreaking scheme $\pi(\cdot, \cdot)$. Recall that for each demand pair $(s, t) \in P$, our tiebreaking scheme fixes an $s \rightsquigarrow t$ shortest path $\pi(s, t)$. We let $\pi(P)$ denote this collection of shortest paths in G , i.e.,

$$\pi(P) = \{\pi(s, t) \mid (s, t) \in P\}.$$

Throughout this section, we will assume that every edge $e \in E(G)$ is contained in a path in $\pi(P)$. (This assumption is without loss of generality because we can safely remove all other edges in G without increasing distances between demand pairs.)

Given graph G and demand pairs P , we define the following quantities.

- We define quantity $\ell = \ell(G, P)$ to be

$$\ell(G, P) := \left\lceil \frac{|E(G)|}{p} \right\rceil.$$

Quantity ℓ roughly corresponds to the average number of edges each path in $\pi(P)$ contributes to $|E(G)|$.

- We define quantity $d = d(G, P)$ to be

$$d(G, P) := \left\lceil \frac{|E(G)|}{n} \right\rceil.$$

Quantity d roughly corresponds to the average degree of nodes in G .

We will assume without loss of generality that ℓ and d are sufficiently large. If $\ell \leq c$ or $d \leq c$ for some constant c , then $|E(G)| \leq cp$ or $|E(G)| \leq cn$, respectively. Thus our assumption that $\ell > c$ and $d > c$ for a sufficiently large constant c will add

$$O(\ell p + dn) = O(p + n)$$

additional edges to our final preserver H . With this assumption, it follows that

$$\ell p = \Theta(|E(G)|) = \Theta(dn).$$

We will use the equality repeatedly in our proof. We now define an analogous quantity $\widehat{\ell}$ that captures how much paths in $\pi(P)$ overlap.

- We define $\widehat{\ell} = \widehat{\ell}(G, P)$ to be

$$\widehat{\ell} := \frac{\sum_{\pi \in \pi(P)} |\pi|}{p}.$$

Quantity $\widehat{\ell}$ roughly corresponds to the average length of the paths in $\pi(P)$. Note that $\widehat{\ell} \gg \ell$ in general, and $\widehat{\ell} = \Theta(\ell)$ when paths in $\pi(P)$ are pairwise edge-disjoint.

We will make the following assumption about paths in $\pi(P)$. For every path $\pi = \pi(s, t)$, we have that

$$\widehat{\ell}/2 \leq |\pi| \leq 2\widehat{\ell}.$$

This assumption holds without loss of generality by a standard bucketing argument. That is, we can bucket the demand pairs $(s, t) \in P$ into buckets $P_0, \dots, P_{\log n}$ so that

$$P_i = \{(s, t) \in P \mid 2^i \leq \text{dist}_G(s, t) \leq 2^{i+1}\}$$

for $i \in [0, \log n]$. Then we can apply our bound on G, P_i for each $i \in [0, \log n]$. This will worsen our upper bound by at most a logarithmic factor.

Ingredient 1: the Dense Low-Diameter Cluster Lemma

The first ingredient in our distance preserver construction will be the following lemma, which roughly states that if our distance preserver is sufficiently dense, then it contains a low-diameter subset of vertices containing many high-degree nodes. We defer the proof of this lemma to Section 2.3.

Lemma 2.12 (Dense Low-Diameter Cluster Lemma). *Let G be an n -node directed unweighted graph with associated set of demand pairs $P \subseteq V(G) \times V(G)$ of size $|P| = p$.² Let $\ell = \ell(G, P)$, $d = d(G, P)$, and $\widehat{\ell} = \widehat{\ell}(G, P)$. If G has average degree d , where $d \geq 10p/n^{1/2}$, then G contains a collection of nodes $S \subseteq V(G)$ such that*

1. $|S| = \Theta(d)$,
2. (Dense.) For all $s \in S$, $\deg_G(s) = \Omega(d)$, and
3. (Low diameter.) For all $s, t \in S$, $\text{dist}_G(s, t) = O\left(\frac{\widehat{\ell}p^2}{nd^2}\right)$.

An important feature of the set of vertices S identified in Lemma 2.12 is that S contains $\Theta(d)$ nodes each of degree $\Omega(d)$ in G . We will see how this property implies that many paths in $\pi(P)$ pass through nodes in S , as we state in the following lemma.

Lemma 2.13. *Let G, P, S and $\ell, d, \widehat{\ell}$ be as described in Lemma 2.12. Let $Q \subseteq P$ be the set of demand pairs $(s, t) \in P$ such that $\pi(s, t)$ contains a node in S . Then*

$$|Q| = \Omega(d^2).$$

Before we can prove Lemma 2.13, we will first need to introduce some new notation and establish several technical claims. Let $\Pi \subseteq \pi(P)$ be a collection of paths in $\pi(P)$, and let $S \subseteq V(G)$ be a set of nodes in G . Let $\Pi[S]$ denote the collection of paths in Π induced on the nodes in set S . Formally, for a path $\pi \in \Pi$, we define the path $\pi[S]$ induced on the set S to be the path obtained by deleting every node in $\pi \setminus S$ from π . Likewise, we let $\Pi[S] = \{\pi[S] \mid \pi \in \Pi\}$. Note that the paths in $\Pi[S]$ do not necessarily correspond to paths in graph G , as $E(\pi[S]) \not\subseteq E(\pi)$, in general.

We quickly observe that if a collection of paths Π is consistent, then the collection of paths $\Pi[S]$ is also consistent.

Claim 2.14. *Let Π be a consistent collection of paths, and let $S \subseteq V(G)$ be a set of nodes. Then the collection of paths $\Pi[S]$ is also consistent.*

We need an additional definition and technical claim before we can prove Lemma 2.13. Given a collection of paths Π and a set of nodes $S \subseteq V(G)$, we define an auxiliary graph $J(S, \Pi)$ and an associated collection of paths Π_J as follows.

²We assume without loss of generality that every edge in $E(G)$ lies on a path in $\pi(P)$ for a consistent tiebreaking scheme $\pi(\cdot, \cdot)$.

Definition 2.15 (Graph J and paths Π_J). Let Π be a collection of paths, and let $S \subseteq V(G)$ be a set of nodes. Then we define a graph $J = J(S, \Pi)$ and associated collection of paths Π_J as follows.

- Let $V(J) = S$,
- Let $\Pi_J = \Pi[S]$, and
- Let the edge set $E(J)$ of J be

$$E(J) = \bigcup_{\pi \in \Pi_J} E(\pi).$$

The definition of graph J and paths Π_J we be useful in the proof of Lemma 2.13, when combined with the following technical claim.

Claim 2.16. *Let G, P, S and $\ell, d, \widehat{\ell}$ be as described in Lemma 2.12. Let $Q \subseteq P$ be the set of demand pairs $(s, t) \in P$ such that $\pi(s, t)$ contains a node in S . Let $J = J(S, \pi(Q))$ and $\Pi_J = \pi(Q)[S]$ be the graph and collection of paths specified in Definition 2.15 with respect to set S and paths $\pi(Q)$. Then*

$$|V(J)| = |S| = \Theta(d) \quad \text{and} \quad |E(J)| = \Omega(d^2) - |Q|.$$

Proof. It is immediate from Definition 2.15 that $|V(J)| = |S| = \Theta(d)$. What remains is to prove that $|E(J)| = \Omega(d^2) - |Q|$.

Let $E' \subseteq E(G)$ be the set of edges in G incident to a node in S . Formally, let

$$E' = \{(u, v) \in E(G) \mid u \in S \text{ or } v \in S\}.$$

By Lemma 2.12, $|E'| = \Omega(d) \cdot |S| = \Omega(d^2)$. We can assume without loss of generality that every edge $e \in E'$ lies on a path π in $\pi(Q)$. We will define a subset $E'' \subseteq E'$ of E' as follows.

$$E'' = \{e \in E' \mid \text{there exists } \pi \in \pi(Q) \text{ and } s, t \in V(\pi) \cap S \text{ such that } e \in \pi[s \rightsquigarrow t]\}.$$

Informally, E'' corresponds to the set of edges in E' that lie between two nodes $s, t \in S$ on a path $\pi \in \pi(Q)$. Notice that for every path $\pi \in \pi(Q)$, among the set of edges $E(\pi) \cap E'$, only the first and last edge in $E(\pi) \cap E'$ to appear on path π are not contained in E'' . Consequently,

$$|E(\pi) \cap E''| \geq |E(\pi) \cap E'| - 2.$$

This immediately gives the bound

$$|E''| \geq |E'| - 2|Q| = \Omega(d^2) - 2|Q|.$$

To prove Claim 2.16, we want to translate our lower bound on $|E''|$ into a lower bound on $|E(J)|$. Towards this end, we will define a mapping $\varphi : E'' \mapsto E(J)$ as follows. Let e be an edge in E'' . Since $e \in E''$, there exists a path $\pi \in \pi(Q)$ and nodes $s, t \in V(\pi) \cap S$ such that $e \in \pi[s \rightsquigarrow t]$. Moreover, we can assume without loss of generality that $(s, t) \in E(J)$, by choosing nodes s and t that are closest to edge e on path π . We let $\varphi(e) = (s, t)$. We can define $\varphi(e)$ in this way for all $e \in E''$.

We claim that for every edge $(s, t) \in E(J)$, the number of edges in the preimage of edge (s, t) is at most two, i.e., $|\varphi^{-1}((s, t))| \leq 2$. Fix an edge $(s, t) \in E(J)$. Let $E''_{(s,t)} \subseteq E''$ denote the set of edges e in E'' such that $\varphi(e) = (s, t)$. If $|E''_{(s,t)}| = 0$, then we are done. Otherwise, there must exist a path $\pi^* \in \pi(P)$ and nodes $s, t \in V(\pi^*) \cap S$ such that node s precedes node t in path π^* . Since the collection of paths $\pi(P)$ is consistent, for every path $\pi \in \pi(P)$ such that $s, t \in V(\pi)$ and node s precedes node t , we have that $\pi[s \rightsquigarrow t] = \pi^*[s \rightsquigarrow t]$. Additionally, since $(s, t) \in E(J)$, path $\pi^*[s \rightsquigarrow t]$ is internally vertex-disjoint from set S . Consequently, only the first and last edge of path $\pi^*[s \rightsquigarrow t]$ are contained in E'' . This implies that $|E''_{(s,t)}| \leq 2$, as claimed.

Putting everything together, we find that

$$|E(J)| \geq \frac{1}{2} \cdot |\varphi^{-1}(E(J))| = \frac{1}{2} \cdot |E''| = \Omega(d^2) - |Q|,$$

as claimed. □

With Definition 2.15 and Claim 2.16 in hand, we are now ready to prove Lemma 2.13.

Proof of Lemma 2.13. Let G, P, S and $\ell, d, \widehat{\ell}$ be as described in Lemma 2.12. Let $Q \subseteq P$ be the set of demand pairs $(s, t) \in P$ such that $\pi(s, t)$ contains a node in S . Let $J = J(S, \pi(Q))$ and $\Pi_J = \pi(Q)[S]$ be the graph and collection of paths specified in Definition 2.15 with respect to set S and paths $\pi(Q)$.

Note that by Claim 2.14, the collection of paths Π_J is consistent. Additionally, by Claim 2.16 $|V(J)| = \Theta(d)$ and $|E(J)| \geq \Omega(d^2) - |Q|$. Now if $|Q| \geq \Omega(d^2)$, then we have finished the proof of Lemma 2.13. Otherwise, we may assume that $|E(J)| = \Omega(d^2)$.

Then Π_J is a consistent collection of $|\Pi_J| = |Q|$ paths, over a vertex set S of size $|S| = \Theta(d)$. Moreover, the number of distinct edges in Π_J is at least $|E(J)| = \Omega(d^2)$. By Theorem 1.1 of [CE06], a consistent tiebreaking scheme for x demand pairs over y nodes has at most $O(yx^{1/2})$ distinct edges.

Plugging in $x \geq |Q|$ and $y = |S| = \Theta(d)$, this implies that graph J has at most $|E(J)| = O(d|Q|^{1/2})$ distinct edges. Putting everything together,

$$\Omega(d^2) = |E(J)| = O(d|Q|^{1/2}),$$

so $|Q| = \Omega(d^2)$, as claimed. $\square$

Ingredient 2: Sourcewise Distance Preservers

The second ingredient in our distance preserver construction will be a sourcewise distance preserver upper bound for a collection of sources S with low diameter. This sourcewise preserver is a direct generalization of the sourcewise preserver of [BV21] to the directed setting, and follows from a similar argument.

Lemma 2.17 (cf. Lemma 8 of [BV21]). *Let G be an n -node directed graph, and let $S \subseteq V(G)$ be a set of $|S| = s$ nodes of weak diameter h (i.e., $\text{dist}_G(s, t) \leq h$ for all $s, t \in S$). Let $P \subseteq S \times V$ be a set of $|P| = p$ sourcewise demand pairs in G . Then there exists a distance preserver of G, P of size*

$$O\left((nsp h)^{1/2} + n\right).$$

Proof. We defer the proof of Lemma 2.17 to Appendix B of [HXX24] due to its similarities with the proof of Lemma 8 in [BV21]. $\square$

Main Construction

In the first part of our main construction, we will use Lemmas 2.12 and 2.17 to construct an exact distance preserver H that is sparse when $\widehat{\ell}$ is small.

Construction of distance preserver H for small $\widehat{\ell}$. Let $\pi(\cdot, \cdot)$ be a consistent tiebreaking scheme of G, P . We may assume without loss of generality that every edge in $E(G)$ is contained in a path in $\pi(P)$. Let φ be a parameter of the construction that we will optimize later. Our choice of φ will satisfy $\varphi \geq 10p/n^{1/2}$. While the average degree of G is at least φ , we will repeat the following process.

By Lemma 2.12 and Lemma 2.13, there exists a set of nodes $S \subseteq V(G)$ of size $|S| = \Theta(\varphi)$ such that:

- The set $Q \subseteq P$ of demand pairs (s, t) such that $\pi(s, t)$ contains a node in S is of size $|Q| = \Omega(\varphi^2)$.
- The set S has weak diameter $O(\widehat{\ell}p^2/(n\varphi^2))$ in G .

We will handle $\Theta(\varphi^2)$ paths in Q using two instances of the sourcewise distance preserver claimed in Lemma 2.17. For each demand pair $(x, y) \in Q$ such that $\pi(x, y)$ contains node $s \in S$, we split demand pair (x, y) into the two demand pairs (x, s) and (s, y) . We let $P_1 \subseteq S \times V$ and $P_2 \subseteq V \times S$ be the two resulting sets of demand pairs. Note that in order to construct a distance preserver of G, Q , it suffices to take the union of a distance preserver of G, P_1 and a distance preserver of G, P_2 . We can apply Lemma 2.17 with parameters $s_1 := |S| = \Theta(\varphi)$, $p_1 := \min(|Q|, \varphi^2) = \Theta(\varphi^2)$, and $h_1 := O(\widehat{\ell}p^2/(n\varphi^2))$ to construct a preserver H_1 of demand pairs Q of size

$$O\left((ns_1p_1h_1)^{1/2} + n\right) = O\left(n^{1/2}\varphi^{3/2} \cdot \frac{\widehat{\ell}^{1/2}p}{n^{1/2}\varphi} + n\right) = O\left(\widehat{\ell}^{1/2}\varphi^{1/2}p + n\right).$$

We add the edges of H_1 to our final distance preserver H of G, P . Since we have handled all demand pairs in Q , we can delete Q from P and repeat the above process. Formally, we update G, P as follows:

- $P \leftarrow P \setminus Q$, and
- $E(G) \leftarrow \bigcup_{(s,t) \in P} E(\pi(s, t))$.

We continue this process until the average degree of G is less than φ .

Since $p_1 = \Theta(\varphi^2)$, we will repeat this process at most $p/p_1 = O(p/\varphi^2)$ times before the average degree of H_1 is less than φ . This phase of the construction contributes at most

$$\frac{p}{p_1} \cdot O\left(\widehat{\ell}^{1/2}\varphi^{1/2}p + n\right) = O\left(\widehat{\ell}^{1/2} \cdot \frac{p^2}{\varphi^{3/2}} + \frac{np}{\varphi^2}\right)$$

edges to H . We finish the construction of H by adding all remaining $O(n\varphi)$ edges in H_1 to H . This yields the following lemma.

Lemma 2.18. *For every $\varphi \geq 10p/n^{1/2}$, there exists a distance preserver H of G, P with size*

$$|E(H)| = \widetilde{O}\left(\widehat{\ell}^{1/2} \cdot \frac{p^2}{\varphi^{3/2}} + \frac{np}{\varphi^2} + n\varphi\right).$$

To complement the distance preserver of Lemma 2.18, we will construct a distance preserver that is sparse when $\widehat{\ell}$ is large.

Distance preserver H for large $\widehat{\ell}$. To preserve distances when $\widehat{\ell}$ is large, we will make use of a theorem of [BCE05], which states that we can preserve distances between vertices that are far apart in H using few edges.

Theorem 2.19 (Theorem 2.24 of [BCE05]). *Given a (possibly directed) unweighted graph G , a D -preserver is a subgraph H of G such that for all $s, t \in V(G)$ with $\text{dist}_G(s, t) \geq D$, we have that*

$$\text{dist}_H(s, t) = \text{dist}_G(s, t).$$

There exists a D -preserver H of G with size $O(n^2/\widehat{\ell})$.

As an immediate consequence of Theorem 2.19, we can obtain a distance preserver H of G, P with size $O(n^2/\widehat{\ell})$, since we can assume all shortest paths π in $\pi(P)$ are of length at least $|\pi| \geq \widehat{\ell}/2$ without loss of generality, by our earlier bucketing argument.

Lemma 2.20. *There exists a distance preserver H of G, P with size $O(n^2/\widehat{\ell})$.*

Finishing the proof of Theorem 2.11.

Claim 2.21. *Together, Lemmas 2.18 and 2.20 imply a distance preserver H of G, P of size*

$$|E(H)| = \widetilde{O}\left(\frac{n^{2/3}p^{4/3}}{\varphi} + \frac{np}{\varphi^2} + n\varphi\right).$$

Proof. We will use the construction from Lemma 2.18 when $\widehat{\ell}$ is small, and we will use the construction from Lemma 2.20 when $\widehat{\ell}$ is large. Specifically, we define our threshold t for $\widehat{\ell}$ to be

$$t = \frac{n^{4/3}\varphi}{p^{4/3}}.$$

We calculate the following:

- When $\widehat{\ell} \leq t$,

$$\widehat{\ell}^{1/2} \cdot \frac{p^2}{\varphi^{3/2}} \leq t^{1/2} \cdot \frac{p^2}{\varphi^{3/2}} = O\left(\frac{n^{2/3}p^{4/3}}{\varphi}\right),$$

so the distance preserver H of Lemma 2.18 has size

$$|E(H)| = \tilde{O}\left(\frac{n^{2/3}p^{4/3}}{\varphi} + \frac{np}{\varphi^2} + n\varphi\right),$$

as desired.

- When $\widehat{\ell} > t$, distance preserver H of Lemma 2.20 has size

$$|E(H)| = \tilde{O}(n^2/\widehat{\ell}) \leq \tilde{O}(n^2/t) = \tilde{O}\left(\frac{n^{2/3}p^{4/3}}{\varphi}\right),$$

as desired. □

We are now ready to prove Theorem 2.11.

Proof of Theorem 2.11. The bounds stated in Theorem 2.11 will follow directly from the bounds achieved in Claim 2.21 after parameter balancing. Initially, we will assume that $n^{1/2} \leq p \leq n$.

Straightforward calculations show that the three terms in Claim 2.21 are balanced when

$$\varphi = \frac{p^{2/3}}{n^{1/6}}.$$

Recall that parameter φ must be at least $\varphi \geq 10p/n^{1/2}$ for Lemma 2.12 to go through. This holds for our choice of $\varphi = \frac{p^{2/3}}{n^{1/6}}$ when $p \leq n$. Plugging this value of φ into Claim 2.21, we obtain the upper bound

$$\tilde{O}\left(n^{5/6}p^{2/3} + \frac{n^{4/3}}{p^{1/3}} + n^{5/6}p^{2/3}\right).$$

We observe that $\frac{n^{4/3}}{p^{1/3}} \leq n^{5/6}p^{2/3}$ when $p \geq n^{1/2}$. We conclude that $|E(H)| = \tilde{O}(n^{5/6}p^{2/3})$ when $n^{1/2} \leq p \leq n$. For the settings where $p < n^{1/2}$ or $p > n$, we observe that the existing upper bounds of $\min(n^{2/3}p + n, np^{1/2})$ for directed distance preservers due to [CE06; Bod21] are smaller than our claimed bound of $\tilde{O}(n^{5/6}p^{2/3} + n)$. □

Proving the Dense Low-Diameter Cluster Lemma

The objective of this section is to prove Lemma 2.12.

Lemma (Dense Low-Diameter Cluster Lemma). *Let G be an n -node directed unweighted graph with associated set of demand pairs $P \subseteq V(G) \times V(G)$ of size $|P| = p$. Let $\ell = \ell(G, P)$, $d = d(G, P)$,*

and $\widehat{\ell} = \widehat{\ell}(G, P)$. If G has average degree d , where $d \geq 10p/n^{1/2}$, then G contains a collection of nodes $S \subseteq V(G)$ such that

1. $|S| = \Theta(d)$,
2. (Dense.) For all $s \in S$, $\deg_G(s) = \Omega(d)$, and
3. (Low diameter.) For all $s, t \in S$, $\text{dist}_G(s, t) = O\left(\frac{\widehat{\ell}p^2}{nd^2}\right)$.

Before proving this lemma, we will need to introduce several new definitions, which will be useful for finding our dense low-diameter cluster. Let G, P, S and $\ell, d, \widehat{\ell}$ be as in the statement of Lemma 2.12.

Recall the definition of a branching event from Definition 2.3. We say that a branching event $(x, y), (x, z)$ is a *high-degree branching event* if $\text{outdeg}_G(x) \geq d/4$. A simple argument will show that G contains many high-degree branching events.

Claim 2.22 ([CE06]). *G contains $\Omega(nd^2)$ high-degree branching events.*

Proof. Since G has average degree d , it follows that

$$\sum_{v \in V(G)} \text{outdegree}_G(v) = nd/2.$$

In particular, we have that

$$\begin{aligned} & \sum_{\{v \in V(G) \mid \text{outdeg}_G(v) \geq d/4\}} \text{outdegree}_G(v) \\ &= \sum_{v \in V(G)} \text{outdegree}_G(v) - \sum_{\{v \in V(G) \mid \text{outdeg}_G(v) < d/4\}} \text{outdegree}_G(v) \\ &\geq nd/2 - nd/4 \geq nd/4. \end{aligned}$$

Then the number of high-degree branching events is at least

$$\begin{aligned} & \sum_{\{v \in V(G) \mid \text{outdeg}_G(v) \geq d/4\}} \text{outdegree}_G^2(v) \\ &\geq \frac{1}{n} \cdot \left(\sum_{\{v \in V(G) \mid \text{outdeg}_G(v) \geq d/4\}} \text{outdegree}_G(v) \right)^2 \\ &\geq \frac{1}{n} \cdot \frac{n^2 d^2}{16} = \Omega(nd^2), \end{aligned}$$

where the second to last inequality follows from Cauchy-Schwarz. $\square$

We will also introduce an auxiliary graph, called the path intersection graph, which will be useful in our proof of Lemma 2.24.

Definition 2.23 (Path Intersection Graph J). We define the vertex set of J to be

$$V(J) = \pi(P),$$

the set of all shortest paths associated with our tiebreaking scheme $\pi(\cdot, \cdot)$. For every $\pi_1, \pi_2 \in V(J)$, we add the edge (π_1, π_2) to J and we assign this edge weight equal to the number of high-degree branching events between π_1 and π_2 .

Lemma 2.24. *There exists a path $\pi^* \in \pi(P)$ and a collection of paths $\pi(Q)$, where $Q \subseteq P$, satisfying the following properties:*

- *Every path $\pi' \in \pi(Q)$ has at least $\Omega(nd^2/p^2)$ high-degree branching events with π^* .*
- *Let $S \subseteq V(\pi^*)$ be the set of nodes in π^* with outdegree at least $d/4$, i.e.,*

$$S = \{v \in V(\pi^*) \mid \text{outdeg}_G(v) \geq d/4\}.$$

The total number of high-degree branching events between path π^ and the paths in $\pi(Q)$ is at least $\Omega(|S|d)$.*

Proof. Let b denote the number of high-degree branching events in G . By Claim 2.22, there is a constant $c > 0$ such that G has at least $b \geq cnd^2$ high-degree branching events. Then

$$\sum_{e \in E(J)} w(e) \geq b \geq cnd^2,$$

where $w(e)$ denotes the weight of edge $e \in E(J)$. We will perform the following procedure on J .

1. Delete all edges $e \in E(J)$ with weight

$$w(e) \leq \frac{cnd^2}{4p^2}.$$

2. While there exists a node $\pi \in V(J)$ such that sum of the edge weights incident to π is small, we will delete node π from $V(J)$. Formally, for each node $\pi \in V(J)$, let $S_\pi \subseteq V(\pi)$ be the set

of nodes in path $\pi \in \pi(P)$ with outdegree at least $d/4$, i.e.,

$$S_\pi = \{v \in V(\pi) \mid \text{outdeg}_G(v) \geq d/4\}.$$

While there exists a node $\pi \in V(J)$ such that the sum of the edge weights incident to π in J is at most

$$\sum_{\{(\pi, \pi') \in E(J) \mid \pi' \in V(J)\}} w((\pi, \pi')) \leq \frac{|S_\pi|d}{16},$$

we will delete node π from J .

We claim that after the above procedure terminates, graph J is nonempty and still contains a constant fraction of its edge weight. After step 1 of the procedure terminates, J will have total edge weight at least

$$\sum_{e \in E(J)} w(e) \geq b - \frac{cnd^2}{4p^2} \cdot p^2 = b - cnd^2/4 = 3/4 \cdot b,$$

since $|E(J)| = p^2$ and $b \geq cnd^2$. Consequently, a large fraction of the edge weights in J survive step 1 of our procedure. Now we will attempt to argue the same is true for the second step of our procedure.

By the definition of high-degree branching events,

$$b \geq \sum_{\pi \in V(J)} |S_\pi| \cdot \frac{d}{4},$$

since there are at each node $v \in S_\pi$, there are at least $d/4$ high-degree branching events between π and other paths in $V(J)$. Then after step 2 of the procedure terminates, J will have total edge weight at least

$$\sum_{e \in E(J)} w(e) \geq 3/4 \cdot b - \sum_{\pi \in V(J)} \frac{|S_\pi|d}{16} = 3/4 \cdot b - \frac{1}{4} \cdot \left(\sum_{\pi \in V(J)} |S_\pi| \cdot \frac{d}{4} \right) \geq b/2.$$

Consequently, after our procedure terminates, the amount of edge weight remaining in $E(J)$ is at least $b/2$, so $E(J)$ and $V(J)$ must be nonempty.

We have shown that after this procedure terminates, graph J remains nonempty. In particular, we have shown that every surviving node in $V(J)$ is incident to edge weight at least $|S_\pi|d/16$, and every surviving edge in $E(J)$ has weight at least $cnd^2/(4p^2)$. Then we can take π^* to be any surviving node $\pi^* \in V(J)$, and we can take $\pi(Q)$ to be the set of all surviving paths adjacent to π^*

in J . The lemma immediately follows. $\square$

We will denote the path described in Lemma 2.24 as π^* , and we will use Q to denote the associated set of demand pairs.

Definition 2.25 (π -close nodes). Fix a path $\pi \in \pi(Q)$, and let

$$B = \{((s_i, t_i), (s_i, u_i))\}_{i \in [1, k]}$$

be the set of all high-degree branching events between π and π^* , written so that

- $(s_i, t_i) \in \pi$ and $(s_i, u_i) \in \pi^*$, and
- If $i < j \in [1, k]$, then node s_i comes before node s_j on path π , and node s_j comes before node s_i on path π^* . Note that nodes s_i and s_j cannot appear in the same order on both π and π^* , by consistency.

We say that a pair of nodes $(x, y) \in V(\pi^*) \times V(\pi^*)$ is π -close if $(x, y) = (s_i, s_{i+1})$ for some $i \in [1, k - 1]$.

Claim 2.26. *If nodes x and y are in π^* , and x and y are π_1 -close and π_2 -close for paths $\pi_1, \pi_2 \in \pi(Q)$, then $\pi_1 = \pi_2$.*

Proof. This claim will follow directly from our choice of a *consistent* tiebreaking scheme $\pi(\cdot, \cdot)$ and our definition of branching events. Let x and y be nodes in π^* that are π_1 -close and π_2 -close, for some $\pi_1, \pi_2 \in \pi(P)$. We may assume wlog that node x comes before node y on path π^* . Since π_1 has branching events with π^* at nodes x and y , by consistency we must have that y comes before x on path π_1 . By an identical argument y comes before x on path π_2 . Then by consistency, it follows that $\pi_1[y, x] = \pi_2[y, x]$.

Let z be the node following y in paths π_1 and π_2 , so that $(y, z) \in \pi$. Let w be the node following y in path π^* . Observe that $(y, z), (y, w)$ is a branching event. Moreover, by the definition of π -close nodes, branching event $(y, z), (y, w)$ is associated with paths π_1, π^* and paths π_2, π^* . However, as stated in Definition 2.3, every branching event is associated with at most one pair of paths. We conclude that $\pi_1 = \pi_2$. $\square$

Definition 2.27 (close nodes). We say that nodes $x, y \in V$ are *close* if

$$\text{dist}_G(x, y) + \text{dist}_G(y, x) = O\left(\frac{\widehat{\ell} p^2}{nd^2}\right).$$

Claim 2.28. Fix a path $\pi = \pi(s, t) \in \pi(Q)$, and let $C \subseteq V(\pi^*) \times V(\pi^*)$ be the set of all pairs of nodes in π^* that are π -close. Then at least $|C|/2$ node pairs $(x, y) \in C$ are close.

Proof. Fix a path $\pi \in \pi(Q)$, and let

$$B = \{((s_i, t_i), (u_i, v_i))\}_{i \in [1, k]}$$

be the set of all high-degree branching events between π and π^* , written so that

- $(s_i, t_i) \in \pi$ and $(u_i, v_i) \in \pi^*$, and
- if $i < j \in [1, k]$, then node s_i comes before node s_j on path π , and node s_j comes before node s_i on path π^* .

Recall from Definition 2.25 that the set of all pairs of π -close nodes in π^* is exactly $C = \{(s_i, s_{i+1})\}_{i \in [1, k-1]}$, where $|C| = k - 1$. Additionally, as we assumed at the beginning of Section 2.3, $|\pi| \leq 2\widehat{\ell}$ and $|\pi^*| \leq 2\widehat{\ell}$. In particular, this implies

$$\sum_{i=1}^{k-1} (\text{dist}_G(s_i, s_{i+1}) + \text{dist}_G(s_{i+1}, s_i)) \leq |\pi| + |\pi^*| \leq 4\widehat{\ell}.$$

Let $C_1 \subseteq C$ be the set of all π -close pairs of nodes (s_i, s_{i+1}) with roundtrip distance at least $16\widehat{\ell}/k$, i.e.,

$$C_1 = \{(s_i, s_{i+1}) \mid \text{dist}_G(s_i, s_{i+1}) + \text{dist}_G(s_{i+1}, s_i) \geq 16\widehat{\ell}/k\}.$$

By the previous inequality, we observe that

$$|C_1| \cdot 16\widehat{\ell}/k \leq |\pi| + |\pi^*| \leq 4\widehat{\ell},$$

so $|C_1| \leq k/4$. Let $C_2 = C \setminus C_1$. It follows that $|C_2| \geq |C|/2$. Additionally, by Lemma 2.24, we know that $|C| = k - 1 = \Omega(nd^2/p^2)$. Then for all $(s_i, s_{i+1}) \in C_2$,

$$\text{dist}_G(s_i, s_{i+1}) + \text{dist}_G(s_{i+1}, s_i) \leq 16\widehat{\ell}/k = O\left(\frac{\widehat{\ell}p^2}{nd^2}\right).$$

We conclude that all pairs of nodes $(x, y) \in C_2$ are close, establishing the claim. $\square$

Claim 2.29. Let $S \subseteq V(\pi^*)$ be the set of nodes v in π^* of degree at least $\deg_G(v) \geq d/4$. There are $\Omega(|S|d)$ distinct pairs of nodes in $S \times S$ that are close.

Proof. Let $\pi(Q) = \{\pi_1, \dots, \pi_k\}$, for some integer k . For $i \in [1, k]$, let $C_i \subseteq V(\pi^*) \times V(\pi^*)$ be the set of all pairs of nodes in π^* that are π_i -close. Note that if $(x, y) \in C_i$, then $\deg_G(x) \geq d/4$ and $\deg_G(y) \geq d/4$ by Definition 2.25, so $(x, y) \in S \times S$. For $i \in [1, k]$, let $C'_i \subseteq C_i$ be the set of all pairs of nodes $(x, y) \in C_i$ that are close. We will need the following statements proved earlier.

- By Claim 2.26, $C_i \cap C_j = \emptyset$ for $i \neq j \in [1, k]$.
- By Claim 2.28, $|C'_i| \geq |C_i|/2$ for $i \in [1, k]$.
- By Lemma 2.24, the total number of high-degree branching events between path π^* and the paths in $\pi(Q)$ is at least $\Omega(|S|d)$. This implies that $\sum_{i \in [1, k]} |C_i| = \Omega(|S|d)$.

Putting it all together, the total number of distinct close pairs in $S \times S$ is at least

$$\left| \bigcup_{i \in [1, k]} C'_i \right| = \sum_{i \in [1, k]} |C'_i| \geq \frac{1}{2} \cdot \sum_{i \in [1, k]} |C_i| = \Omega(|S|d).$$

□

We are now ready to prove Lemma 2.12.

Proof of Lemma 2.12. Let $S \subseteq V(\pi^*)$ be the set of nodes referenced in Claim 2.29. By Claim 2.29, each node $u \in S$ is close to $\Omega(d)$ distinct nodes in S on average. Then there must exist a node $v \in S$ that is close to $\Omega(d)$ distinct nodes in S . Let T be the set of nodes in S close to v . Observe that for all $x, y \in T$,

$$\text{dist}_G(x, y) \leq \text{dist}_G(x, v) + \text{dist}_G(v, y) = O\left(\frac{\widehat{\ell} p^2}{nd^2}\right).$$

Moreover, by our choice of set S in Claim 2.29, $\deg_G(x) = \Omega(d)$ for all $x \in S$. We conclude that set S satisfies all properties described in the statement of Lemma 2.12. □

2.4 Directed Approximate Preserver Lower Bound

Theorem 2.30. *For any n and $\alpha = \alpha(n)$, there exist n -node directed weighted graphs G with demand pairs P of size $|P| = p$, with $p \in [n^{1/2}, n^2]$, such that any α -approximate preserver on G, P has at least*

$$\Omega(n^{2/3} p^{2/3})$$

edges.

Construction

Let $\mathcal{X}$ and $\mathcal{L}$ be an arrangement of $|\mathcal{X}| = n$ points and $|\mathcal{L}| = p$ distinct lines in $\mathbb{R}^2$ with $\varphi = \Omega(n^{2/3}p^{2/3})$ point-line incidences (e.g., take $\mathcal{X}$ and $\mathcal{L}$ to be the arrangement of points and lines Elekes' construction in [Ele01]). We may assume that every line in $\mathcal{L}$ has finite slope wlog (e.g., by rotating the arrangement of points and lines). Given $\mathcal{X}$ and $\mathcal{L}$, we will construct a directed graph G with associated demand pairs P . We choose the vertices of G to be the points in $\mathcal{X}$, i.e.,

$$V(G) = \mathcal{X}.$$

For each line $L \in \mathcal{L}$, we will add edges to $E(G)$ and demand pairs to P as follows.

- Let $x_1, \dots, x_k$ denote the points lying on L , ordered so that the first coordinates of points are strictly increasing.
- Add directed edge (x_i, x_{i+1}) to $E(G)$ for $i \in [1, k-1]$.
- Add demand pair (x_1, x_k) to P .

What remains is to assign weights to the edges of G . Let $L_1, \dots, L_p$ denote the lines in $\mathcal{L}$, ordered by increasing slope (breaking ties arbitrarily). If edge $(x, y) \in E(G)$ satisfies $x, y \in L_i$, then we assign edge (x, y) weight

$$w((x, y)) = (2\alpha n)^i.$$

Weight $w((x, y))$ is well-defined because distinct lines intersect on at most one point.

Analysis

Claim 2.31. *Fix a line $L \in \mathcal{L}$. Let $x_1, \dots, x_k$ denote the points lying on L , ordered by increasing first coordinate. Then $(x_1, \dots, x_k)$ is the unique α -approximate shortest path between x_1 and x_k in G .*

Proof. Let $L = L_i$ for some $i \in [1, p]$. Let $\pi = (x_1, \dots, x_k)$ denote the (x_1, x_k) -path in G traveling along line L . Observe that

$$w(\pi) = \sum_{j=1}^{k-1} w((x_j, x_{j+1})) = k \cdot (2\alpha n)^i \leq 2^i \alpha^i n^{i+1}.$$

Let π' be an (x_1, x_k) -path such that $\pi' \neq P$. Let $(x, y) \in E(G)$ be the first edge on path π' such that $(x, y) \in E(\pi') \setminus E(\pi)$. If $x, y \in L_j$ for some $j > i$, then

$$w(\pi') \geq w((x, y)) = (2\alpha)^j n^j \geq (2\alpha)^{i+1} n^{i+1} > \alpha \cdot w(\pi),$$

so π' is not an α -approximate shortest path.

Otherwise, $x, y \in L_j$ for some $j < i$. Consider the line L' passing through points y and x_k . Since $x, y \in L_j$ for some $j < i$, the slope of line L' is strictly greater than the slope of line L_i . Additionally, by construction, the first coordinates of the nodes on path π' are strictly increasing. Then the subpath $\pi'[y, x_k]$ of π' must contain an edge $(x', y') \in E(G)$ such that $x', y' \in L_{j'}$ for some $j' > i$. We conclude that π' is not an α -approximate shortest path. $\square$

Claim 2.32. Any α -approximate preserver on G, P has at least $\varphi - p = \Omega(n^{2/3} p^{2/3})$ edges.

Proof. By Claim 2.31, every edge $(x, y) \in E(G)$ lies on a unique α -approximate shortest path between a demand pair $(s, t) \in P$. Then any α -approximate preserver of G must contain every edge in $E(G)$, and must be of size at least

$$|E(G)| = \sum_{L \in \mathcal{L}} |\{x \in \mathcal{X} \mid x \in L\}| - 1 = \varphi - p.$$

$\square$

Chapter 3

Strong Connectivity Preservers

This chapter develops new upper bounds for fault-tolerant strong connectivity preservers and higher-order connectivity preservers in directed graphs.¹

3.1 Preliminaries

Let $G = (V, E)$ be a directed graph with n vertices and m edges. For any edge set $F \subseteq V \times V$, we use $G - F$ to denote the graph formed by removing edges in F from G , and we use $G + F$ to denote the graph formed by adding edges in F to G . When $F = \{e\}$ is a singleton set, we abbreviate this to $G - e$ or $G + e$. For a vertex set $S \subseteq V$, we use $G[S]$ to denote the induced subgraph of S in G .

For any vertex $v \in V$, we use $\delta_G^+(v)$ and $\delta_G^-(v)$ to denote the set of outgoing edges from v and incoming edges to v respectively. When $(u, v) \in E$, we say v is an out-neighbor of u , and u is an in-neighbor of v . We denote the set of out- and in- neighbors of $v \in V$ in G by $N_G^+(v)$ and $N_G^-(v)$.

We define a cut as a partition $(S, V - S)$ of the vertex set V into two disjoint subsets, where $S \subseteq V$. We use $\delta_G^+(S)$ to denote the set of outgoing edges from S , and $\delta_G^-(S)$ to denote the set of incoming edges to S . For two disjoint sets $X, Y \subseteq V$, we say that $(S, V - S)$ is an (X, Y) -cut if $X \subseteq S$ and $Y \subseteq V - S$. In particular, if $X = \{x\}$ and $Y = \{y\}$, we call $(S, V - S)$ an (x, y) -cut. We typically refer to the size of a cut $(S, V - S)$ as the number of outgoing edges from S , $|\delta_G^+(S)|$. In particular, we say that an (X, Y) -cut $(S, V - S)$ is a minimum (X, Y) -cut if $|\delta_G^+(S)| \leq |\delta_G^+(S')|$ for every (X, Y) -cut $(S', V - S')$.

We define $\text{flow}(G, s, t)$ to be the maximum number of pairwise edge-disjoint paths from s to t in G . Additionally, we define the *symmetric* connectivity $\lambda_G(s, t)$ between s and t to be $\lambda_G(s, t) = \min(\text{flow}(G, s, t), \text{flow}(G, t, s))$. We define the k -bounded connectivity to be $\lambda_G^k(s, t) = \min(\lambda_G(s, t), k)$. For terminal sets X, Y , we define $\text{flow}(G, X, Y)$ to be the maximum

¹The contents of this chapter originally appeared in [HSW25] and are based on joint work with Thatchaphol Saranurak and Benyu Wang that appeared in FOCS 2025.

number of pairwise edge-disjoint (X, Y) -paths.

Farthest minimum cuts and important cuts

We need to define a class of well-structured (X, Y) -cuts, which we call reachable cuts.

Definition 3.1 (Reachable Cuts). We say that an (X, Y) -cut $(S, V - S)$ is *out-reachable* if for every $s \in S$ there exists an $x \in X$ such that x can reach s in $G - \delta_G^+(S)$. Likewise, we say that $(S, V - S)$ is *in-reachable* if for every $s \in S$ there exists an $x \in X$ such that s can reach x in $G - \delta_G^-(S)$.

In some sense, reachable cuts are without loss of generality.

Claim 3.2. *Let $(S, V - S)$ be an (X, Y) -cut. Then there exists an out-reachable (X, Y) -cut $(S', V - S')$ such that $\delta_G^+(S') \subseteq \delta_G^+(S)$. Likewise, there exists an in-reachable (X, Y) -cut $(S'', V - S'')$ such that $\delta_G^-(S'') \subseteq \delta_G^-(S)$.*

Proof. Consider the out-reachable case. For any (X, Y) -cut $(S, V - S)$, we let S' be the set that X can reach in $G - \delta_G^+(S)$. It follows that S' is out-reachable and $\delta_G^+(S') \subseteq \delta_G^+(S)$. The in-reachable case can be proved in the same way. $\square$

A classic result of Ford-Fulkerson implies the existence of a unique min-cut that is “farther” than every other min-cut.

Lemma 3.3 (Farthest Minimum Cuts [FF62]). *Given a directed graph G , we say that an (X, Y) -cut $(S, V - S)$ is a farthest minimum (X, Y) -cut in G if $(S, V - S)$ is an out-reachable, minimum (X, Y) -cut, and there does not exist an out-reachable, minimum (X, Y) -cut $(S', V - S')$ such that $S \subset S'$. For disjoint vertex sets $X, Y \subseteq V(G)$, the unique farthest minimum (X, Y) -cut always exists and can be computed in max-flow time.*

We denote the farthest minimum (X, Y) -cut in G by $\text{FMC}(G, X, Y)$. Important cuts generalize the notion of farthest minimum cuts to cuts that are not min-cuts.

Definition 3.4 (Important Cuts [Mar11]). We say that an (X, Y) -cut $(S, V - S)$ is an important (X, Y) -cut if $(S, V - S)$ is out-reachable, and there does not exist an out-reachable (X, Y) -cut $(S', V - S')$ with size at most $|\delta_G^+(S')| \leq |\delta_G^+(S)|$ such that $S \subset S'$.

The definition of important cuts is well-known in the parametrized complexity community. We also state the following claim, which is clear from definition:

Claim 3.5. *If $(S, V - S)$ is an out-reachable (X, Y) -cut of size $|\delta_G^+(S)| \leq k$, then there exists an important (X, Y) -cut $(S', V - S')$ of size $|\delta_G^+(S')| \leq k$ such that $S \subseteq S'$.*

Proof. If $(S, V - S)$ is important, then we let $S' = S$ and the claim holds. Otherwise, since $(S, V - S)$ is not important, we can find $(S', V - S')$ that $|\delta_G^+(S')| \leq |\delta_G^+(S)|$ and $S \subset S'$. We can set $S = S'$ and repeat this procedure until $(S, V - S)$ is important. $\square$

Unbreakable sets and directed expander hierarchies

We begin this section by defining (q, k) -unbreakable sets.

Definition 3.6 (unbreakable). Let G be a directed graph, and let $U \subseteq V$. Let $q \geq k$ be positive integers. We say that U is (q, k) -unbreakable in G if for every cut $(S, V - S)$ in G such that $|\delta_G^+(S)| \leq k$ or $|\delta_G^-(S)| \leq k$, we have either $|S \cap U| \leq q$ or $|U - S| \leq q$.

Unbreakable sets have the useful property that if we remove a small number of edges from the graph, then almost all of the vertices in the unbreakable set will lie in the same strongly connected component in the resulting graph.

Claim 3.7 (Giant component). *Let G be a directed graph, let $U \subseteq V(G)$ be a (q, k) -unbreakable terminal set in G , and let $F \subseteq E(G)$ be a set of edges with size at most k . Then $G - F$ contains a strongly connected component C containing at least $|C \cap U| \geq |U| - 2q$ of the nodes in U . We refer to C as a giant component of U in $G - F$.*

Proof of Theorem 3.7. Let $C_1, \dots, C_r \subseteq V(G)$ be the strongly connected components in $G - F$. We may assume without loss of generality that for each edge $e \in E(G - F)$, there exist strongly connected components C_i, C_j of $G - F$ such that $e \in C_i \times C_j$, where $i \leq j$. This follows from the fact that the condensation graph of $G - F$ is acyclic.

Let $C_{\leq i} = C_1 \cup \dots \cup C_i$ for each $i \in [1, r]$. Suppose towards contradiction that for each $i \in [1, r]$, SCC C_i in $G - F$ satisfies $|C_i \cap U| < |U| - 2q$. Let $i \in [1, r]$ be the largest index such that $|C_{\leq i} \cap U| \leq q$. Then $|C_{\leq i+1} \cap U| > q$. Moreover, since $|C_{i+1} \cap U| < |U| - 2q$, it follows that

$$|(V - C_{\leq i+1}) \cap U| = |U| - |C_{\leq i} \cap U| - |C_{i+1} \cap U| > |U| - q - (|U| - 2q) = q.$$

Then the cut $(C_{\leq i+1}, V - C_{\leq i+1})$ will satisfy $|C_{\leq i+1} \cap U| > q$ and $|(V - C_{\leq i+1}) \cap U| > q$. Additionally, since $\delta_{G-F}^-(C_{\leq i+1}) = \emptyset$ by our earlier discussion, it follows that $|\delta_G^-(C_{\leq i+1})| \leq |F| \leq k$. However, this implies that set U is not (q, k) -unbreakable in G , a contradiction.

We conclude that there exists a strongly connected component C in $G - F$ such that $|C \cap U| \geq |U| - 2q$, as claimed. $\square$

Our upper bounds for fault-tolerant symmetric connectivity preservers will rely on the following directed expander hierarchy decomposition for unbreakable sets, which is proven in its entirety in [HSW25].

Lemma 3.8 (Directed Expander Hierarchy for Unbreakable Sets). *Let G be an n -node directed graph, and let q, k be positive integers with $q \geq k/\phi$, where $0 < \phi \leq 1$. Then there exists a partition $\{V_1, V_2, \dots, V_\ell\}$ of $V(G)$ with the following properties:*

1. *Fix an $i \in [1, \ell]$. Let $V_{\leq i} = V_1 \cup \dots \cup V_i$, and let $C \subseteq V(G)$ be a strong component in $G[V_{\leq i}]$. Then the set $V_i \cap C$ is (q, k) -unbreakable in $G[C]$.*
2. *The number of sets in the partition is at most $\ell = O(\log n)$.*

Moreover, when $\phi = 1/2$, this partition can be computed in exponential time, and when $\phi = O(1/\sqrt{\log n})$, this partition can be computed in polynomial time.

A version of expander hierarchies in *undirected* graphs was first introduced in the context of oblivious routing [Rac02; BKR03; HHR03] and is called *tree flow sparsifiers* or *Räcke tree*. It later finds many more applications for devising almost-linear time flow algorithms [RST14; Abr+16; Gor+21b; Van+24; LRW25].

In 2007, a weaker variant of the expander hierarchy was introduced by Patrascu and Thorup [PT07] for connectivity oracles under edge faults. It turns out, in contrast to tree flow sparsifiers, that this weaker variant naturally generalizes from the usual edge expansion to vertex expansion [LS22b; LPS25]. Recently, [Ber+24] introduced a directed expander hierarchy as a further generalization to directed expansion. They also presented a fast construction algorithm that is highly complicated.

3.2 Fault-Tolerant Connectivity Preservers

We now present our new results for fault-tolerant connectivity preservers. In section 3.2, we prove our existential upper bound for fault-tolerant connectivity preservers (Theorem 3.11).

Throughout this section, we will repeatedly use the following lemma about important cuts, which is proved in full generality in Theorem 6.1 of [HSW25].

Lemma 3.9 (Important cut container). *Let k be a positive integer, let G be an n -vertex directed graph, and let $X, Y \subseteq V(G)$ be disjoint sets. There exists an (X, Y) -cut $(S, V - S)$ of size at most $|\delta^+(S)| \leq 2^{k-1}$ such that every important (X, Y) -cut $(S', V - S')$ of size at most $|\delta^+(S')| \leq k$ satisfies $S' \subseteq S$. Cut $(S, V - S)$ can be computed in $O(k2^k m + k^2 4^k)$ time.*

We define some new notation that we will use when we apply Theorem 3.9.

- We will let $\text{ImpCut}(G, X, Y, k, +)$ denote an (X, Y) -cut $(S, V - S)$ in G of size at most $|\delta^+(S)| \leq 2^{k-1}$ such that every out-reachable (X, Y) -cut $(S', V - S')$ of size $|\delta^+(S')| \leq k$ satisfies $S' \subseteq S$.
- We will let $\text{ImpCut}(G, X, Y, k, -)$ denote an (X, Y) -cut $(S, V - S)$ in G of size at most $|\delta^-(S)| \leq 2^{k-1}$ such that every in-reachable (X, Y) -cut $(S', V - S')$ of size $|\delta^-(S')| \leq k$ satisfies $S' \subseteq S$.

By Lemma 3.9 and Claim 3.5, $\text{ImpCut}(G, X, Y, k, +)$ and $\text{ImpCut}(G, X, Y, k, -)$ are well-defined and can be computed in $O(k2^k m + k^2 4^k)$ time.

We will also make use of the so-called anti-isolation lemma, which is directly implied by Lemma 3.9. See Section 6 of [HSW25] for a complete proof of Lemma 3.10.

Lemma 3.10 (Anti-isolation lemma). *Let $s, t_1, \dots, t_r$ be vertices in a directed graph G , and let $F_1, \dots, F_r \subseteq E(G)$ be edge sets each of size at most k such that for all $i, j \in [1, r]$ there is an (s, t_j) -path in $G - F_i$ if and only if $i = j$. Then $r \leq 2^k$.*

Additionally, throughout this section we will construct k -FT single-source connectivity preservers as a subroutine in our algorithms. These k -FT single-source connectivity preservers maintain the strong connectivity between a source node s and the rest of the vertex set $V(G)$ under k edge faults. We use $\text{SSCP}(G, s, k)$ to denote a k -FT single-source connectivity preserver of G rooted at source vertex s with $O(2^k |V(G)|)$ edges. Subgraph $\text{SSCP}(G, s, k)$ always exists by [BCR18].

Existential upper bound for fault-tolerant connectivity preservers

The goal of this section is to present our existential upper bound for fault-tolerant connectivity preservers.

Theorem 3.11. *For every positive integer k , every n -vertex directed graph admits a k -fault-tolerant connectivity preserver with $O(k4^k n \log n)$ edges.*

We will prove Theorem 3.11 in two steps. In the first step, we will show how to construct sparse fault-tolerant *sourcewise* connectivity preservers between a well-connected set of source vertices U and the rest of the graph (Lemma 3.13). In the second step, we combine our sourcewise preservers with a directed expander hierarchy (Lemma 3.8) to construct a sparse fault-tolerant connectivity preserver of the input graph.

3.2.0.1 Fault-tolerant sourcewise connectivity preservers for unbreakable sets

We begin with a formal definition of fault-tolerant sourcewise connectivity preservers.

Definition 3.12 (Fault-tolerant sourcewise connectivity preservers). Let G be an n -vertex directed graph, and let $S \subseteq V(G)$ be a set of source vertices. We say that a subgraph $H \subseteq G$ is a k -FT sourcewise connectivity preserver of G with respect to set S if for any pair of vertices $(s, v) \in S \times V(G)$ and any edge fault set $F \subseteq E(G)$ of size at most $|F| \leq k$, s and v are strongly connected in $G - F$ if and only if s and v are strongly connected in $H - F$.

Our fault-tolerant sourcewise connectivity preservers can be summarized with the following lemma.

Lemma 3.13. *Let G be an n -vertex directed graph, and let $U \subseteq V(G)$ be a (q, k) -unbreakable set in G . There exists a k -FT sourcewise connectivity preserver H of graph G with respect to source vertices U of size $|E(H)| = O(4^k qn)$.*

We will construct our fault-tolerant sourcewise connectivity preserver using a decremental greedy strategy that is standard in the area of graph spanners and preservers.

Greedy Algorithm. Let $G = (V, E)$ be an n -vertex directed graph, and let $U \subseteq V$ be a (q, k) -unbreakable set in G . We may assume G is strongly connected, without loss of generality. We will initialize our fault-tolerant sourcewise connectivity preserver H to be the input graph, so $H \leftarrow G$. While there exists an edge $e \in E(H)$ such that $H - e$ is a k -FT sourcewise connectivity preserver of G with respect to set U , we will remove edge e from H , so that $H \leftarrow H - e$. This algorithm terminates and outputs H when no such edge $e \in E(H)$ exists.

Correctness. By construction, H is a k -FT sourcewise connectivity preserver of G with respect to set U . Specifically, H is an *edge-minimal* k -FT sourcewise connectivity preserver of G with respect to U , by the terminating condition of the greedy algorithm.

Time Complexity. There are $n^{O(k)}$ fault sets $F \subseteq E(G)$ of size $|F| \leq k$. Consequently, we can check if a subgraph $H \subseteq G$ is a k -FT sourcewise connectivity preserver of G with respect to U in $n^{O(k)}$ time. We conclude that the greedy algorithm can be implemented in $n^{O(k)}$ time.

Size Analysis. Before we can bound the size of $E(H)$, we need to define a subgraph $J \subseteq H$ that is useful for our analysis. Let U' be an arbitrary subset $U' \subseteq U$ of size $|U'| = \min(2q + 1, |U|)$. Let

$$J = \bigcup_{u \in U'} \text{SSCP}(H, u, k)$$

be the subgraph of H obtained by unioning $|U'|$ many k -FT single-source connectivity preservers, each rooted at a distinct vertex in U' . The following claim states that subgraph J preserves strong connectivity between pairs of vertices $(u, v) \in U \times V(G)$ for a certain category of fault sets.

Claim 3.14 (Large SCC fault sets). *Let $F \subseteq E(G)$ be a fault set of size $|F| \leq k$, and let $C \subseteq V(G)$ be an SCC of $G - F$. If $|C \cap U| > 2q$, then vertices in C are pairwise strongly connected in $J - F$.*

Proof. By Theorem 3.7, if $|C \cap U| > 2q$, then $|C \cap U| \geq |U| - 2q$. Since $|U'| = \min(2q + 1, |U|)$, it follows that $C \cap U' \neq \emptyset$. Let $x \in C \cap U'$ be a vertex in the nonempty intersection of U' and C .

Now we make the following sequence of observations. First, vertices in C are pairwise strongly connected in $G - F$. Second, since H is a k -FT sourcewise connectivity preserver of G with respect to U , and $C \cap U \neq \emptyset$, it follows that vertices in C are pairwise strongly connected in $H - F$. Finally, subgraph J contains $\text{SSCP}(H, x, k)$, where $x \in C$, so we conclude that vertices in C are pairwise strongly connected in $J - F$ as well. $\square$

Since J is the union of $O(q)$ many k -FT single-source connectivity preservers, $|E(J)| = O(2^k qn)$ by Theorem 1.1 of [BCR18]. Then to prove $|E(H)| = O(4^k qn)$, it will be sufficient to prove that $|E(H) - E(J)| = O(4^k qn)$. Before proceeding with the proof, we will need the following structural claim about the edges in H .

Claim 3.15. *For each edge $(s, t) \in E(H)$, there exists a vertex $u \in U$ and a fault set $F \subseteq E(H)$, $|F| \leq k$, such that:*

1. *Vertices s , t , and u are strongly connected in $H - F$, and*
2. *$H - F$ contains either*
 - *an out-reachable (u, t) -cut $(S, V - S)$ such that $\delta_{H-F}^+(S) = \{(s, t)\}$, or*
 - *an in-reachable (u, s) -cut $(S, V - S)$ such that $\delta_{H-F}^-(S) = \{(s, t)\}$.*

Proof. If $(s, t) \in E(H)$, then there exists a fault set F , with $|F| \leq k$, such that $(H - (s, t)) - F$ does not preserve the strong connectivity of a pair of vertices u, v in $H - F$, where $u \in U$ and $v \in V$. Specifically, vertices u, v are strongly connected in $H - F$, but not strongly connected in $(H - (s, t)) - F$. This implies that edge (s, t) is inside the SCC containing u and v in $H - F$, so vertices s, t , and u are all strongly connected in $H - F$, as claimed.

Since u and v are strongly connected in $H - F$ and not strongly connected in $(H - (s, t)) - F$, there must exist a (u, v) -cut $(S, V - S)$ in H such that $\delta_{H-F}^+(S) = \{(s, t)\}$ or $\delta_{H-F}^-(S) = \{(s, t)\}$. If $\delta_{H-F}^+(S) = \{(s, t)\}$, then by Theorem 3.2 there exists an out-reachable (u, t) -cut $(S', V - S')$

such that $\delta_{H-F}^+(S') = \{(s, t)\}$. Else if $\delta_{H-F}^-(S) = \{(s, t)\}$, then by Theorem 3.2 there exists an in-reachable (u, s) -cut $(S', V - S')$ such that $\delta_{H-F}^-(S') = \{(s, t)\}$. $\square$

For each edge $e = (s, t) \in E(H) - E(J)$ we will associate a label $\ell(e) \in U \times V$ as follows. By Theorem 3.15, there exist a vertex $u \in U$ and a fault set $F \subseteq E(H)$, $|F| \leq k$, such that vertices s, t , and u are all strongly connected in $H - F$, but not all strongly connected in $(H - (s, t)) - F$.

- If H has an out-reachable (u, t) -cut $(S, V - S)$ such that $\delta_H^+(S) \subseteq F \cup \{(s, t)\}$, then we let $\ell(e) = (u, t)$.
- If H has an in-reachable (u, s) -cut $(S, V - S)$ such that $\delta_H^-(S) \subseteq F \cup \{(s, t)\}$, then we let $\ell(e) = (u, s)$.

By Claim 3.15, $\ell(e)$ is well-defined for all $e \in E(H)$.² Let

$$\mathcal{L} = \{\ell(e) \mid e \in E(H) - E(J)\}.$$

The set of labels $\mathcal{L}$ is useful because we can use it to upper bound the size of $E(H)$.

Claim 3.16. $|E(H)| \leq 2^{k+1} \cdot |\mathcal{L}| + O(2^k qn)$

Proof. Since $|E(J)| = O(2^k qn)$ by an earlier discussion, it will be sufficient to show that $|E(H) - E(J)| \leq 2^{k+1} \cdot |\mathcal{L}|$. Suppose towards contradiction that $|E(H) - E(J)| > 2^{k+1} \cdot |\mathcal{L}|$. Then there exists a vertex pair $(u, v) \in U \times V$ such that $|\{e \in E(H) - E(J) \mid \ell(e) = (u, v)\}| > 2^{k+1}$. Note that every edge $e \in E(H)$ with label $\ell(e) = (u, v)$ is of the form $e = (w, v)$ or $e = (v, w)$ for some vertex $w \in V$. Let $E_1 = \{(w, v) \in E(H) - E(J) \mid \ell((w, v)) = (u, v)\}$, and let $E_2 = \{(v, w) \in E(H) - E(J) \mid \ell((v, w)) = (u, v)\}$. Either $|E_1| > 2^k$ or $|E_2| > 2^k$.

Suppose that $|E_1| > 2^k$. By the definition of $\ell(e)$, for each edge $(w, v) \in E_1$, there exists an out-reachable (u, v) -cut $(S_w, V - S_w)$ of size $|\delta_H^+(S_w)| \leq k + 1$ such that $(w, v) \in \delta_H^+(S_w)$. Let $(S', V - S') = \text{ImpCut}(H, u, v, k + 1, +)$. Then by Lemma 3.9, $(S', V - S')$ is a (u, v) -cut such that $S_w \subseteq S'$, so $(w, v) \in \delta_H^+(S')$. This implies that $E_1 \subseteq \delta_H^+(S')$. However, by the important cut lemma (Lemma 3.9), $|\delta_H^+(S')| \leq 2^k$, a contradiction.

If instead $|E_2| > 2^k$, then we let $(S', V - S') = \text{ImpCut}(H, u, v, k + 1, -)$. By a symmetric argument as before, we can argue that $E_2 \subseteq \delta_H^-(S')$. However, by Lemma 3.9, $|\delta_H^-(S')| \leq 2^k$, a contradiction. $\square$

²If there are multiple valid choices of a label $\ell(e)$, then we select one arbitrarily.

By Theorem 3.16, it is sufficient to upper bound $|\mathcal{L}|$ in order to upper bound $|E(H)|$. We prove that $|\mathcal{L}| = O(2^k qn)$ in Theorem 3.19 by appealing to our important cut lemma (Lemma 3.9). The proof of Theorem 3.19 uses the following structural claim about the edges in $E(H) - E(J)$.

Claim 3.17. *Let $(s, t) \in E(H) - E(J)$. By Claim 3.15, there exists a vertex $u \in U$ and a fault set $F \subseteq E(H)$, $|F| \leq k$, such that vertices s , t , and u are strongly connected in $H - F$, but not strongly connected in $(H - (s, t)) - F$. Let C be the SCC in $H - F$ containing vertices s , t , and u . Then $|C \cap U| \leq 2q$.*

Proof. Suppose, towards a contradiction, that $|C \cap U| > 2q$. Let C' be the SCC in $G - F$ containing vertices s , t , and u . Since H is a k -FT sourcewise connectivity preserver of G with respect to U and $|F| \leq k$, it must follow that $C = C'$.

Then $|C' \cap U| > 2q$, and by Theorem 3.14, vertices s , t , and u are strongly connected in $J - F$. Because $J \subseteq H$ and s and t are not strongly connected in $(H - (s, t)) - F$, it follows that s and t are not strongly connected in $(J - (s, t)) - F$ as well. Since s and t are strongly connected in $J - F$ but not strongly connected in $(J - (s, t)) - F$, we conclude that $(s, t) \in E(J)$. This contradicts our assumption that $(s, t) \in E(H) - E(J)$. $\square$

Additionally, our proof of Theorem 3.19 will need the following general claim about set systems.

Claim 3.18. *Let $\mathcal{S} = \{S_u\}_{u \in U}$ be a family of sets indexed by a set U such that $u \in S_u$ and $|S_u \cap U| \leq q$ for all $S_u \in \mathcal{S}$. Then there exists a subset $\mathcal{S}_1 \subseteq \mathcal{S}$ of size $|\mathcal{S}_1| \geq |\mathcal{S}|/(2q)$ such that for all distinct sets $S_u, S_{u'} \in \mathcal{S}_1$, $u' \notin S_u$ and $u \notin S_{u'}$.*

Proof. We define a directed graph $G_{\mathcal{S}}$ with vertex set $V(G_{\mathcal{S}}) = \mathcal{S}$. We add the directed edge $(S_u, S_w) \in \mathcal{S} \times \mathcal{S}$ to $G_{\mathcal{S}}$ if $w \in S_u$. Since $|S_u \cap U| \leq q$, each vertex $S_u \in \mathcal{S}$ has out degree at most $q - 1$ in graph $G_{\mathcal{S}}$.

We say that a (possibly directed) graph is α -degenerate if every subgraph of the graph contains a vertex of total degree at most α . Since every vertex of $G_{\mathcal{S}}$ has outdegree at most $q - 1$, every subgraph of $G_{\mathcal{S}}$ has average degree at most $2q - 2$. Consequently, graph $G_{\mathcal{S}}$ is $(2q - 2)$ -degenerate. By the standard greedy vertex coloring algorithm for bounded-degeneracy graphs, graph $G_{\mathcal{S}}$ has a proper vertex coloring with at most $2q - 1$ colors.

In particular, this implies that graph $G_{\mathcal{S}}$ has an independent set $\mathcal{S}_1 \subseteq \mathcal{S}$ of size at least $|\mathcal{S}|/(2q - 1) \geq |\mathcal{S}|/(2q)$. Since set $\mathcal{S}_1 \subseteq \mathcal{S}$ is an independent set in $G_{\mathcal{S}}$, it follows that for all distinct sets $S_u, S_{u'} \in \mathcal{S}_1$, $u' \notin S_u$ and $u \notin S_{u'}$, as claimed. $\square$

We are now ready to prove Theorem 3.19.

Claim 3.19. $|\mathcal{L}| \leq 2^{k+4}qn$.

Proof. Suppose towards contradiction that $|\mathcal{L}| > 2^{k+4}qn$. Then there exists a vertex $v \in V$ such that $|\mathcal{L} \cap (U \times \{v\})| > 2^{k+4}q$. Let $\mathcal{L}_v = \mathcal{L} \cap (U \times \{v\})$. For each $(u, v) \in \mathcal{L}_v$, let $e_u = (s_u, t_u) \in E(H) - E(J)$ be an edge such that $\ell(e_u) = (u, v)$. By Theorem 3.15 and the definition of $\ell(e_u)$, there exists a fault set $F_u \subseteq E(H)$, $|F_u| \leq k$, such that vertices v and u are strongly connected in $H - F_u$, but not strongly connected in $(H - e_u) - F_u$. Let C_u denote the SCC in $H - F_u$ containing vertices v and u . By Claim 3.17, $|C_u \cap U| \leq 2q$.

We claim that for each $(u, v) \in \mathcal{L}_v$, there exists a cut $(W_u, V - W_u)$ in H such that

1. $C_u \subseteq W_u$,
2. $|W_u \cap U| \leq 4q$, and
3. either $|\delta_H^+(W_u)| \leq k$ or $|\delta_H^-(W_u)| \leq k$.

By Claim 3.7, there is an SCC C in $G - F_u$ such that $|C \cap U| \geq |U| - 2q$. Since H is a k -FT sourcewise connectivity preserver of G with respect to U , set C is an SCC in $H - F_u$ as well. If $C_u = C$, then $|U| \leq |C_u \cap U| + 2q \leq 4q$, so we can let $W_u = V$. Otherwise, C_u and C are distinct SCC's in $H - F_u$. Then there exists an (C_u, C) -cut $(W_u, V - W_u)$ in $H - F_u$ such that either $\delta_{H-F_u}^+(W_u) = \emptyset$ or $\delta_{H-F_u}^-(W_u) = \emptyset$, so $|\delta_H^+(W_u)| \leq k$ or $|\delta_H^-(W_u)| \leq k$. Moreover, $|W_u \cap U| \leq 2q$, since $|S \cap U| \geq |U| - 2q$. We conclude that cut $(W_u, V - W_u)$ satisfies the claim.

Let $\mathcal{W} = \{W_u \mid (u, v) \in \mathcal{L}_v\}$. Let $\mathcal{W}^+ = \{W_u \in \mathcal{W} \mid \delta_{H-F_u}^+(W_u) = \emptyset\}$, and let $\mathcal{W}^- = \{W_u \in \mathcal{W} \mid \delta_{H-F_u}^-(W_u) = \emptyset\}$. Observe that $|\mathcal{W}^+| + |\mathcal{W}^-| \geq |\mathcal{L}_v|$. We now split our analysis into two cases.

- **Case 1:** $|\mathcal{W}^+| \geq |\mathcal{L}_v|/2 > 2^{k+3}q$. By Claim 3.18, there exists a subset $\mathcal{W}_1^+ \subseteq \mathcal{W}^+$ of size $|\mathcal{W}_1^+| \geq |\mathcal{W}^+|/(8q) > 2^k$ such that for all distinct $W_u, W_{u'} \in \mathcal{W}_1^+$, $u \notin W_{u'}$ and $u' \notin W_u$. Let $U_1^+ = \{u \in U \mid W_u \in \mathcal{W}_1^+\}$, and note that $|U_1^+| = |\mathcal{W}_1^+| > 2^k$. For each $u \in U_1^+$, we observe the following:

1. $|\delta_H^+(W_u)| \leq k$,
2. v can reach u in $H - \delta_H^+(W_u)$, since $u, v \in C_u \subseteq W_u$, and
3. v cannot reach any vertex $u' \in U_1^+ - \{u\}$ in $H - \delta_H^+(W_u)$, since $u' \notin W_u$ if $u \neq u'$.

We can apply the anti-isolation lemma (Theorem 3.10) with respect to source vertex v , sink vertices U_1^+ , and edge sets $\{\delta_H^+(W_u) \mid u \in U_1^+\}$ to argue that $|U_1^+| \leq 2^k$. However, this contradicts our earlier claim that $|U_1^+| > 2^k$. We conclude that $|\mathcal{L}| \leq 2^{k+3}qn$.

- **Case 2:** $|\mathcal{W}^-| \geq |\mathcal{L}_v|/2 > 2^{k+3}q$. This case is symmetric to Case 1. Specifically, we can obtain a contradiction in this case by considering the reverse graph of H and applying an identical argument as in Case 1. $\square$

We finish our size analysis of H with the following claim.

Claim 3.20. $|E(H)| = O(4^k qn)$.

Proof. $|E(H)| \leq 2^{k+1} \cdot |\mathcal{L}| + O(2^k qn) \leq 2^{k+1} \cdot 2^{k+4}qn + O(2^k qn) = O(4^k qn)$, where the first inequality follows from Claim 3.16 and the second inequality follows from Claim 3.19. $\square$

3.2.0.2 Applying the directed expander hierarchy

We will combine our fault-tolerant sourcewise connectivity preservers from Lemma 3.13 with the existential directed expander hierarchy of Lemma 3.8 to complete the construction of our fault-tolerant connectivity preserver claimed in Theorem 3.11.

Theorem 3.11. *For every positive integer k , every n -vertex directed graph admits a k -fault-tolerant connectivity preserver with $O(k4^k n \log n)$ edges.*

Proof. Let G be an n -vertex directed graph, and let k be a positive integer. Let $\{V_1, \dots, V_\ell\}$ be the directed expander hierarchy for $(2k, k)$ -unbreakable sets specified in Lemma 3.8. For each $i \in [1, \ell]$, let $V_{\leq i} = V_1 \cup \dots \cup V_i$, and let $C_i^1, \dots, C_i^{j_i} \subseteq V_{\leq i}$ be the SCCs of $G[V_{\leq i}]$, for some $j_i \in [1, n]$. Additionally, for each $i \in [1, \ell]$ and $j \in [1, j_i]$, we let $U_i^j = C_i^j \cap V_i$.

For each $i \in [1, \ell]$ and $j \in [1, j_i]$ we build a k -FT sourcewise connectivity preserver H_i^j of graph $G[C_i^j]$ with respect to source vertices U_i^j using Lemma 3.13. We let $H_i = \cup_{j \in [1, j_i]} H_i^j$, and let $H = \cup_{i \in [1, \ell]} H_i$. We claim that H is a k -FT connectivity preserver of G with size $|E(H)| = O(k4^k \cdot n \log n)$.

First, we verify that $|E(H)| = O(k4^k \cdot n \log n)$. By Lemma 3.8, set U_i^j is $(2k, k)$ -unbreakable in $G[C_i^j]$ for all $i \in [1, \ell]$ and $j \in [1, j_i]$. Then by Lemma 3.13, $|E(H_i^j)| = O(4^k k |C_i^j|)$. Then we can upper bound the number of edges in H as follows:

$$|E(H_i)| \leq \sum_{j \in [1, j_i]} |E(H_i^j)| \leq O(k4^k) \cdot \sum_{j \in [1, j_i]} |C_i^j| = O(k4^k n).$$

Since there are at most $\ell = O(\log n)$ levels in our directed expander hierarchy, we conclude that

$$|E(H)| \leq \sum_{i \in [1, \ell]} |E(H_i)| \leq O(k4^k n \log n).$$

We now verify that H is a k -FT connectivity preserver of G . Fix a pair of vertices $s, t \in V$ and a fault set $F \subseteq E(G)$ of size $|F| \leq k$ such that s and t are strongly connected in $G - F$. Let $C \subseteq V(G)$ be the SCC in $G - F$ containing s and t . Let $i \in [1, \ell]$ be the largest index such that $V_i \cap C \neq \emptyset$, and let $x \in V_i \cap C$. Since $C \subseteq V_{\leq i}$, set C is contained in some SCC C_i^j in $G[V_{\leq i}]$, where $j \in [1, j_i]$. It follows that $x \in U_i^j$, since $V_i \cap C \subseteq V_i \cap C_i^j = U_i^j$. Recall that subgraph H_i^j is a sourcewise connectivity preserver of $G[C_i^j]$ with respect to source vertices U_i^j . Then since $x \in U_i^j$ is strongly connected to $s, t \in C_i^j$ in $G[C_i^j] - F$, it follows that x is strongly connected to s and t in $H_i^j - F$. We conclude that s and t are strongly connected in $H_i^j - F \subseteq H - F$, as claimed. $\square$

3.3 k -Connectivity Preservers

In this section, we present our upper bound on k -connectivity preservers in Theorem 3.21.

Theorem 3.21. *For every positive integer k , every n -vertex directed graph admits a k -connectivity preserver with $O(n\sqrt{nk})$ edges.*

Recall that a subgraph $H \subseteq G$ is a k -connectivity preserver of G if for all $s, t \in V(G)$, $\lambda_H^k(s, t) = \lambda_G^k(s, t)$, where $\lambda_G^k(s, t) = \min(\lambda_G(s, t), k)$ is the k -bounded connectivity between s, t in G .

Lemma 3.22 (transitivity). *For three graphs $H' \subseteq H \subseteq G$, if H' is a k -connectivity preserver of H , and H is a k -connectivity preserver of G , then H' is a k -connectivity preserver of G .*

Proof. For each pair of vertices $s, t \in V(G)$, by the definition of k -connectivity preservers we have $\lambda_{H'}^k(s, t) = \lambda_H^k(s, t) = \lambda_G^k(s, t)$. Therefore, H' is a k -connectivity preserver of G . $\square$

Our proof of Theorem 3.21 requires the following definition.

Definition 3.23 (k -critical graphs). We say that a graph H is k -critical if no strict edge subgraph $H' \subset H$ of H is a k -connectivity preserver of H .

We prove Theorem 3.21 by establishing the following upper bound on the number of edges in an n -vertex, k -critical directed graph.

Theorem 3.24. *Let H be an n -vertex, k -critical directed graph. Then $|E(H)| = O(k^{1/2}n^{3/2})$.*

We quickly verify that Theorem 3.24 implies Theorem 3.21.

Proof of Theorem 3.21. Let $G = (V, E)$ be an n -vertex directed graph, and let $H \subseteq G$ be an edge-minimal k -connectivity preserver of G . By Theorem 3.22 and the minimality of H , there does not exist a subgraph $H' \subsetneq H$ that is a k -connectivity preserver of H . Then H is an n -vertex, k -critical graph. By Theorem 3.24, $|E(H)| = O(k^{1/2}n^{3/2})$. $\square$

We will prove Theorem 3.24 using a two-level decomposition with regard to (q, k) -unbreakability. Let H be an n -vertex, k -critical directed graph. Let $q \in [1, n]$ be a parameter of the decomposition that we will optimize later. We maintain a partition $\mathcal{S}$ of vertex set V and a collection of cuts $\mathcal{C}$ as follows. Initially, we let $\mathcal{S} = \{V\}$ and $\mathcal{C} = \emptyset$. While there exists a set $S \in \mathcal{S}$ and a cut (L, R) in H of size at most $\min(|\delta_H^+(L)|, |\delta_H^-(L)|) \leq k$ such that $|S \cap L| \geq q$ and $|S \cap R| \geq q$, we remove set S from $\mathcal{S}$ and we add sets $S \cap L$ and $S \cap R$ to $\mathcal{S}$. Additionally, we add cut (L, R) to collection $\mathcal{C}$. Note that $\mathcal{S}$ remains a partition of V after this operation. We repeat this process until no such set $S \in \mathcal{S}$ and cut (L, R) exists.

Claim 3.25. *This process must terminate. After the above process terminates, every set $U \in \mathcal{S}$ is (q, k) -unbreakable in H . Moreover, $|\mathcal{C}| \leq n/q$.*

Proof. This process must terminate, since $|\mathcal{S}|$ increases by one after each iteration. By the terminating condition of this procedure, every set $U \in \mathcal{S}$ is (q, k) -unbreakable in H . Additionally, every set $U \in \mathcal{S}$ has size at least $|U| \geq q$. We conclude that $|\mathcal{S}| \leq n/q$, and therefore $|\mathcal{C}| \leq |\mathcal{S}| \leq n/q$. $\square$

We will bound the number of edges inside each set in $\mathcal{S}$ in Theorem 3.26, and we will bound the number of edges crossing sets in $\mathcal{S}$ using Theorem 3.29.

Lemma 3.26. *Consider a k -critical directed graph H . For every subset $U \subseteq V(H)$, if U is (q, k) -unbreakable in H , then the induced subgraph $H[U]$ has at most $O((q + k)|U|)$ edges.*

Proof. Since H is k -critical, for each edge $e \in E(H)$, graph $H - e$ is not a k -connectivity preserver of H . Therefore, there exists a pair of nodes $s, t \in V(H)$ such that $\lambda_{H-e}^k(s, t) < \lambda_H^k(s, t)$. Thus, for each edge $e \in E(H)$, there exists a minimum (s, t) -cut (L_e, R_e) of size $|\delta_H^+(L_e)| \leq k$ such that $e \in \delta_H^+(L_e)$.

Now we will show that for each edge $e = (u, v) \in H[U]$,

$$\min(\text{outdeg}_{H[U]}(u), \text{indeg}_{H[U]}(v)) \leq q + k.$$

Suppose towards contradiction that $\min(\text{outdeg}_{H[U]}(u), \text{indeg}_{H[U]}(v)) > q + k$. Then since (L_e, R_e) is a cut of size at most $|\delta_H^+(L_e)| \leq k$ and $e \in \delta_H^+(L_e)$, this implies that

$$|L_e \cap S| \geq |N_{H[U]}^+(u) \cap L_e| \geq |N_{H[U]}^+(u)| - k > q$$

and

$$|R_e \cap U| \geq |N_{H[U]}^-(v) \cap R_e| \geq |N_{H[U]}^-(v)| - k > q.$$

However, this contradicts our assumption that set U is (q, k) -unbreakable in H . We conclude that for each edge $(u, v) \in H[U]$, $\min(\text{outdeg}_{H[U]}(u), \text{indeg}_{H[U]}(v)) \leq q + k$, as claimed.

Using this inequality, it becomes straightforward to prove that the total number of edges in $H[U]$ is at most $|E(H[U])| \leq 2(q + k)|U|$. For each edge $(u, v) \in E(H[U])$, if $\text{outdeg}_{H[U]}(u) \leq q + k$, then we charge edge (u, v) to vertex $u \in U$. Otherwise, if $\text{indeg}_{H[U]}(v) \leq q + k$, then we charge edge (u, v) to vertex $v \in U$. Under this charging scheme each vertex in U is charged at most $2(q + k)$ times, so the total number of edges in $H[U]$ is at most $|E(H[U])| \leq 2(q + k)|U|$. $\square$

We also need the next lemma, which reduces the task of constructing an all-pairs k -connectivity preserver to preserving k -connectivity between just $O(n)$ demand pairs. This lemma follows from well-known results about Gomory-Hu trees [GH61; CH91].

Lemma 3.27. *For any n -vertex directed graph $H = (V, E)$, there exists a collection of vertex demand pairs $P \subseteq V \times V$ of size $|P| = O(n)$ such that for any edge subgraph $H' \subseteq H$, H' is a k -connectivity preserver of H if and only if for every demand pair $(s, t) \in P$, $\lambda_{H'}^k(s, t) = \lambda_H^k(s, t)$.*

Proof. The “only if” direction is trivial, so we focus on proving the “if” direction below. Let C_H be a complete undirected weighted graph with vertex set V . We assign each edge $(u, v) \in E(C_H)$ weight $\lambda_H(u, v)$. Let T be a maximum spanning tree of C_H .

For any edge $(u, v) \in E(C_H)$, consider the (u, v) -path $Q = (u_0, u_1, \dots, u_\ell)$ in T , where $u = u_0$ and $v = u_\ell$. We claim that $\lambda_H(u, v) = \min_{(s, t) \in Q} \lambda_H(s, t)$. On one side, $\lambda_H(u, v) \geq \min_{(s, t) \in Q} \lambda_H(s, t)$, by the triangle inequality of symmetric connectivity. On the other side, $\lambda_H(u, v) \leq \min_{(s, t) \in Q} \lambda_H(s, t)$, since T is a maximum spanning tree in C_H .

We define our set of vertex demand pairs $P \subseteq V \times V$ to be the edges of tree T , so that $P = E(T)$. Now suppose that $H' \subseteq H$ is an edge subgraph of H that satisfies $\lambda_{H'}^k(s, t) = \lambda_H^k(s, t)$ for each demand pair $(s, t) \in P$. Then by the triangle inequality of symmetric connectivity, we have

$$\lambda_{H'}^k(u, v) \geq \min_{(s, t) \in Q} \lambda_{H'}^k(s, t) = \min_{(s, t) \in Q} \lambda_H^k(s, t) = \lambda_H^k(u, v).$$

Therefore, H' preserves the k -bounded connectivity for all vertex pairs (u, v) in H . $\square$

We will need the following lemma in order to eventually bound the number of edges between components in $\mathcal{S}$.

Lemma 3.28. *Let H be an n -vertex, k -critical directed graph. For every cut (L, R) with size $|\delta^+(L)| \leq k$, we have $|\delta^-(L)| = O(kn)$.*

Proof. Let $P \subseteq V(H) \times V(H)$ be the set of $|P| = O(n)$ demand pairs specified in Theorem 3.27. For each demand pair $(s, t) \in P$, there exists a collection $Q_{s,t}$ of $\lambda_H^k(s, t)$ pairwise edge-disjoint paths from s to t in H , and there exists a collection $Q_{t,s}$ of $\lambda_H^k(s, t)$ pairwise edge-disjoint paths from t to s in H . For each path $Q \in Q_{s,t} \cup Q_{t,s}$, it must pass through edges in $\delta_H^+(L)$ and edges in $\delta_H^-(L)$ interleavingly, which means the number of edges in $\delta_H^-(L)$ on the path is at most the number of edges in $\delta_H^+(L)$ on the path, plus one. Since the paths in $Q_{s,t}$ are pairwise edge-disjoint, the paths in $Q_{s,t}$ contain at most $|\delta_H^+(L)| + \lambda_H^k(s, t) \leq 2k$ edges in $\delta_H^-(L)$. An identical argument implies that the paths in $Q_{t,s}$ contain at most $2k$ edges in $\delta_H^-(L)$ as well.

Let $H' \subseteq H$ be the subgraph of H obtained by unioning all the paths in all collections $Q_{s,t} \cup Q_{t,s}$ for all $(s, t) \in P$, so that $H' = \cup_{\{Q \in Q_{s,t} \cup Q_{t,s} \mid (s,t) \in P\}} Q$. We observe that $\lambda_{H'}^k(s, t) = \lambda_H^k(s, t)$. Graph H' is the union of $2|P|$ collections of paths $Q_{s,t}$, and the paths in $Q_{s,t}$ contain at most $2k$ edges in $\delta_H^-(L)$, by the earlier discussion. We conclude that there are at most $|\delta_{H'}^-(L)| \leq 2k \cdot 2|P| = O(kn)$ edges in $\delta_{H'}^-(L)$. Now observe that for any demand pair $(s, t) \in P$, $\lambda_{H'}^k(s, t) = \lambda_H^k(s, t)$, so H' is a k -connectivity preserver of H by Theorem 3.27. Finally, since H is k -critical, we know $H' = H$ and thus $|\delta_H^-(L)| = |\delta_{H'}^-(L)| = O(kn)$. $\square$

We can now bound the number of cross-component edges in $H[S]$.

Lemma 3.29. *Let $E' = \{(u, v) \in E(H[S]) \mid (u, v) \in S_1 \times S_2, S_1 \neq S_2 \in \mathcal{S}\}$ be the set of edges in $H[S]$ between different components in $\mathcal{S}$. Then $|E'| = O(kn^2/q)$.*

Proof. Recall that $\mathcal{C}$ is the family of all cuts (L, R) that we considered in our construction of partition $\mathcal{S}$. By our construction of $\mathcal{S}$, every edge $e \in E'$ crosses a cut (L, R) in cut family $\mathcal{C}$ (i.e., $e \in \delta^+(L) \cup \delta^-(L)$ for some cut $(L, R) \in \mathcal{C}$).

By Theorem 3.25, $|\mathcal{C}| \leq n/q$, and by Theorem 3.28, we have that $|\delta^+(L) \cup \delta^-(L)| = O(kn)$ for each cut $(L, R) \in \mathcal{C}$. Therefore,

$$|E'| \leq \sum_{(L,R) \in \mathcal{C}} |\delta^+(L) \cup \delta^-(L)| \leq O((n/q) \cdot kn) = O(kn^2/q).$$

$\square$

Proof of Theorem 3.24. By Theorem 3.25 and Theorem 3.26, we know that the total number of

edges in $\cup_{S \in \mathcal{S}} H[S]$ is at most

$$\left| \bigcup_{S \in \mathcal{S}} H[S] \right| \leq \sum_{S \in \mathcal{S}} O((q+k)|S|) = O((q+k)n).$$

Additionally, by Theorem 3.29, $|E'| = O(kn^2/q)$. We conclude that

$$|E(H)| \leq |\cup_{S \in \mathcal{S}} H[S]| + |E'| = O((q+k)n + kn^2/q) = O(k^{1/2}n^{3/2}),$$

where the final inequality follows by choosing $q = \sqrt{nk}$ and observing that $k \leq n$. □

Chapter 4

Hopsets and Shortcut Sets

This chapter presents the lower bounds proved for exact hopsets and for shortcut sets.¹

4.1 Preliminaries

We use the following notations:

- For a path π , we use $|\pi|$ to denote the number of **nodes** in π . This is 1 different from the (unweighted) length of π . In weighted graphs, we write $w(\pi)$ for the sum of edge weights in π .
- We write $\text{dist}_G(s, t)$ for the shortest path distance from node s to node t in graph G (counting edge weights, if G is weighted). We write $\text{hopdist}_G(s, t)$ for the least number of edges contained in any $s \rightsquigarrow t$ shortest path.
- We use $\langle \cdot, \cdot \rangle$ to denote the standard Euclidean inner product.

4.2 Exact Hopsets

In this section we will prove the following theorem.

Theorem 4.1. *For any parameter $p \in [1, n^2]$, there exists an n -node weighted undirected graph $G = (V, E, w)$ such that for any exact hopset H of size $|H| \leq p$ where $p \in [1, n^2]$, the graph $G \cup H$ must have hopbound $\Omega\left(\frac{n}{p^{1/2} \log^{1/2} n}\right)$.*

We will prove this via a construction of the following type:

¹The contents in this section originally appeared in [BH23a] and are based on joint work with Greg Bodwin that appeared in FOCS 2023.

Lemma 4.2. *For any $p \in [1, n^2]$, there is an infinite family of n -node undirected weighted graphs $G = (V, E, w)$ and sets Π of $|\Pi| = p$ paths in G with the following properties:*

- *G has $\ell = \Theta\left(\frac{n}{p^{1/2} \log^{1/2} n}\right)$ layers, and each path in Π starts in the first layer, ends in the last layer, and contains exactly one node in each layer.*
- *Each path in Π is the unique shortest path between its endpoints in G .*
- *For any two nodes $u, v \in V$, there are at most $\frac{\ell}{\text{hopdist}_G(u, v)}$ paths in Π that contain both u and v .*
- *Each node $v \in V$ lies on at most $O\left(\frac{\ell p}{n}\right)$ paths in Π .*

Next we show how Lemma 4.2 implies Theorem 4.1.

Proving Theorem 4.1 using Lemma 4.2

Fix an n and $p \in [1, n^2]$. Let $G = (V, E, w)$ be the graph in Lemma 4.2 with associated set Π of $|\Pi| = 2p$ paths in G . Let H be an exact hopset of size $|H| \leq p$. Let $P \subseteq V \times V$ be the set of node pairs that are the endpoints of paths in Π . We define the following potential function over hopsets H , which simply sums hopdistances over critical pairs:

$$\phi(H) := \sum_{(s, t) \in P} \text{hopdist}_{G \cup H}(s, t)$$

Observe that by Lemma 4.2, we have $\phi(\emptyset) = \sum_{(s, t) \in P} \text{hopdist}_G(s, t) = |\Pi| \cdot (\ell - 1) = 2p(\ell - 1)$. Now fix a pair of nodes $(x, y) \in V \times V$, and let $\Pi' \subseteq \Pi$ be the set of paths $\pi \in \Pi$ such that $x, y \in \pi$. We make the following observations.

- For all $(s, t) \in P$, if the unique shortest $s \rightsquigarrow t$ -path $\pi \in \Pi$ in G is not in Π' , then

$$\text{hopdist}_{G \cup \{(x, y)\}}(s, t) = \text{hopdist}_G(s, t).$$

- For all $(s, t) \in P$, if the unique shortest $s \rightsquigarrow t$ -path $\pi \in \Pi$ in G is in Π' , then

$$\text{hopdist}_G(s, t) - \text{hopdist}_{G \cup \{(x, y)\}}(s, t) \leq \text{hopdist}_G(x, y).$$

Then by Lemma 4.2, $\phi(\emptyset) - \phi(\{(x, y)\}) \leq |\Pi'| \cdot \text{hopdist}_G(x, y) \leq \ell$. We obtain the following sequence of inequalities:

$$\phi(\emptyset) - \phi(H) \leq \sum_{(x,y) \in H} (\phi(\emptyset) - \phi(\{(x, y)\})) \leq \ell \cdot |H| \leq \ell p.$$

Rearranging, we find that

$$\phi(H) \geq \phi(\emptyset) - \ell p \geq 2p(\ell - 1) - \ell p = (\ell - 2)p.$$

Thus, over the $|P| = 2p$ pairs of path endpoints in P , the average hopdistance in $G \cup H$ is $\Theta(\ell)$, and so there must be a pair $(s, t) \in P$ such that $\text{hopdist}_{G \cup H}(s, t) = \Theta(\ell)$ by the pigeonhole principle.

Constructing G

Our goal is now to prove Lemma 4.2. Let n be a sufficiently large positive integer, and let $p \in [n, n^2]$.² For simplicity of presentation, we will frequently ignore issues related to non-integrality of expressions that arise in our construction; these issues affect our bounds only by lower-order terms. Initially, all edges (u, v) in G will be directed from u to v ; we will convert G into an undirected graph in the final step of our construction.

Vertex Set V .

- Let ℓ be a positive integer parameter of the construction to be specified later. Our graph G will have ℓ layers $L_1, \dots, L_\ell$, and each layer will have n/ℓ nodes, ordered from 1 to n/ℓ . Initially, we will label the j th node in layer L_i with tuple (i, j) . We will interpret the node labeled (i, j) as a point in $\mathbb{R}^2$ with integer coordinates. These n nodes arranged in ℓ layers will be the node set V of graph G .
- We now perform the following random operation on the node labels of V . For each layer L_i , $i \in [1, \ell]$, uniformly sample a random real number in the interval $(0, 1)$ and call it ε_i . Now for each node in layer L_i of G labeled (i, j) , relabel this node with the label

$$\left(i, j + \sum_{k=1}^j \varepsilon_k\right).$$

²We will handle the case where $p \in [1, n]$ later.

Again, we interpret the resulting labels for nodes in V as points in $\mathbb{R}^2$. In a slight abuse of notation, we will treat $v \in V$ as either a node in G or a point in $\mathbb{R}^2$, depending on the context. Less formally: for each layer i , this step shifts the nodes in layer i vertically upwards to be ε_i higher than the previous layer (and thus, these vertical shifts compound across the layers).

Edge Set E .

- All the edges in E will be between consecutive layers L_i, L_{i+1} of G . We will let E_i denote the set of edges in G between layers L_i and L_{i+1} .
- Just as our nodes in V correspond to points in $\mathbb{R}^2$, we can interpret the edges E in G as vectors in $\mathbb{R}^2$. In particular, for every edge $e = (v_1, v_2) \in E$, we identify e with the corresponding vector $\vec{u}_e := v_2 - v_1$. Note that since all edges in E are between adjacent layers L_i and L_{i+1} , the first coordinate of $\vec{u}_e$ is 1 for all $e \in E$. We will use u_e to denote the 2nd coordinate of $\vec{u}_e$, i.e., for all $e \in E$, we write $\vec{u}_e = (1, u_e)$.
- We begin our construction of E by defining the following set C of vectors:

$$C := \left\{ (1, x) \mid x \in \left[0, \frac{n}{4\ell^2} \right] \right\}.$$

We will refer to the vectors in C as *edge vectors*.

- For each $i \in [1, \ell - 1]$, let

$$C_i := \{ \vec{c} + (0, \varepsilon_{i+1}) \mid \vec{c} \in C \}.$$

Intuitively: we want the edge vectors in C to point between nodes in adjacent layers, and due to the random vertical shifts between layers applied to the nodes, we need to apply a similar shift to C at each layer to adjust for this.

- For each $v \in L_i$ and edge vector $\vec{c} \in C_i$, if $v + \vec{c} \in V$, then add edge $(v, v + \vec{c})$ to E_i . After adding these edges to E_i , we will have that

$$C_i = \{ \vec{u}_e \mid e \in E_i \}.$$

Additionally, note that the case $v + \vec{c} \notin V$ only occurs if $v + \vec{c} = (i + 1, j)$ for some j that is higher than any point in the $(i + 1)$ st layer; that is, $j > \frac{n}{\ell} + \sum_{k=1}^{i+1} \varepsilon_k$.

- For each $e \in E$, if $\vec{u}_e = (1, u_e)$, then we assign edge e the weight $w(e) := u_e^2$.

This completes the construction of our graph $G = (V, E, w)$.

Direction Vectors, Critical Pairs, and Critical Paths

Our next step is to generate a set of *critical pairs* $P \subseteq V \times V$, as well as a set of *critical paths* Π . Specifically, there will be one critical path $\pi_{s,t} \in \Pi$ going between each critical pair $(s, t) \in P$, and we will show that $\pi_{s,t}$ is the unique shortest (weighted) $s \rightsquigarrow t$ path in G . We will identify our critical pairs and paths by first constructing a set of vectors $D \subseteq \mathbb{R}^2$ that we call *direction vectors*, which we define next.

Direction Vectors D .

- Let $q \in \mathbb{Z}_{\geq 0}$ be a sufficiently large integer parameter to be specified later. The size of q will roughly correspond to the maximum number of edges shared between any two critical paths in G .
- We choose our set of direction vectors D to be ³

$$D := \left\{ \left(1, x + \frac{y}{q} \right) \mid x \in \left[1, \frac{n}{4\ell^2} - 1 \right] \text{ and } y \in [0, q] \right\}.$$

Note that there are $q + 1$ direction vectors between adjacent vectors $(1, x), (1, x + 1) \in C$ for $x \neq 0$. Additionally, adjacent direction vectors in D differ only by $1/q$ in their second coordinate.

Proposition 4.3. *With probability 1, for every $i \in [1, \ell - 1]$ and every direction vector $\vec{d} = (1, d) \in D$, there is a unique vector $(1, c) \in C_i$ that minimizes $|c - d|$ over all choices of $(1, c) \in C_i$.*

Proof. There are only finitely many choices of $\varepsilon_{i+1} \in (0, 1)$ that result in there being two distinct vectors $(1, c_1), (1, c_2) \in C_i$ such that $|c_1 - d| = |c_2 - d|$. We conclude that the claim holds with probability 1. $\square$

In the following we assume that this event holds, i.e., there is a unique minimizing vector in C_i for all $\vec{d} \in D$. Each of our critical paths π in Π will have an associated direction vector $\vec{d} \in D$, and for all $i \in [1, \ell - 1]$, path π will take an edge vector in C_i that is closest to $\vec{d}$ in the sense of Proposition 4.3.

Critical Pairs P and Critical Paths Π .

³Note that if $(1, x) \in D$, then $x \in [1, \frac{n}{4\ell^2}]$. However, if $(1, x) \in C$, then $x \in [0, \frac{n}{4\ell^2}]$. This $+1$ gap between C and D is needed to accommodate the ε -shifting operation used to obtain C_i , and is relevant in the proof of Lemma 4.8.

- We first define a set $S \subseteq L_1$ containing half of the nodes in the first layer L_1 of G :

$$S := \left\{ (1, j + \varepsilon_1) \in L_1 \mid j \in \left[1, \frac{n}{2\ell}\right] \right\}.$$

We will choose our set of demand pairs P so that $P \subseteq S \times L_\ell$. For every node $s \in S$ and direction vector $\vec{d} \in D$, we will create a critical pair $(s, t) \in S \times L_\ell$ and a corresponding critical path $\pi_{s,t}$ to add to P and Π .

- Let $v_1 \in S$, and let $\vec{d} = (1, d) \in D$. The associated path π has start node v_1 . We iteratively grow π , layer-by-layer, as follows. Suppose that currently $\pi = (v_1, \dots, v_i)$, for $i < \ell$, with each $v_i \in L_i$. To determine the next node $v_{i+1} \in L_{i+1}$, let $E_i^{v_i} \subseteq E_i$ be the edges in E_i incident to v_i , and let

$$e_i := \operatorname{argmin}_{e \in E_i^{v_i}} (|u_e - d|).$$

By definition, e_i is an edge whose first node is v_i ; we define $v_{i+1} \in L_{i+1}$ to be the other node in e_i , and we append v_{i+1} to π .

- This completes our construction of P and Π . Note that
 - we will show that the paths generated in this way have distinct endpoints (with high probability), and therefore $|P| = |S||D| \geq \frac{n^2 q}{16\ell^3}$, and
 - every path $\pi_{s,t} \in \Pi$ contains one node in each layer, and therefore its number of nodes is $|\pi_{s,t}| = \ell$.

An important feature for correctness of our construction is that, when we iteratively generate paths, we never reach a point v_i such that $v_i + C_i \not\subseteq L_{i+1}$ (i.e. $v_i + \vec{c} \notin L_{i+1}$ for some $\vec{c} \in C_i$). This follows by straightforward counting, based on the maximum second coordinate used in our edge vectors C and also on our choice of start nodes S as only the “lower half” of the nodes in the first layer. The following proposition expresses this correctness in a particular way, pointing out that for any node v lying on a generated path π , none of the edges from v to the following layer are omitted from the graph due to falling off the top of the grid with a too-high second coordinate.

Proposition 4.4. *Let $v \in L_i \cap \pi$ for some $i \in [1, \ell - 1]$ and $\pi \in \Pi$. Then $\{\vec{u}_e \mid e \in E_i^v\} = C_i$.*

Proof. Let $v =: (i, j)$, and let $(s, t) \in P$ be the endpoints of π . Since $s \in S$, we have $s = (1, s_2) \in \mathbb{R}^2$, where

$$s_2 \leq \frac{n}{2\ell} + \varepsilon_1.$$

Moreover, since for all $e \in E_i$ the corresponding vector $\vec{u}_e = (1, u_e)$ satisfies $\varepsilon_{i+1} \leq u_e \leq \frac{n}{4\ell^2} + \varepsilon_{i+1}$, we have

$$j \leq \frac{n}{2\ell} + (i-1) \cdot \frac{n}{4\ell^2} + \sum_{k=1}^i \varepsilon_k \leq \frac{3n}{4\ell} + \sum_{k=1}^i \varepsilon_k.$$

Then observe that for all $\vec{c} = (1, c) \in C_i$, we have that $v + \vec{c} = (i+1, j+c)$, where

$$j+c \leq \left(\frac{3n}{4\ell} + \sum_{k=1}^i \varepsilon_k \right) + \left(\frac{n}{4\ell^2} + \varepsilon_{i+1} \right) \leq \frac{n}{\ell} + \sum_{k=1}^{i+1} \varepsilon_k.$$

Thus $(i+1, j+c) \in L_{i+1}$, and so we have $(v, v + \vec{c}) \in E_i^v$ for all $\vec{c} \in C_i$. It follows that $\{\vec{u}_e \mid e \in E_i^v\} = C_i$. $\square$

Critical paths are unique shortest paths

We now verify that graph G and paths Π have the unique shortest path property as stated in Lemma 4.2.

Lemma 4.5 (Unique shortest paths). *With probability 1, for every $(s, t) \in P$, path $\pi_{s,t} \in \Pi$ is a unique shortest (weighted) $s \rightsquigarrow t$ -path in G .*

We begin with a technical proposition:

Proposition 4.6. *Let $b, x_1, \dots, x_k \in \mathbb{R}$. Now consider $\hat{x}_1, \dots, \hat{x}_k$ such that*

- $|\hat{x}_i - b| \leq |x_i - b|$ for all $i \in [1, k]$, and
- $\sum_{i=1}^k x_i = \sum_{i=1}^k \hat{x}_i$.

Then

$$\sum_{i=1}^k x_i^2 \geq \sum_{i=1}^k \hat{x}_i^2,$$

with equality only if $|\hat{x}_i - b| = |x_i - b|$ for all $i \in [1, k]$.

Proof. We will prove the equivalent statement $\sum_{i=1}^k (\hat{x}_i^2 - x_i^2) \leq 0$. Fix an $i \in [1, k]$. First we will show that

$$\hat{x}_i^2 - x_i^2 \leq 2b(\hat{x}_i - x_i).$$

We split our analysis into four cases:

- **Case 1:** $b \leq \hat{x}_i \leq x_i$. In this case, $\hat{x}_i^2 - x_i^2 = (\hat{x}_i + x_i)(\hat{x}_i - x_i) \leq 2b(\hat{x}_i - x_i)$.

- **Case 2:** $x_i \leq \hat{x}_i \leq b$. In this case, $\hat{x}_i^2 - x_i^2 = (\hat{x}_i + x_i)(\hat{x}_i - x_i) \leq 2b(\hat{x}_i - x_i)$.
- **Case 3:** $\hat{x}_i \leq b \leq x_i$. In this case, $\hat{x}_i^2 - x_i^2 = (\hat{x}_i + x_i)(\hat{x}_i - x_i) \leq 2b(\hat{x}_i - x_i)$, since $b - \hat{x}_i \leq x_i - b$.
- **Case 4:** $x_i \leq b \leq \hat{x}_i$. In this case, $\hat{x}_i^2 - x_i^2 = (\hat{x}_i + x_i)(\hat{x}_i - x_i) \leq 2b(\hat{x}_i - x_i)$, since $\hat{x}_i - b \leq b - x_i$.

Then

$$\sum_{i=1}^k (\hat{x}_i^2 - x_i^2) \leq 2b \sum_{i=1}^k (\hat{x}_i - x_i) = 0.$$

This inequality is strict if $|\hat{x}_i - b| < |x_i - b|$ for some $i \in [1, k]$. □

Using Proposition 4.6, we can now prove Lemma 4.5.

Proof of Lemma 4.5. Fix an $(s, t) \in P$, and let $(1, x) \in D$ be the direction vector associated with $\pi_{s,t}$. Let $\hat{x}_1, \dots, \hat{x}_{\ell-1} \in \mathbb{R}$ be real numbers such that the i th edge of $\pi_{s,t}$ has the corresponding vector $(1, \hat{x}_i) \in C_i$ for $i \in [1, \ell - 1]$. Now consider an arbitrary $s \rightsquigarrow t$ -path π in G , where $\pi \neq \pi_{s,t}$. Since all edges in G are directed from L_i to L_{i+1} , it follows that π has $\ell - 1$ edges and the i th edge of π is in E_i . Let $x_1, \dots, x_{\ell-1} \in \mathbb{R}$ be real numbers such that the i th edge of π has the corresponding vector $(1, x_i) \in C_i$ for $i \in [1, \ell - 1]$. Now observe that since π and $\pi_{s,t}$ are both $s \rightsquigarrow t$ -paths, it follows that

$$\sum_{i=1}^{\ell-1} \hat{x}_i = \sum_{i=1}^{\ell-1} x_i.$$

Additionally, by our construction of $\pi_{s,t}$, it follows that

$$|\hat{x}_i - x| \leq |x_i - x|$$

for all $i \in [1, \ell - 1]$. In particular, since $\pi \neq \pi_{s,t}$, there must be some $j \in [1, \ell - 1]$ such that $\hat{x}_j \neq x_j$, and so by Proposition 4.3, $|\hat{x}_j - x| < |x_j - x|$ with probability 1. Then by Proposition 4.6,

$$w(\pi_{s,t}) = \sum_{e \in \pi_{s,t}} w(e) = \sum_{i=1}^{\ell-1} \hat{x}_i^2 < \sum_{i=1}^{\ell-1} x_i^2 = \sum_{e \in \pi} w(e) = w(\pi).$$

Path $\pi_{s,t}$ is a unique shortest $s \rightsquigarrow t$ -path in G , as desired. □

Critical Paths Intersection Properties

Before finishing our proof of Lemma 4.2, we will need to establish several properties of the critical paths in Π .

Proposition 4.7. *Let $\pi_1, \pi_2 \in \Pi$ be two critical paths with the same corresponding direction vector $\vec{d} \in D$. Then $\pi_1 \cap \pi_2 = \emptyset$.*

Proof. Let $v_i^j \in L_i$ denote the i th node of π_j , where $j \in \{1, 2\}$. Note that since π_1 and π_2 share the same direction vector $\vec{d}$, the edges (v_i^1, v_{i+1}^1) and (v_i^2, v_{i+1}^2) have the same corresponding vector $\vec{u}_i \in C_i$ for all $i \in [1, \ell - 1]$ by Proposition 4.4. By our construction of Π , for each node in the first layer $v \in L_1$, v belongs to at most one path $\pi \in \Pi$ with direction vector $\vec{d}$, so $v_1^1 \neq v_1^2$. Then for all $i \in [1, \ell]$,

$$v_i^1 = v_1^1 + \sum_{i=1}^{i-1} \vec{u}_i \neq v_1^2 + \sum_{i=1}^{i-1} \vec{u}_i = v_i^2.$$

□

Let $\pi_1, \pi_2 \in \Pi$ be two critical paths, and let $v \in V$ be a node in G . We say that paths π_1 and π_2 *split* at v if $v \in \pi_1 \cap \pi_2$ and the node following v in π_1 is distinct from the node following v in π_2 , and we simply say that π_1 and π_2 *split* if there exists some $v \in V$ such that they split at v . Note that since $\pi_1, \pi_2 \in \Pi$ are unique shortest paths in G , paths π_1 and π_2 can split at most once.

Lemma 4.8. *Fix a node $v \in L_i$, where $i \in [1, \ell - 1]$, and let $\pi_1, \pi_2 \in \Pi$ be critical paths with direction vectors $(1, d_1), (1, d_2) \in D$ such that $v \in \pi_1$ and $v \in \pi_2$. Then paths π_1 and π_2 split at v with probability at least $\min\{|d_1 - d_2|, 1\}$.⁴*

Proof. By Proposition 4.7, $d_1 \neq d_2$, and assume wlog that $d_1 < d_2$. Let F be the event that the random variable ε_{i+1} was sampled so that

$$(\mathbb{Z} + \varepsilon_{i+1} + 1/2) \cap (d_1, d_2) \neq \emptyset,$$

(where $(d_1, d_2) \subseteq \mathbb{R}$ is the open interval with endpoints d_1 and d_2). Our proof strategy is to show that F implies that π_1, π_2 split at v , and then to show that F occurs with the claimed probability.

⁴For the sake of completeness, let us be more precise here about the probability claim being made in this lemma. Consider any two paths π_1, π_2 , indexed by two start nodes and two direction vectors, and consider a node $v \in L_i$. The event that we generate π_1, π_2 in such a way that $v \in \pi_1 \cap \pi_2$ depends only on the random choices of $\varepsilon_1, \dots, \varepsilon_i$. If $v \in \pi_1 \cap \pi_2$, then the event that π_1, π_2 split at v depends only on the random choice of ε_{i+1} . The claim is: conditional on the event that $\{\varepsilon_1, \dots, \varepsilon_i\}$ are selected in such a way that $v \in \pi_1 \cap \pi_2$, the probability that ε_{i+1} is selected such that π_1, π_2 split at v is at least $\min\{|d_1 - d_2|, 1\}$.

F implies that π_1, π_2 split at v . Assume that F occurs. By construction there is a nonnegative integer $c \in \mathbb{Z}$ such that $c + \varepsilon_{i+1} + 1/2$ is in the interval (d_1, d_2) . Since $(d_1, d_2) \subseteq \left(1, \frac{n}{4\ell^2}\right)$, it follows that vectors $(1, c + \varepsilon_{i+1}), (1, c + \varepsilon_{i+1} + 1) \in \mathbb{R}^2$ are in C_i , because $0 \leq c \leq \frac{n}{4\ell^2} - 1$. More generally, by our choice of sets C and D there are vectors $(1, c_1), (1, c_2) \in C_i$ such that

$$c_1 \leq d_1 \leq c_1 + 1 \quad \text{and} \quad c_2 \leq d_2 \leq c_2 + 1.$$

Now we claim that

$$\operatorname{argmin}_{(1,x) \in C_i} |x - d_1| \neq \operatorname{argmin}_{(1,x) \in C_i} |x - d_2|.$$

To see this, suppose for the sake of contradiction that there is a vector $(1, y) \in C_i$ such that

$$(1, y) = \operatorname{argmin}_{(1,x) \in C_i} |x - d_1| = \operatorname{argmin}_{(1,x) \in C_i} |x - d_2|.$$

Then using our assumption that $d_1 < c + \varepsilon_{i+1} + 1/2 < d_2$, we obtain

$$y - d_1 \leq |y - d_1| \leq \min\{|c_1 - d_1|, |(c_1 + 1) - d_1|\} \leq 1/2 < (c + \varepsilon_{i+1} + 1) - d_1$$

and

$$d_2 - y \leq |y - d_2| \leq \min\{|c_2 - d_2|, |(c_2 + 1) - d_2|\} \leq 1/2 < d_2 - (c + \varepsilon_{i+1}).$$

Together, these two sequences of inequalities imply that $c + \varepsilon_{i+1} < y < c + \varepsilon_{i+1} + 1$. But this contradicts our assumption that $(1, y) \in C_i$, so we conclude that

$$\operatorname{argmin}_{(1,x) \in C_i} |x - d_1| \neq \operatorname{argmin}_{(1,x) \in C_i} |x - d_2|.$$

By Proposition 4.4, $\{\vec{u}_e \mid e \in E_i^v\} = C_i$, so we have also shown that

$$\operatorname{argmin}_{\{\vec{u}_e \mid e \in E_i^v\}} |u_e - d_1| \neq \operatorname{argmin}_{\{\vec{u}_e \mid e \in E_i^v\}} |u_e - d_2|.$$

Then π_1 and π_2 must split at v by our construction of the critical paths in Π .

F happens with good probability. Since ε_{i+1} is sampled uniformly at random from the interval $(0, 1)$, it follows that $(\mathbb{Z} + \varepsilon_{i+1} + 1/2) \cap (d_1, d_2) \neq \emptyset$ with probability at least $\min\{d_2 - d_1, 1\}$. $\square$

We will use Lemma 4.8 to prove the following two lemmas, which capture the key properties of our graph G .

Lemma 4.9 (Low path overlap). *Let $\pi_1, \pi_2 \in \Pi$ be critical paths with distinct associated direction vectors $(1, d_1), (1, d_2) \in D$. Then:*

- *If $|d_1 - d_2| < 1$, then with probability at least $1 - n^{-8}$, we have⁵ $|\pi_1 \cap \pi_2| \leq \frac{8 \log n}{|d_1 - d_2|}$.*
- *If $|d_1 - d_2| \geq 1$, then $|\pi_1 \cap \pi_2| \leq 1$ (deterministically).*

Proof. We begin with the first point; suppose $|d_1 - d_2| < 1$. Suppose we iteratively generate π_1, π_2 one layer at a time. Each time we choose a node v that lies in both π_1 and π_2 , by Lemma 4.8, π_1 and π_2 split at v with probability at least $|d_1 - d_2|$ (over the random choice of ε_{i+1}). Moreover, since π_1 and π_2 are unique shortest paths in G and G is acyclic, it follows that $\pi_1 \cap \pi_2$ is a contiguous subpath of π_1 and π_2 ; thus, once they split, they can no longer intersect in later layers. The number of nodes in the intersection $|\pi_1 \cap \pi_2|$ is 1 more than the number of consecutive nodes at which π_1, π_2 intersect but do not split. So by the above discussion, we have

$$\begin{aligned} \Pr \left[|\pi_1 \cap \pi_2| > \frac{8 \log n}{|d_1 - d_2|} \right] &\leq (1 - |d_1 - d_2|)^{\frac{8 \log n}{|d_1 - d_2|}} \\ &\leq e^{-|d_1 - d_2| \cdot \frac{8 \log n}{|d_1 - d_2|}} \\ &\leq e^{-8 \log n} \\ &= n^{-8} \end{aligned}$$

For the second point of the lemma: if $|d_1, d_2| \geq 1$, then by Lemma 4.8, if there is a node $v \in \pi_1 \cap \pi_2$, then π_1 and π_2 split at v with probability 1, and then they can no longer intersect in later layers. So we have $|\pi_1 \cap \pi_2| \leq 1$. $\square$

Since $|\Pi| = p \leq n^2$, we can argue by a union bound that Lemma 4.9 holds for all $\pi_1, \pi_2 \in \Pi$ simultaneously with probability at least $1 - n^{-4}$. From now on, we will assume that this property holds for our constructed graph G .

Once we specify our construction parameters ℓ and q , the following lemma will immediately imply the third property of G as stated in Lemma 4.2.

Lemma 4.10. *Let $x, y \in V$ be distinct nodes in G , and let $z = \text{hopdist}_G(x, y)$. Let*

$$\Pi' := \{\pi \in \Pi \mid x, y \in \pi\}.$$

⁵Formally, we consider any two paths π_1, π_2 indexed by two start nodes and direction vectors. When we iteratively generate these paths, the number of nodes in their intersection (possibly 0) depends only on the random choices of $\varepsilon_1, \dots, \varepsilon_\ell$. The probability claim in this lemma is with respect to these random choices.

Then $|\Pi'| \leq \max \left\{ \frac{16q \log n}{z}, 1 \right\}$.

Proof. Let $\Pi' = \{\pi_1, \dots, \pi_k\}$ and let $(1, d_i) \in D$ be the direction vector associated with π_i for $i \in [1, k]$. Since the paths in Π' all intersect, by Proposition 4.7 we must have $d_i \neq d_j$ for $i \neq j$. Let $a = \min_{i \in [1, k]} d_i$ and let $b = \max_{i \in [1, k]} d_i$. Then

$$d_b - d_a \geq \frac{k-1}{q},$$

since $|d_i - d_j| \geq 1/q$ for all $(1, d_i), (1, d_j) \in D$ such that $i \neq j$. Thus, by Lemma 4.9 we must have $\frac{k-1}{q} < 1$, since we have at least two nodes $x, y \in \pi_a \cap \pi_b$. So by Lemma 4.9,

$$|\pi_a \cap \pi_b| \leq \frac{8 \log n}{d_b - d_a} \leq \frac{8q \log n}{k-1}.$$

Since $x, y \in \pi_a \cap \pi_b$ and π_a and π_b are unique shortest paths in G , it follows that they coincide on their subpaths $\pi_a[x, y] = \pi_b[x, y]$. Moreover, since the hopdistance from x to y in G is z , it follows that $|\pi_a[x, y]| = |\pi_b[x, y]| = z + 1$. Then taken together we have

$$z \leq |\pi_a \cap \pi_b| \leq \frac{8q \log n}{k-1}.$$

Rearranging, we get

$$k \leq \frac{8q \log n}{z} + 1.$$

If $\frac{8q \log n}{z} \geq 1$, then this implies $k \leq \frac{16q \log n}{z}$. Otherwise, if $\frac{8q \log n}{z} < 1$, then this implies that $k \leq 1$ since $k \in \mathbb{Z}$. $\square$

Finishing the proof of Lemma 4.2

We note that Theorem 4.1 is trivial in the parameter regime $p = \Omega(n^2/\log n)$, since its lower bound on hopbound is $\Omega(1)$. So we will assume $p = O(n^2/\log n)$ in the following, with as small of an implicit constant as needed. Let

$$\ell = \frac{n}{2^{10} p^{1/2} \log^{1/2} n} \quad \text{and} \quad q = \frac{\ell}{2^{10} \log n}.$$

We now quickly verify that graph G and associated critical paths Π satisfy the properties of Lemma 4.2:

- By construction, G has $\ell = \Theta\left(\frac{n}{p^{1/2} \log^{1/2} n}\right)$ layers, and each path in Π travels from the first

layer to the last layer.

- Proposition 4.7 implies that each vertex has at most $|D| = O(qn/\ell^2)$ paths passing through it. By our choice of construction parameters ℓ and q , we conclude that $|D| = O(\ell \cdot p/n)$.
- Each path $\pi \in \Pi$ is a unique shortest path between its endpoints in G by Lemma 4.5.
- Since $16q \log n \leq \ell$ and $\text{hopdist}_G(u, v) \leq \ell$, Lemma 4.10 immediately implies that for all $u, v \in V$, there are at most

$$\max \left\{ \frac{16q \log n}{\text{hopdist}_G(u, v)}, 1 \right\} \leq \frac{\ell}{\text{hopdist}_G(u, v)}$$

paths in Π that contain both u and v .

- For all critical pairs $(s, t) \in P$, the hopdistance from s to t in G is $\ell - 1 \gg 16q \log n$. Then by Lemma 4.10, each of the $|\Pi|$ paths constructed in Section 4.2 have distinct endpoints and thus are distinct. Then

$$|\Pi| = |S||D| \geq \frac{n}{2\ell} \cdot \frac{nq}{8\ell^2} \geq \frac{n^2q}{16\ell^3} \geq p.$$

We have shown that our *directed* graph G satisfies the properties of Lemma 4.2 in the regime of $p \in [n, n^2]$. Moreover, our construction still goes through even in the extended regime of $p \in [n/c, n^2]$ for any constant $c > 0$. All that remains is to extend our construction to the entire regime of $p \in [1, n]$ and make G undirected.

Extending the construction to $p \in [1, n]$. We can extend our construction to the regime of $p \in [1, n]$ with a simple modification to G that was previously used in the prior work of [KP22d]. We will sketch the modification here and defer the proof of correctness to Lemma 6 in Appendix A of [BH23a].

Let $G(n, p)$ denote an instance of our originally constructed graph G with input parameters n and $p \in [n, n^2]$. Let n be a sufficiently large integer and let $p \in [1, n]$. Let $G_1 := G(p_1, p)$ where $p_1 = \Theta(p)$ and p_1 divides n . Now for each node v in G_1 , replace v with a directed path $\pi_v = (u_1^v, \dots, u_k^v)$ with $k = n/p_1$ nodes. For all $v \in V$, assign weight 0 to all edges in π_v . For each edge (v_1, v_2) originally in G_1 , add edge $(u_k^{v_1}, u_1^{v_2})$ to the graph. Let G_2 be the resulting graph, and let Π_2 be the updated set of critical paths. This completes the modification.

Lemma 4.11. *The n -node graph G_2 and the set Π_2 of $|\Pi_2| \geq p$ paths satisfy the properties of Lemma 4.2.*

Proof. We defer the proof to Appendix A of [BH23a], as it largely follows our earlier analysis. $\square$

Making G undirected. To make G undirected, we use the following standard simple blackbox reduction. Let W be the sum of all edge weights in G , i.e., $W = \sum_{e \in E} w(e)$. For every edge $e \in E$, add $+W$ to the weight $w(e)$ of e , and treat e as an undirected edge. Call the resulting graph G' .

We now argue correctness: in particular, we need to argue that for all $s, t \in V$ such that t is reachable from s in G , π is a shortest weighted (directed) $s \rightsquigarrow t$ -path in G if and only if π is a shortest weighted (undirected) $s \rightsquigarrow t$ -path in G' .

- First, note that for all $s \rightsquigarrow t$ -paths π' in G' , the number of nodes in π' satisfies $|\pi'| \geq |\pi|$ by the construction of G and G' .
- Moreover, if $|\pi'| > |\pi|$, then π' has one more edge than π . Thus, its weighted length in G' satisfies

$$\begin{aligned} w(\pi') &> (|\pi| + 1) \cdot W \\ &= W|\pi| + \sum_{e \in E(G)} w(e) \\ &> w(\pi) \end{aligned}$$

and so π' is not a shortest path. We conclude that if π' is a shortest $s \rightsquigarrow t$ -path in G' , then $|\pi'| = |\pi|$.

- Any $s \rightsquigarrow t$ -path π' in G' with $|\pi|$ edges must use exactly one node in each layer, and thus it respects the original edge directions in G . We conclude that π' is a shortest weighted $s \rightsquigarrow t$ -path in G' if and only if π' is a shortest weighted (directed) $s \rightsquigarrow t$ -path in G .

Lemma 4.2 is immediate from the above discussion.

4.3 Shortcut Sets

In this section we will prove the following theorem.

Theorem 4.12. *For any parameter $p \in [1, n^{5/4}]$, there exists an n -node unweighted directed graph $G = (V, E)$ such that for any shortcut set H of size $|H| \leq p$, the graph $G \cup H$ must have diameter D , where*

$$D = \begin{cases} \Omega\left(\frac{n}{p^{3/4} \log n}\right) & \text{for } p \in [1, n/(\alpha \log n)], \text{ where } \alpha > 0 \text{ is a sufficiently large constant} \\ \Omega\left(\frac{n^{5/4}}{p \log^{9/4} n}\right) & \text{for } p \in [n/(\alpha \log n), n^{5/4}]. \end{cases}$$

In particular, when $p = O(n)$, $G \cup H$ must have diameter $D = \Omega\left(\frac{n^{1/4}}{\log^{9/4} n}\right)$.

We will prove this via a construction of the following type:

Lemma 4.13. *For any $p \in [1, n/(\alpha \log n)]$, where $\alpha > 0$ is a sufficiently large constant, there is an infinite family of n -node directed unweighted graphs $G = (V, E, w)$ and sets Π of $|\Pi| = p$ paths in G with the following properties:*

- G has $\ell = \Theta\left(\frac{n}{p^{3/4} \log n}\right)$ layers. Each path in Π starts in the first layer, ends in the last layer, and contains exactly one node in each layer.
- Each path in Π is the unique path between its endpoints in G .
- For any two nodes $u, v \in V$, there are at most $\frac{\ell}{\text{hopdist}_G(u,v)}$ paths in Π that contain both u and v .

We quickly verify that Lemma 4.13 implies Theorem 4.12.

Proof of Theorem 4.12 using Lemma 4.13. Fix an n and $p \in [1, n/(\alpha \log n)]$. Let $G = (V, E, w)$ be the graph in Lemma 4.13 with associated set Π of $|\Pi| = 2p$ paths in G . Let H be a shortcut set of size $|H| \leq p$. Let $P \subseteq V \times V$ be the set of node pairs that are the endpoints of paths in Π . Since all paths in Π are *unique* paths between their endpoints in G , it follows that

$$\text{diameter}(G \cup H) \geq \max_{(s,t) \in P} \text{dist}_{G \cup H}(s,t) = \max_{(s,t) \in P} \text{hopdist}_{G \cup H}(s,t).$$

Then when $p \in [1, n/(\alpha \log n)]$, we can achieve the bounds in Theorem 4.12 using the same potential function argument as in Section 4.2.

In order to extend our shortcut set lower bound to $p \in [n/(\alpha \log n), n^{5/4}]$, we will appeal to a more general property of the extremal functions of exact hopsets and shortcut sets, which we defer to Lemma 13 in Appendix B of [BH23a]. $\square$

Constructing the strongly convex vector set $W(q)$

In our construction of the graph G , we will implicitly use the following lemma from [BL98; BH22] that establishes the existence of a dense set of vectors that each extend the farthest in their own direction.⁶

⁶This is a slightly stronger property than convexity, and hence is sometimes referred to as “strong convexity” in the area.

Lemma 4.14 (Theorem 1 of [BL98]; Lemma 7 of [BH22]). *For sufficiently large $r \in \mathbb{Z}_{\geq 0}$, there exists a strongly convex set of integer vectors $W(r) \subseteq \mathbb{Z}^2$ of size $|W(r)| = \Theta(r^{2/3})$, such that*

- *for all $\vec{w} \in W(r)$, $\|\vec{w}\| \leq r$,*
- *every $\vec{w} \in W(r)$ lies in the first quadrant, i.e., both coordinates are positive, and*
- *for all distinct $\vec{w}_1, \vec{w}_2 \in W(r)$, $\langle \vec{w}_1, \vec{w}_2 \rangle < \langle \vec{w}_1, \vec{w}_1 \rangle$.*

In our construction of G , we will use a vector set W from this lemma to help generate edge and direction vectors. We will make use of the following technical property of the vectors in W :

Proposition 4.15. *Let $W = \{\vec{w}_1, \dots, \vec{w}_q\}$ be a set of vectors as described in Lemma 4.14, with its vectors ordered counterclockwise. For all $\vec{w}_i, \vec{w}_j, \vec{w}_k \in W$ with $i < j < k$, the following inequalities hold:*

$$\langle \vec{w}_i, \vec{w}_k \rangle < \langle \vec{w}_j, \vec{w}_k \rangle < \langle \vec{w}_k, \vec{w}_k \rangle \quad \text{and} \quad \langle \vec{w}_i, \vec{w}_k \rangle < \langle \vec{w}_i, \vec{w}_j \rangle < \langle \vec{w}_i, \vec{w}_i \rangle.$$

Proof. We will only prove here that $\langle \vec{w}_i, \vec{w}_k \rangle < \langle \vec{w}_j, \vec{w}_k \rangle < \langle \vec{w}_k, \vec{w}_k \rangle$; the other set of inequalities follows from an identical argument. By Lemma 4.14 we already have $\langle \vec{w}_j, \vec{w}_k \rangle < \langle \vec{w}_k, \vec{w}_k \rangle$, so it remains only to show that $\langle \vec{w}_i, \vec{w}_k \rangle < \langle \vec{w}_j, \vec{w}_k \rangle$.

Let ψ_1 be the inner angle between $\vec{w}_i$ and $\vec{w}_j$ and let ψ_2 be the inner angle between $\vec{w}_j$ and $\vec{w}_k$; thus the inner angle between $\vec{w}_i$ and $\vec{w}_k$ is $\psi_1 + \psi_2$. We first establish a useful inequality:

$$\begin{aligned} \langle \vec{w}_i, \vec{w}_j \rangle &< \langle \vec{w}_j, \vec{w}_j \rangle && \text{by Lemma 4.14} \\ \|\vec{w}_i\| \|\vec{w}_j\| \cos \psi_1 &< \|\vec{w}_j\|^2 && \text{cosine formula} \\ \|\vec{w}_i\| \|\vec{w}_k\| \cos \psi_1 \cos \psi_2 &< \|\vec{w}_j\| \|\vec{w}_k\| \cos \psi_2. && \text{follows algebraically from previous line} \end{aligned}$$

We will next use the trigonometric identity

$$\begin{aligned} \cos \psi_1 \cos \psi_2 &= \sin \psi_1 \sin \psi_2 + \cos(\psi_1 + \psi_2) \\ &> \cos(\psi_1 + \psi_2) && \text{since } \psi_1, \psi_2 \in (0, \pi/2) \text{ by Lemma 4.14.} \end{aligned}$$

We are now ready to show:

$$\begin{aligned}
\langle \vec{w}_i, \vec{w}_k \rangle &= \|\vec{w}_i\| \|\vec{w}_k\| \cos(\psi_1 + \psi_2) && \text{cosine formula} \\
&< \|\vec{w}_i\| \|\vec{w}_k\| \cos \psi_1 \cos \psi_2 && \text{second inequality} \\
&< \|\vec{w}_j\| \|\vec{w}_k\| \cos \psi_2 && \text{first inequality} \\
&< \langle \vec{w}_j, \vec{w}_k \rangle. && \text{cosine formula } \square
\end{aligned}$$

Constructing G

We next construct the graph that will be used for Lemma 4.13. Let n be a sufficiently large positive integer, and let $p = n/(\alpha_0 \log n)$ for a sufficiently large constant $\alpha_0 > 0$ to be chosen later (we will extend our construction to other choices of $p \in [1, n^{5/4}]$ later). For simplicity of presentation, we will frequently ignore issues related to non-integrality of expressions that arise in our construction; these issues affect our bounds only by lower-order terms. All edges (u, v) in G will be directed from u to v .

Vertex Set V .

- Let r be a positive integer construction parameter to be specified later. Our graph G will have $\ell := \frac{n^{1/3}}{r^{2/3}}$ layers $L_1, \dots, L_\ell$, and each layer will have $n/\ell = n^{2/3}r^{2/3}$ nodes.
- We will label each node in layer L_i , $i \in [1, \ell]$, with a distinct triple in $\{i\} \times [1, n^{1/3}r^{1/3}] \times [1, n^{1/3}r^{1/3}]$. We will interpret the node in L_i labeled (i, j, k) as an integer point $(i, j, k) \in \mathbb{Z}^3$.
- These n nodes arranged in ℓ layers will compose our node set

$$V = [1, \ell] \times [1, n^{1/3}r^{1/3}] \times [1, n^{1/3}r^{1/3}] \subseteq \mathbb{Z}^3$$

of graph G . In a slight abuse of notation, we will treat $v \in V$ as either a node in G or a point in $\mathbb{Z}^3$, depending on the context.

Edge Set E .

- All the edges in E will be between consecutive layers L_i, L_{i+1} of G . We will let E_i denote the set of edges in G between layers L_i and L_{i+1} .
- Just as our nodes in V correspond to points in $\mathbb{R}^3$, we can interpret the edges E in G as vectors in $\mathbb{R}^3$. In particular, for every edge $e = (v_1, v_2) \in E$, we identify e with the corresponding

vector $\vec{u}_e := v_2 - v_1$. Note that since all edges in E are between adjacent layers L_i and L_{i+1} , the first coordinate of $\vec{u}_e$ is 1 for all $e \in E$. We will use u_e^i to denote the i th coordinate of $\vec{u}_e$ for $i \in \{2, 3\}$ (i.e. for all $e \in E$, we write $\vec{u}_e = (1, u_e^2, u_e^3)$).

- We begin our construction of E by defining the set of vectors $W := W(r^3) \subseteq \mathbb{Z}^2$, where $W(r^3)$ is the strongly convex set of vectors defined in Lemma 4.14. Let $W = \{\vec{w}_1, \dots, \vec{w}_q\}$, where vectors $\vec{w}_i$ are ordered counterclockwise and $q := |W| = \Theta(r^2)$. We may assume wlog that $q \leq r^2$ (e.g. by removing vectors from W until this is true).
- Using W , we define our set of *edge vectors* $C \subseteq \mathbb{Z}^2$ as:

$$C := \{\vec{w}_{i+1} - \vec{w}_i \mid i \in [1, q-1]\}.$$

Let $\vec{c}_i := \vec{w}_{i+1} - \vec{w}_i$ denote the i th vector of C .

- For each layer L_i , $i \in [1, \ell-1]$, sample a random integer from $[1, q-1]$ and call it λ_i . We define the set $C_i \subseteq \mathbb{Z}^2$ as

$$C_i := \{(0, 0), \vec{c}_{\lambda_i}\},$$

where $\vec{c}_{\lambda_i}$ is the λ_i th vector of C . Note that C_i contains exactly two vectors, the zero vector and a randomly chosen vector $\vec{c}_{\lambda_i}$ from C . Intuitively: for each layer L_i , we are sampling two adjacent vectors $\vec{w}_{\lambda_i}$ and $\vec{w}_{\lambda_i+1}$ from C and adding $-\vec{w}_{\lambda_i}$ to each of them to obtain C_i . The purpose of adding the normalizing vector $-\vec{w}_{\lambda_i}$ to $\vec{w}_{\lambda_i}$ and $\vec{w}_{\lambda_i+1}$ is to reduce the magnitude of the vectors in C_i , as we will formalize in Proposition 4.16.

- The vectors in set C_i will define the edges in E_i . Specifically, for all $(i, v_1, v_2) \in L_i$ and for all $(c_1, c_2) \in C_i$ such that

$$(i+1, v_1 + c_1, v_2 + c_2) \in L_{i+1},$$

we add the edge $((i, v_1, v_2), (i+1, v_1 + c_1, v_2 + c_2))$ to E_i .

This completes the construction of our graph $G = (V, E)$. We now verify that the vectors in C_i have small magnitude in expectation.

Proposition 4.16. *For all $i \in [1, \ell-1]$, $\mathbb{E}[\|\vec{c}_{\lambda_i}\|] \leq \frac{2^4 r^3}{q} = \Theta(r)$.*

Proof. Note that the vectors in C correspond to sides of a convex polygon whose vertices are the vectors in W . Since this polygon is contained in a ball of radius r^3 in $\mathbb{R}_{>0}^2$ by Lemma 4.14, it follows

that $\sum_{\vec{c} \in C} \|\vec{c}\| \leq 2\pi r^3$. Note that $|C| = q - 1 \geq q/2$ for sufficiently large q . Then

$$\mathbb{E}[\|\vec{c}_{\lambda_i}\|] = \sum_{\vec{c} \in C} \frac{1}{|C|} \cdot \|\vec{c}\| \leq \frac{2\pi r^3}{|C|} \leq \frac{2^4 r^3}{q}.$$

□

The vectors $\vec{w}_i \in W$ have magnitude roughly r^3 , whereas the vectors in C_i have expected magnitude at most $\Theta(r)$ by Proposition 4.16. Since each edge in our critical path corresponds to a vector in C_i , ensuring that the vectors in C_i have small magnitude (at least in expectation) will be essential for guaranteeing that the paths $\pi \in \Pi$ are long.

Let us comment here on a discrepancy between this construction and the technical overview. In the technical overview, we stated that we would use the convex vectors W to generate the edge vectors C . Instead, we are using the *difference* between adjacent convex vectors W to generate C . This is an optimization: our plan is to sample *one* edge vector from C , and use it together with the zero vector $(0, 0)$ as the two available vectors between pairs of adjacent layers. This is equivalent to sampling *two adjacent* edge vectors from W , as advertised in the technical overview, and then applying an appropriate translation of the next layer in space. Our strategy lets us use vectors of length $\Theta(r)$, instead of $\Theta(r^3)$, and these shorter vectors ultimately lead to a stronger lower bound.

Direction Vectors, Critical Pairs, and Critical Paths

Our next step is to generate a set of *critical pairs* $P \subseteq V \times V$, as well as a set of *critical paths* Π . Specifically, there will be one critical path $\pi_{s,t} \in \Pi$ going between each critical pair $(s, t) \in P$, and we will show that $\pi_{s,t}$ is the unique $s \rightsquigarrow t$ path in G .

Direction Vectors D . We choose our set of direction vectors D to be $D := W$, where W is our strongly convex set of $q = \Theta(r^2)$ vectors. We will let $D = \{\vec{d}_1, \dots, \vec{d}_q\}$ be our list of direction vectors, and we will let the i th vector $\vec{d}_i$ of D correspond to the i th vector $\vec{w}_i$ of W , i.e. $\vec{d}_i := \vec{w}_i$, for $i \in [1, q]$. We will simply use the name $\vec{d}_i$ when we wish to emphasize the role of these vectors as direction vectors.

Note that since $D = W$, Proposition 4.15 also holds for D . That is, if $i, j, k \in [1, q - 1]$ and $i < j < k$, then

$$\langle \vec{d}_i, \vec{d}_k \rangle < \langle \vec{d}_j, \vec{d}_k \rangle < \langle \vec{d}_k, \vec{d}_k \rangle \quad \text{and} \quad \langle \vec{d}_i, \vec{d}_k \rangle < \langle \vec{d}_i, \vec{d}_j \rangle < \langle \vec{d}_i, \vec{d}_i \rangle.$$

Critical Pairs P and Critical Paths Π .

- We first define a set $S \subseteq L_1$, containing a subset of the nodes in the first layer L_1 of G :

$$S := \left\{ (1, j, k) \mid (j, k) \in \left[\frac{1}{3}n^{1/3}r^{1/3}, \frac{2}{3}n^{1/3}r^{1/3} \right] \times \left[\frac{1}{3}n^{1/3}r^{1/3}, \frac{2}{3}n^{1/3}r^{1/3} \right] \right\}.$$

Informally, S is a middle square patch of the nodes in L_1 . The key property of S is that all nodes in S are of distance at least $\frac{1}{3}n^{1/3}r^{2/3}$ from the sides of the square grid corresponding to layer L_1 .

We will choose our set of demand pairs P so that $P \subseteq S \times V$. For every node $s \in S$ and direction vector $\vec{d} \in D$, we will choose a critical pair $(s, t) \in S \times V$ and a corresponding critical path $\pi_{s,t}$ to add to P and Π .

- Let $v_1 \in S$, and let $\vec{d} = (1, d) \in D$. The associated path π has start node v_1 . We iteratively grow π , layer-by-layer, as follows. Suppose that currently $\pi = (v_1, \dots, v_i)$ with each $v_i \in L_i$. To determine the next node $v_{i+1} \in L_{i+1}$, let $E_i^{v_i} \subseteq E_i$ be the edges in E_i incident to v_i , and let $\vec{u}_i = (u_i^1, u_i^2) \in C_i$ be

$$\vec{u}_i := \operatorname{argmax}_{\vec{c} \in C_i} \langle \vec{c}, \vec{d} \rangle.$$

If $(1, u_i^1, u_i^2) = \vec{u}_i$ for some $e_i \in E_i^{v_i}$, then by definition, e_i is an edge whose first node is v_i ; we define $v_{i+1} \in L_{i+1}$ to be the other node in e_i , and we append v_{i+1} to π . Otherwise, if there is no such edge e_i in $E_i^{v_i}$, then we terminate our construction of path π (i.e. v_i will be the final node in π).

This completes our construction of P and Π . We will show that the paths generated in this way have distinct endpoints (with high probability), and therefore $|P| = |S||D| \geq \frac{n^{2/3}r^{2/3}q}{2^5}$, where $q = \Theta(r^2)$.

An important feature for correctness of our construction is that, when we iteratively generate paths, if we reach a point v_i such that $v_i + C_i \not\subseteq L_{i+1}$ (i.e. $v_i + \vec{c} \notin L_{i+1}$ for some $\vec{c} \in C_i$), then we end our path at v_i . As a consequence, our critical paths in Π may not travel through all ℓ layers of G . However, with nonzero probability, paths in Π travel through a constant fraction of layers, as we prove in the following proposition.

Proposition 4.17. *Let $\hat{\ell} := \ell \cdot \frac{q}{2^7 r^2} = \Theta(\ell)$. With probability at least $1/2$, for all $\pi \in \Pi$, $|\pi| \geq \hat{\ell}$.*

Proof. Each critical path starts at a node s in S . Each edge of the path corresponds to a vector $(1, c_1, c_2) \in \mathbb{Z}^3$ such that $(c_1, c_2) \in C_i$ for some $i \in [1, \ell - 1]$. Our path ends when we reach the boundary of our vertex set $V = [1, \ell] \times [1, n^{1/3}r^{1/3}] \times [1, n^{1/3}r^{1/3}] \subseteq \mathbb{Z}^3$. We must show that before

any such path travels through $\hat{\ell}$ nodes before reaching the boundary. Note that $\hat{\ell} = \ell \cdot \frac{q}{2^7 r^2} \leq \ell$, since $q \leq r^2$.

Let x be the random variable defined as $x := \sum_{i=1}^{\hat{\ell}} \vec{c}_{\lambda_i}$. Then by Proposition 4.16 and the linearity of expectation,

$$\mathbb{E}[x] \leq \hat{\ell} \cdot \frac{2^4 r^3}{q} = \ell \cdot \frac{r}{2^3} = \frac{1}{2^3} n^{1/3} r^{1/3},$$

where the final equality follows from the fact that $\ell = n^{1/3} r^{-2/3}$. Then by Markov's inequality,

$$\Pr \left[x \leq \frac{1}{4} n^{1/3} r^{1/3} \right] \geq 1/2.$$

Now we claim that if $x \leq \frac{1}{4} n^{1/3} r^{1/3}$, then for all $\pi \in \Pi$, $|\pi| \geq \hat{\ell}$. Let $\pi_{s,t} \in \Pi$ be a critical path for critical pair $(s, t) \in P$. Let $s = (s_1, s_2, s_3) \in \mathbb{Z}^3$ and let $t = (t_1, t_2, t_3) \in \mathbb{Z}^3$. By our construction of critical paths Π , either $t_1 = \ell$ or $t + C_{t_1} \notin V$. In the first case, $|\pi| = \ell \geq \hat{\ell}$, since path π traveled through all ℓ layers L_i . In the second case, we must have that $\|(t_2, t_3) - (s_2, s_3)\| \geq \frac{1}{3} n^{1/3} r^{1/3}$ by our choice of set S . But since $x \leq \frac{1}{4} n^{1/3} r^{1/3}$, we conclude that $|\pi| \geq \hat{\ell}$. The claim follows. $\square$

We have shown that with nonzero probability, all our paths in Π travel through the first $\hat{\ell}$ layers of G . However, we cannot guarantee that paths in Π travel to layers L_i with $i > \hat{\ell}$. Because of this, we choose to remove all layers L_i , $i > \hat{\ell}$, from G . We replace all critical paths $\pi \in \Pi$ with the truncated subpath of π containing only the first $\hat{\ell}$ nodes of π , and we update our critical pairs $P \subseteq V \times V$ to be the set of all pairs of endpoints of the updated paths in Π .

Critical paths are unique paths

We now verify that graph G and paths Π have the unique path property as stated in Lemma 4.13. This will follow straightforwardly from the properties of our set of direction vectors D , particularly Proposition 4.15.

Lemma 4.18 (Unique paths). *For every $(s, t) \in P$, path $\pi_{s,t}$ is a unique $s \rightsquigarrow t$ -path in G .*

Proof. Fix a direction vector $\vec{d}_j$ in D . We claim that for all $i \in [1, \ell - 1]$, there is a unique vector $\vec{c} \in C_i$ such that maximizes $\langle \vec{c}, \vec{d}_j \rangle$. Recall that $C_i \subseteq \mathbb{Z}^2$ contains exactly two vectors: the zero vector $(0, 0)$ and the vector $\vec{c}_{\lambda_i} = \vec{w}_{\lambda_i+1} - \vec{w}_{\lambda_i}$.

Now assume that $j \geq \lambda_i + 1$ and observe the following sequence of equivalent inequalities:

$$\begin{aligned} \langle \vec{w}_{\lambda_i}, \vec{w}_j \rangle &< \langle \vec{w}_{\lambda_i+1}, \vec{w}_j \rangle && \text{by Lemma 4.14 and Proposition 4.15} \\ \langle \vec{0}, \vec{w}_j \rangle &< \langle \vec{w}_{\lambda_i+1} - \vec{w}_{\lambda_i}, \vec{w}_j \rangle \\ \langle \vec{0}, \vec{d}_j \rangle &< \langle \vec{c}_{\lambda_i+1}, \vec{d}_j \rangle. \end{aligned}$$

When $j < \lambda_i + 1$, an identical argument shows that $\langle \vec{0}, \vec{d}_j \rangle > \langle \vec{c}_{\lambda_i+1}, \vec{d}_j \rangle$. In either case, there is a unique vector $\vec{c} \in C_i$ maximizing $\langle \vec{c}, \vec{d}_j \rangle$ for all $i \in [1, \ell - 1]$.

Now fix a critical pair $(s, t) \in P$ that has $\vec{d}_j \in D$ as its associated direction vector and $\pi_{s,t} = (v_1, \dots, v_k)$ as its critical path (where $s = v_1$ and $t = v_k$). Let $f : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be the function that projects each point in $\mathbb{R}^3$ onto the subspace corresponding to the last two coordinates of $\mathbb{R}^3$, i.e. $f(x, y, z) = (y, z)$ for all $(x, y, z) \in \mathbb{R}^3$.

Let π be an arbitrary $s \rightsquigarrow t$ -path, and note that $|\pi| = |\pi_{s,t}|$ since G is a layered directed graph. Let $\pi = (v'_1, \dots, v'_k)$, where $s = v'_1$ and $t = v'_k$. By our construction of path $\pi_{s,t}$, we must have that for all $i \in [1, k - 1]$,

$$\langle f(v'_{i+1} - v'_i), \vec{d}_j \rangle \leq \langle f(v_{i+1} - v_i), \vec{d}_j \rangle.$$

Now suppose for the sake of contradiction that $\pi \neq \pi_{s,t}$. Then $v'_{i+1} - v'_i \neq v_{i+1} - v_i$ for some $i \in [1, k - 1]$. Then $\langle f(v'_{i+1} - v'_i), \vec{d}_j \rangle < \langle f(v_{i+1} - v_i), \vec{d}_j \rangle$ by the above discussion. But then since π and $\pi_{s,t}$ are both $s \rightsquigarrow t$ -paths,

$$\langle f(t - s), \vec{d}_j \rangle = \sum_{i=1}^{k-1} \langle f(v'_{i+1} - v'_i), \vec{d}_j \rangle < \sum_{i=1}^{k-1} \langle f(v_{i+1} - v_i), \vec{d}_j \rangle = \langle f(t - s), \vec{d}_j \rangle.$$

This is a contradiction, so we conclude that $\pi = \pi_{s,t}$. Then path $\pi_{s,t}$ is a unique $s \rightsquigarrow t$ -path in G . $\square$

Critical Paths Intersection Properties

Before finishing our proof of Lemma 4.13, we will need to establish several properties of the critical paths in Π .

Proposition 4.19. *Let $\pi_1, \pi_2 \in \Pi$ be two critical paths with the same corresponding direction vector $\vec{d} \in D$. Then $\pi_1 \cap \pi_2 = \emptyset$.*

Proof. Let $k = \min\{|\pi_1|, |\pi_2|\}$. Let $v_i^j \in L_i$ denote the i th node of π_j , where $j \in \{1, 2\}$ and $i \in [1, k]$. Note that since π_1 and π_2 share the same direction vector $\vec{d}$, edges (v_i^1, v_{i+1}^1) and (v_i^2, v_{i+1}^2) have the same corresponding vector $\vec{u}_i \in C_i$ for all $i \in [1, k - 1]$ by our construction of π_1 and π_2 .

Also, for each node $v \in L_1$, v belongs to at most one path $\pi \in \Pi$ with direction vector $\vec{d}$, so $v_1^1 \neq v_1^2$. Then for all $i \in [1, k]$,

$$v_i^1 = v_1^1 + \sum_{i=1}^{i-1} \vec{u}_i \neq v_1^2 + \sum_{i=1}^{i-1} \vec{u}_i = v_i^2.$$

□

Let $\pi_1, \pi_2 \in \Pi$ be two critical paths, and let $v \in V$ be a node in G . We say that paths π_1 and π_2 *split* at v if $v \in \pi_1 \cap \pi_2$ and the node following v in π_1 is distinct from the node following v in π_2 , and we simply say that π_1 and π_2 *split* if there exists some $v \in V$ such that π_1 and π_2 split at v . Note that since $\pi_1, \pi_2 \in \Pi$ are unique paths in G , paths π_1 and π_2 can split at most once.

Lemma 4.20. *Fix a node $v \in L_i$, where $i \in [1, \ell - 1]$, and let $\pi_1, \pi_2 \in \Pi$ be critical paths with direction vectors $\vec{d}_j$ and $\vec{d}_k \in D$, $j, k \in [1, q]$, such that $v \in \pi_1$ and $v \in \pi_2$. Then paths π_1 and π_2 split at v with probability at least $\frac{|j-k|}{q}$.*

Proof. Fix a node $v \in L_i$, where $i \in [1, \ell - 1]$, and let $\pi_1, \pi_2 \in \Pi$ be critical paths with direction vectors $\vec{d}_j$ and $\vec{d}_k \in D$, $j, k \in [1, q]$, such that $v \in \pi_1$ and $v \in \pi_2$. By Proposition 4.19, $j \neq k$, and assume wlog that $j < k$. Let F be the event that the random variable λ_i was sampled so that

$$j \leq \lambda_i < k.$$

Our proof strategy is to show that F implies that π_1, π_2 split at v , and then to show that F occurs with the claimed probability.

F implies that π_1, π_2 split at v . Assume that F occurs. Then $j \leq \lambda_i < \lambda_i + 1 \leq k$. Now observe the following sequence of equivalent inequalities:

$$\begin{aligned} \langle \vec{w}_j, \vec{w}_{\lambda_i} \rangle &> \langle \vec{w}_j, \vec{w}_{\lambda_i+1} \rangle && \text{by Lemma 4.14 and Proposition 4.15} \\ \langle \vec{w}_j, \vec{0} \rangle &> \langle \vec{w}_j, \vec{w}_{\lambda_i+1} - \vec{w}_{\lambda_i} \rangle \\ \langle \vec{d}_j, \vec{0} \rangle &> \langle \vec{d}_j, \vec{c}_{\lambda_i} \rangle. \end{aligned}$$

Since $C_i = \{\vec{0}, \vec{c}_{\lambda_i}\}$, by our construction of the critical paths in Π , the above inequality $\langle \vec{d}_j, \vec{0} \rangle > \langle \vec{d}_j, \vec{c}_{\lambda_i} \rangle$ implies that path π_1 takes an edge in E_i corresponding to vector $\vec{0}$. An identical argument will show that $\langle \vec{d}_k, \vec{0} \rangle < \langle \vec{d}_k, \vec{c}_{\lambda_i} \rangle$, so path π_2 takes an edge in E_i corresponding to vector $\vec{c}_{\lambda_i}$. Since paths π_1 and π_2 take different edges in E_i , they must split at v .

F happens with good probability. Random variable λ_i is sampled uniformly from $[1, q-1]$. Then the event F occurs with probability $\frac{|j-k|}{q-1} \geq \frac{|j-k|}{q}$. $\square$

We will use Lemma 4.21 to prove the following two lemmas, which capture key properties of our graph G .

Lemma 4.21. *Let $\pi_1, \pi_2 \in \Pi$ be critical paths with associated direction vectors $\vec{d}_j, \vec{d}_k \in D$. Then $|\pi_1 \cap \pi_2| \leq \frac{8q \log n}{|j-k|}$ with probability at least $1 - n^{-8}$.*

Proof. If $\pi_1 \cap \pi_2 = \emptyset$, then the claim is immediate, so assume there is a node $v \in \pi_1 \cap \pi_2$. Suppose $v \in L_i$. By Lemma 4.20, π_1 and π_2 split at v with probability at least $\frac{|j-k|}{q}$. Moreover, conditioning on $v \in \pi_1 \cap \pi_2$, the event that π_1 and π_2 split at v given that $v \in \pi_1$ and $v \in \pi_2$ depends only on our choice of λ_i and is independent of λ_j for $j \neq i$.

Since π_1 and π_2 are unique paths in G , it follows that $\pi_1 \cap \pi_2$ is a contiguous subpath of π_1 and π_2 . The number of nodes in the intersection $|\pi_1 \cap \pi_2|$ is 1 more than the number of consecutive nodes at which π_1, π_2 intersect but do not split. So by the above discussion, we have

$$\begin{aligned} \Pr \left[|\pi_1 \cap \pi_2| > \frac{8q \log n}{|j-k|} \right] &\leq \left(1 - \frac{|j-k|}{q} \right)^{\frac{8q \log n}{|j-k|}} \\ &\leq e^{-\frac{|j-k|}{q} \cdot \frac{8q \log n}{|j-k|}} \\ &\leq e^{-8 \log n} \\ &= n^{-8}. \end{aligned}$$

$\square$

Since $|\Pi| = p \leq n^2$, we can argue by a union bound that Proposition 4.17 holds and Lemma 4.9 holds for all $\pi_1, \pi_2 \in \Pi$ simultaneously with probability at least $1 - 1/2 - n^{-4} > 0$. From now on, we will assume that this property holds for our constructed graph G .

Once we specify our construction parameters ℓ and q , the following lemma will immediately imply the third property of G as stated in Lemma 4.13.

Lemma 4.22. *Let $x, y \in V$ be nodes in G such that the unweighted distance from x to y in G is z , where $0 < z < \infty$. Let $\Pi' \subseteq \Pi$ be the following set of critical paths:*

$$\Pi' := \{\pi \in \Pi \mid x, y \in \pi\}.$$

Then $|\Pi'| \leq \max \left\{ \frac{16q \log n}{z}, 1 \right\}$.

Proof. Let $\Pi' = \{\pi_1, \dots, \pi_k\}$ and let $\vec{d}_{\sigma_i} \in D$ be the direction vector associated with π_i for $i \in [1, k]$. By Proposition 4.7, $\sigma_i \neq \sigma_j$ for $i \neq j$. Let $a = \min_{i \in [1, k]} \sigma_i$ and let $b = \max_{i \in [1, k]} \sigma_i$. Then $b - a \geq k - 1$, so by Lemma 4.21,

$$|\pi_a \cap \pi_b| \leq \frac{8q \log n}{|b - a|} \leq \frac{8q \log n}{k - 1}.$$

Additionally, since $x, y \in \pi_a \cap \pi_b$ and π_a and π_b are unique paths, it follows that $\pi_a[x, y] = \pi_b[x, y]$. Moreover, since the unweighted distance from x to y in G is z , it follows that $|\pi_a \cap \pi_b| \geq |\pi_a[x, y]| = |\pi_b[x, y]| = z + 1$. Then taken together we have

$$z \leq |\pi_a \cap \pi_b| \leq \frac{8q \log n}{k - 1}.$$

Rearranging, we get that

$$k \leq \frac{8q \log n}{z} + 1 \leq \max \left\{ \frac{16q \log n}{z}, 1 \right\}.$$

(We obtain the final inequality from the following observation: if $\frac{8q \log n}{z} \geq 1$, then this implies $k \leq \frac{16q \log n}{z}$. Otherwise, if $\frac{8q \log n}{z} < 1$, then this implies that $k \leq 1$ since $k \in \mathbb{Z}$.) $\square$

Finishing the proof of Lemma 4.13

Let

$$r = \frac{n^{1/8}}{2^6 \log^{3/8} n}.$$

Now recall that $q = \Theta(r^2)$, and let $\alpha_1 > 0$ be a constant such that $\alpha_1 r^2 \leq q \leq 2\alpha_1 r^2$. Then

$$\ell = \frac{n^{1/3}}{r^{2/3}} \geq n^{1/4} \log^{1/4} n, \quad \hat{\ell} = \frac{q}{2^7 r^2} \cdot \ell \geq \frac{\alpha_1}{2^7} \cdot \ell, \quad \text{and} \quad q \leq 2\alpha_1 r^2 \leq \frac{\hat{\ell}}{16 \log n}.$$

We now quickly verify that graph G and associated critical paths Π satisfy the properties of Lemma 4.13:

- By construction, G has $\hat{\ell} = \Theta(n^{1/4} \log^{1/4} n)$ layers, and each path in Π travels from the first layer to the $\hat{\ell}$ th layer.
- Each path $\pi \in \Pi$ is a unique path between its endpoints in G by Lemma 4.18.
- Since $16q \log n \leq \ell$ and $\text{hopdist}_G(u, v) \leq \ell$, Lemma 4.22 immediately implies that for all

$u, v \in V$, there are at most

$$\max \left\{ \frac{16q \log n}{\text{hopdist}_G(u, v)}, 1 \right\} \leq \frac{\ell}{\text{hopdist}_G(u, v)}$$

paths in Π that contain both u and v .

- For all critical pairs $(s, t) \in P$, the unweighted distance from s to t in G is $\ell - 1 \gg 16q \log n$. Then by Lemma 4.22, each of the $|\Pi|$ paths constructed in Section 4.3 have distinct endpoints and thus are distinct. Then

$$|\Pi| = |S||D| \geq \frac{n^{2/3} r^{2/3} q}{2^5} \geq \frac{\alpha_1 n^{2/3} r^{8/3}}{2^5} \geq \frac{\alpha_1 n}{2^{21} \log n}.$$

Recall that we initially let $p = n/(\alpha_0 \log n)$, for some unspecified constant $\alpha_0 > 0$. Choose α_0 so that $p \leq |\Pi|$.

We have shown that our directed graph G satisfies the properties of Lemma 4.13 when $p = n/(\alpha_0 \log n)$. All that remains is to extend our construction to the regime $p \in [1, n/(\alpha_0 \log n)]$.

Extending the construction to $p \in [1, n/(\alpha_0 \log n)]$. We can extend our construction to the regime of $p \in [1, n/(\alpha_0 \log n)]$ using the same modification to G that we performed on our exact hopset construction and that was previously used in the prior work of [KP22d]. We will sketch the modification here. The proof of correctness follows from an argument identical to the proof of Lemma 6 in Appendix A of [BH23a].

We use $G(n, p)$ denote an instance of our originally constructed graph G with input parameters n and $p = n/(\alpha_0 \log n)$. Let n be a sufficiently large integer and let $p \in [1, n]$. Let $G_1 := G(p_1, p_1/(\alpha_0 \log p_1))$ where $p = \Theta(p_1/\log p_1)$ and p_1 divides n . Now for each node v in G_1 , replace v with a directed path $\pi_v = (u_1^v, \dots, u_k^v)$ with $k = n/p_1$ nodes. For all $v \in V$, assign weight 0 to all edges in π_v . For each edge (v_1, v_2) originally in G_1 , add edge $(u_k^{v_1}, u_1^{v_2})$ to the graph. Let G_2 be the resulting graph, and let Π_2 be the updated set of critical paths. This completes the modification.

Chapter 5

DAG Covers

This chapter develops the results on DAG covers and DAG embeddings, including the near-linear-size upper bounds and the accompanying lower bounds.¹

5.1 Preliminaries

Throughout this section, we will be working exclusively with directed graphs. For a directed graph G , we let $TC(G) \subseteq V(G) \times V(G)$ denote the transitive closure of G . For a node $v \in V(G)$, we will use $\deg_G(v)$ to denote the number of edges incident to v in G . Additionally, we will use $\text{indeg}_G(v)$ and $\text{outdeg}_G(v)$ to denote the the number of incoming edges and outgoing edges, respectively, that are incident to v in G .

We will frequently use π to denote a directed path and Π to denote a collection of paths. We will let $V(\pi)$ and $E(\pi)$ denote the nodes and edges of path π , respectively. Given a path π and a set of nodes S , we will let $\pi \cap S$ denote the sequence of nodes in $V(\pi) \cap S$, listed in the order they appear in path π . We will interpret the ordered set $\pi \cap S$ as a (possibly non-contiguous) subpath of π restricted to the nodes in S . Note that the edges $E(\pi \cap S)$ of subpath $\pi \cap S$ may not be a subset of the set edges $E(\pi)$ of the original path π ; however, instead we can guarantee that the edges in $\pi \cap S$ are contained in the transitive closure of π , i.e., $E(\pi \cap S) \subseteq TC(\pi)$.

Given a collection of nodes V and a collection of paths Π where the paths in π may contain nodes that are not in V , we will define a corresponding graph which we denote as $\text{Graph}(V, \Pi)$ over the vertex set V . This graph will be the (non-disjoint) union of the set of paths in Π restricted to the

¹The contents in this section originally appeared in [AHW25b] and are based on joint work with Sepehr Assadi and Nicole Wein.

nodes in V . Formally, we let $\text{Graph}(V, \Pi)$ be the graph

$$\text{Graph}(V, \Pi) = \left(V, \bigcup_{\pi \in \Pi} E(\pi \cap V) \right).$$

5.2 DAG covers with $f(n)$ additional edges

In this section, we present a lower bound and an upper bound for DAG covers with $f(n)$ additional edges, where function $f(n)$ depends only on the number of nodes n in the input graph. All of our bounds in this section are tight up to $n^{o(1)}$ factors.

A lower bound for DAG covers with $n^{2-o(1)}$ additional edges

The goal of this subsection is to prove the following theorem.

Theorem 5.1. *There exists a family of n -node directed graphs G such that any reachability-preserving DAG cover of G with at most $n^{2-o(1)}$ additional edges requires at least $n^{1-o(1)}$ DAGs.*

Our lower bound construction will make use of the following lemma, which is immediate from existing lower bounds on reachability preservers in [AB18; AB24b].

Lemma 5.2 (cf. Theorem 13 of [AB24b]). *There exists an n -node directed acyclic graph $G = (V, E)$ and a collection of directed paths Π in G with the following properties:*

1. $|\Pi| = n^{2-o(1)}$,
2. each path $\pi \in \Pi$ has at least $|\pi| \geq 10 \log n$ edges,
3. paths in Π are pairwise edge-disjoint, and
4. each path $\pi \in \Pi$ is the unique directed path between its endpoints in G .

Let $G = (V, E)$ and Π be the n -node directed graph and associated collection of paths referenced in Lemma 5.2, respectively. We define a random graph G^* as follows:

1. Replace each node v in V with a copy of the directed clique graph K_2 with bidirectional edges. Denote the clique replacing node v as K_2^v .
2. Fix an edge $(u, v) \in E$. Let $u^* \in V(K_2^u)$ be a node sampled uniformly at random from K_2^u . Likewise, let $v^* \in V(K_2^v)$ be a node sampled uniformly at random from K_2^v . Add edge (u^*, v^*) to graph G^* . Repeat this random process for every edge in E .

This completes the construction of random graph $G^* = (V^*, E^*)$. We now define an associated set of paths Π^* in G^* . Each path in Π^* will be an image of a path in Π . Each edge $(u, v) \in E$ corresponds to a unique edge (u^*, v^*) in E^* between cliques K_2^u and K_2^v . Let $\phi : E \mapsto E^*$ denote this injective mapping. Now fix a path $\pi \in \Pi$, and let $\pi = (v_1, \dots, v_k)$. For each edge $(v_i, v_{i+1}) \in E(\pi)$, let $e_i := \phi((v_i, v_{i+1}))$. Note that $e_i := (x_i, y_i) \subseteq V(K_2^{v_i}) \times V(K_2^{v_{i+1}})$, so $y_i, x_{i+1} \in V(K_2^{v_{i+1}})$, and in particular, either $y_i = x_{i+1}$ or $(y_i, x_{i+1}) \in E(K_2^{v_{i+1}})$. Then we define the image path π^* in G^* as

$$\pi^* = (x_1, y_1, x_2, y_2, \dots, x_i, y_i, \dots, x_{k-1}, y_{k-1}).$$

We will use $\psi(\pi)$ to denote the image path π^* of π in G^* . Given a subset $\Pi' \subseteq \Pi$, we will use $\psi[\Pi']$ to denote the image of set Π' under function ψ , i.e., $\psi[\Pi'] = \{\psi(\pi) \mid \pi \in \Pi'\}$. We define Π^* as the collection of image paths $\psi(\pi)$ of paths π in Π , so $\Pi^* = \psi[\Pi]$.

We now define a family of DAGs $\mathcal{D}$ associated with our distribution of random graphs G^* that will be useful in our analysis.

DAG Family $\mathcal{D}$. We begin by defining a graph $\hat{G} = (V^*, \hat{E})$ over the same vertex set as G^* , and containing every possible edge in G^* . Formally, we define edge set $\hat{E}$ to be

$$\hat{E} = \{V(K_2^u) \times V(K_2^v) \mid (u, v) \in E\} \cup \{E(K_2^v) \mid v \in V\}.$$

Note that $G^* \subseteq \hat{G}$ for each graph G^* in the support of our random graph distribution.

Our DAG family $\mathcal{D}$ will be the set of all edge-maximal acyclic subgraphs of $\hat{G}$. We define a DAG D of $\hat{G}$ as follows. For each clique K_2^v , fix a total order on its 2 vertices, and delete the edge in K_2^v that does not respect this total order. After repeating this process for all cliques K_2^v , the resulting graph will be a DAG, since the original graph G is a DAG. Define $\mathcal{D}$ to be the family of DAGs of G^* that can be constructed from the above process. Note that since there are only 2 choices of total orderings for each clique K_2 , it follows that the size of $\mathcal{D}$ is

$$|\mathcal{D}| = 2^{|V|} = 2^n.$$

For each DAG $D \in \mathcal{D}$, the graph $D \cap G^*$ is an acyclic subgraph of random graph G^* .² This allows us to reason about DAG covers of G^* using the DAGs in graph family $\mathcal{D}$. We note that while G^* is a random graph, the DAGs in $\mathcal{D}$ are fixed and depend only on graph G . The next claim follows directly from the structure of graph G^* as a DAG of strongly connected components K_2^v .

²We use $D \cap G^*$ to denote the graph with vertex set V^* and edge set $E(D) \cap E(G^*)$.

Claim 5.3. *For each acyclic subgraph D of G^* , there is DAG $D' \in \mathcal{D}$ such that $D \subseteq D' \cap G^*$.*

Proof. Fix an acyclic subgraph D of G^* . DAG D has an associated total ordering σ of the vertices $V(G^*)$ of G^* . This total ordering induces a total ordering $\sigma[V(K_2^v)]$ on the vertices $V(K_2^v)$ for each $v \in V(G^*)$. By our construction of family $\mathcal{D}$, there exists a DAG $D' \in \mathcal{D}$ such that D' contains every edge in $E(K_2^v)$ that respects total order $\sigma[V(K_2^v)]$ for all $v \in V(G^*)$. Then for every edge $e \in E(D)$ with $e \in E(K_2^v)$ for some $v \in V(G^*)$, it follows that $e \in E(D') \cap E(G^*)$. Additionally, note that every edge in $E(G^*)$ between distinct strongly connected components K_2^u and K_2^v in G^* , where $u \neq v$, is preserved in D' . We conclude that $D \subseteq D' \cap G^*$. $\square$

We now verify that every DAG $D \in \mathcal{D}$ inherits the unique directed path property of our original graph G (Property 4 of Lemma 5.2).

Claim 5.4. *Let $D \in \mathcal{D}$ be a DAG in $\mathcal{D}$. Fix an $s^* \rightsquigarrow t^*$ path $\pi^* \in \Pi^*$. Then s^* can reach t^* in $D \cap G^*$ only if $\pi^* \subseteq D \cap G^*$.*

Proof. Recall that path $\pi^* \in \Pi^*$ is the image of an $s \rightsquigarrow t$ path $\pi \in \Pi$. This claim will follow straightforwardly from the property that path π is the unique path between s and t in graph G . Consider an arbitrary $s^* \rightsquigarrow t^*$ path π_1^* in $D \cap G^*$. Contract each clique K_2^v in G^* into a single node v . The resulting graph will be identical to our original graph G . Let π' (respectively, π_1') denote the image of path π^* (respectively, path π_1^*) after this contraction operation. Since π is the unique $s \rightsquigarrow t$ path in G , we conclude that $\pi' = \pi_1' = \pi$.

As a consequence, we know that paths π^* and π_1^* in $D \cap G^*$ travel through the same sequence of cliques K_2^v in $D \cap G^*$. Let $K_2^{v_1}, \dots, K_2^{v_k}$ denote the sequence of cliques passed through by paths π^* and π_1^* in G^* , where $v_1 = s$ and $v_k = t$. By our construction of G^* , there is exactly one edge between clique $K_2^{v_i}$ and $K_2^{v_{i+1}}$ in G^* . Formally,

$$|E(G^*) \cap (V(K_2^{v_i}) \times V(K_2^{v_{i+1}}))| = 1,$$

for all $i \in [1, k-1]$. As a consequence, paths π^* and π_1^* must use exactly the same edge in $V(K_2^{v_i}) \times V(K_2^{v_{i+1}})$ between cliques $K_2^{v_i}$ and $K_2^{v_{i+1}}$. Finally, since paths π^* and π_1^* are contained in DAG D , we conclude that paths π^* and π_1^* are identical when restricted to any clique $K_2^{v_i}$. Formally,

$$\pi^* \cap V(K_2^{v_i}) = \pi_1^* \cap V(K_2^{v_i}),$$

for all $i \in [1, k]$. We conclude that $\pi_1^* = \pi^*$, a contradiction. $\square$

Our eventual goal is to argue that with high probability, for each DAG $D \in \mathcal{D}$, the DAG subgraph $D \cap G^*$ of G^* cannot preserve reachability between many pairs of vertices in G^* . We

begin with the following simple claim about the probability a DAG $D \cap G^*$ preserves a short path in G^* (see Section 5.2 for a visualization).

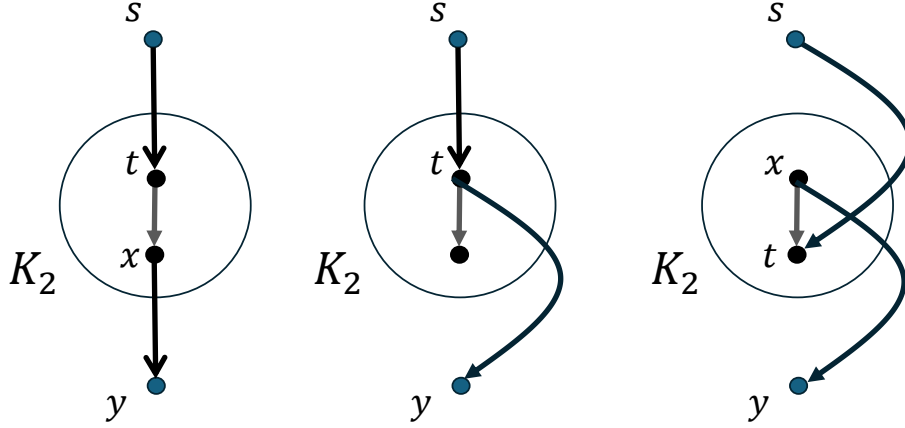


Figure 5.1: A visualization of the three cases in Theorem 5.5. These cases correspond to three possibilities of how edges entering and exiting a clique K_2^v are reattached to vertices in K_2^v .

Claim 5.5. Fix a DAG $D \in \mathcal{D}$. Fix two edges $(u, v), (v, w) \in E$ in graph G , and let $\phi(u, v) = (s, t)$ and $\phi(v, w) = (x, y)$. Then path $\pi^* = (s, t, x, y)$ is contained in DAG $D \cap G^*$ with probability at most $3/4$.

Proof. We observe that edges (s, t) and (x, y) are always contained in graph $D \cap G^*$. Then path π^* is contained in $D \cap G^*$ if and only if $t = x$ or $t <_D x$, where $<_D$ is the total order of DAG D .

Let $V(K_2^v) = v_1^*, v_2^*$, where $v_1^* <_D v_2^*$. Then the probability that $t = x$ is

$$\Pr[t = x] = \Pr[t = x = v_1^*] + \Pr[t = x = v_2^*] = 1/2.$$

Likewise, the probability that $t <_D x$ is

$$\Pr[t <_D x] = \Pr[t = v_1^* \text{ and } x = v_2^*] = 1/4.$$

Then path π^* is contained in $D \cap G^*$ with probability at most $3/4$. □

In Claim 5.5, we showed that certain short paths in G^* survive in $D \cap G^*$ for a fixed DAG $D \in \mathcal{D}$ with probability $3/4$. In the following claim, we will use Claim 5.5 to argue that $D \cap G^*$ contains many paths in Π^* with very low probability.

Claim 5.6. Fix a DAG $D \in \mathcal{D}$, and fix a subset $\Pi_1^* \subseteq \Pi^*$ of size $|\Pi_1^*| = k$. The probability that $D \cap G^*$ contains all k paths in Π_1^* as subgraphs is at most $\left(\frac{3}{4}\right)^{10k \log n}$.

Proof. Fix a path $\pi^* \in \Pi_1^*$ that is the image of a path $\pi \in \Pi$. Let e_i be the i th edge of path π for $i \in [1, |\pi|]$, and let $(u_i, v_i) = \phi(e_i)$. Fix an $i \in [1, |\pi|]$, and let A_i be the event that path $(u_i, v_i, u_{i+1}, v_{i+1})$ is contained in DAG $D \cap G^*$. We will need the following observations:

- Path π^* is contained in DAG $D \cap G^*$ only if event A_i holds for every $i \in [1, |\pi|]$.
- By Claim 5.5, $\Pr[A_i] \leq 3/4$.
- Events A_i and A_j are independent when $|i - j| \geq 2$.

Combining these observations, we get that

$$\Pr[\pi^* \subseteq D \cap G^*] \leq \Pr[\cap_i A_i] \leq \left(\frac{3}{4}\right)^{|\pi|/2} \leq \left(\frac{3}{4}\right)^{5 \log n},$$

where the final inequality follows from the fact that $|\pi| \geq 10 \log n$ by Lemma 5.2. Also by Lemma 5.2, the paths in Π are pairwise-edge disjoint. This implies that for distinct paths $\pi_1^*, \pi_2^* \in \Pi^*$, the event that $\pi_1^* \subseteq D \cap G^*$ is independent of the event that $\pi_2^* \subseteq D \cap G^*$. We conclude that

$$\Pr[\pi^* \subseteq D \cap G^* \text{ for all } \pi^* \in \Pi_1^*] = \prod_{\pi^* \in \Pi_1^*} \Pr[\pi^* \subseteq D \cap G^*] \leq \left(\frac{3}{4}\right)^{5k \log n}.$$

□

We will now combine the very strong probability bound proved in Claim 5.6 with a union bound to argue that no DAG subgraph $D \cap G^*$, where $D \in \mathcal{D}$, can contain many paths in Π^* as subgraphs.

Lemma 5.7. *With nonzero probability, for each DAG $D \in \mathcal{D}$, graph $D \cap G^*$ contains at most n distinct paths in Π^* as subgraphs.*

Proof. By theorem 5.6 and the union bound, the probability that there exists a DAG $D \in \mathcal{D}$ and a set $\Pi' \subseteq \Pi^*$ of size $|\Pi'| = k$ such that each path in Π' is a subgraph of $D \cap G^*$ is at most

$$|\mathcal{D}| \cdot \binom{|\Pi^*|}{k} \cdot \left(\frac{4}{3}\right)^{-5k \log n} \leq 2^n \cdot n^{2k} \cdot \left(\frac{4}{3}\right)^{-5k \log n} < 1,$$

when $k \geq n$. Note that if a graph $D \cap G^*$ contains greater than k distinct paths in Π^* as subgraphs, then it must contain a set $\Pi' \subseteq \Pi^*$ of size $|\Pi'| = k$, so the claim follows. □

From now on, we treat graph G^* as a fixed graph that satisfies Theorem 5.7. This is without loss of generality, since Theorem 5.7 holds with nonzero probability. Intuitively, it should be difficult to construct a DAG cover for graph G^* because any DAG subgraph of G^* can only preserve at most n paths in Π^* by Theorem 5.7. We are ready to prove theorem 5.1.

Theorem 5.1. *There exists a family of n -node directed graphs G such that any reachability-preserving DAG cover of G with at most $n^{2-o(1)}$ additional edges requires at least $n^{1-o(1)}$ DAGs.*

Proof. Let $D_1, \dots, D_g$ be a reachability-preserving DAG cover of G^* with at most $|\Pi^*|/2 = n^{2-o(1)}$ additional edges. Let E' be the set of additional edges used in the DAG cover. From the definition of a DAG cover, we require that $E' \subseteq TC(G^*)$. In particular, we may assume without loss of generality $E' \subseteq TC(G^*) \setminus E(G^*)$. This implies that every edge $e \in E'$ is between distinct clique subgraphs K_2^u and K_2^v in G^* , where $u \neq v$.

We define a subset $\Pi_1^* \subseteq \Pi^*$ of paths in Π^* as follows:

$$\Pi_1^* = \{\pi^* \in \Pi^* \mid TC(\pi^*) \cap E' = \emptyset\}.$$

We observe that $|\Pi_1^*| \geq |\Pi^*| - |E'| \geq |\Pi^*|/2$, since every edge e in E' is contained in $TC(\pi^*)$ for at most one path $\pi^* \in \Pi^*$ by Property 3 of Lemma 5.2, and since $|E'| \leq |\Pi^*|/2$.

Fix an $i \in [1, g]$. By Theorem 5.3, there exists a DAG $D'_i \in \mathcal{D}$ such that $D_i - E' \subseteq D'_i \cap G^*$. Additionally, fix an $s^* \rightsquigarrow t^*$ path $\pi^* \in \Pi_1^*$ such that s^* can reach t^* in D_i . Since $\pi^* \in \Pi_1^*$, we know that s^* can reach t^* in $D_i - E'$. Since $D_i - E' \subseteq D'_i \cap G^*$, we know that s^* can reach t^* in $D'_i \cap G^*$ as well. Finally, by Theorem 5.4, this implies that path π^* is contained in graph $D'_i \cap G^*$, i.e., $\pi^* \subseteq D'_i \cap G^*$.

We have shown that for each $i \in [1, g]$, there exists a DAG $D'_i \in \mathcal{D}$ such that for every $s^* \rightsquigarrow t^*$ path $\pi^* \in \Pi_1^*$, if s^* can reach t^* in DAG D_i , then path π^* is a subgraph of $D'_i \cap G^*$. However, by Theorem 5.7, there are at most n distinct paths in Π^* that are subgraphs of $D'_i \cap G^*$. We conclude that each DAG D_i in our DAG cover preserves the reachability between the endpoints of at most n distinct paths in Π_1^* .

There are $|\Pi_1^*| = |\Pi^*|/2 = n^{2-o(1)}$ distinct paths in Π_1^* . Then the number of DAGs in our DAG cover is at least

$$g \geq \frac{|\Pi_1^*|}{n} = \frac{n^{2-o(1)}}{n} = n^{1-o(1)}.$$

□

An upper bound for DAG covers with $O\left(\frac{n^2(\log \log n)^4}{\log^2 n}\right)$ additional edges

The goal of this section is to prove the following theorem.

Theorem 5.8. *Every n -node directed graph G admits an exact distance-preserving DAG cover of G with $O\left(\frac{n^2(\log \log n)^4}{\log^2 n}\right)$ additional edges and $n^{o(1)}$ DAGs.*

We will rely on the following theorem about the existence of small exact hopsets due to [BRR10b].

Lemma 5.9 (Theorem 5.1 of [BRR10b]). *Let G be an n -node directed, weighted graph. For every $d \in [1, n]$, there exists a collection of edges $H \subseteq TC(G)$ satisfying the following properties:*

1. $|H| = O(n^2/d^2)$, and
2. *for every pair of nodes $(s, t) \in TC(G)$, there exists a $s \rightsquigarrow t$ path in $G \cup H$ with at most d edges.*

With this lemma, we are ready to prove Theorem 5.8.

Proof of Theorem 5.8. Given an input graph G on n nodes, construct a collection of edges H as in Lemma 5.9 with parameter $d = \frac{\log n}{(\log \log n)^2}$. Let Σ denote the collection of all total orders on the vertices $V(G)$ of G . Let $\sigma_1, \dots, \sigma_k \in \Sigma$ be total orders of $V(G)$ sampled from Σ uniformly with replacement. We choose k to be $k = (d+1)! \cdot 10 \log n$. For each σ_i , let $D_i \subseteq G \cup H$ be the DAG obtained by keeping only the edges in $G \cup H$ that respect total order σ_i .

Fix a pair of nodes $(s, t) \in TC(G)$. Let π be an $s \rightsquigarrow t$ shortest path in $G \cup H$ with at most d edges. Such a path must exist by Lemma 5.9. Path π survives in DAG D_i if total ordering σ_i respects the ordering of the $|\pi| + 1 \leq d + 1$ vertices of path π . Then path π survives in D_i with probability at least $\frac{1}{(d+1)!}$. Then the probability that π does not survive in any of our k total orders is at most

$$\left(1 - \frac{1}{(d+1)!}\right)^k = \left(1 - \frac{1}{(d+1)!}\right)^{(d+1)! \cdot 10 \log n} \leq e^{-\frac{(d+1)!}{(d+1)!} \cdot 10 \log n} \leq n^{-5}.$$

By union bounding over all $O(n^2)$ pairs of nodes $(s, t) \in TC(G)$, we conclude that DAG cover $\{D_i\}_{i \in [1, k]}$ is an exact distance-preserving DAG cover of G , with high probability. Finally, we observe that our DAG cover contains

$$k = (d+1)! \cdot 10 \log n = 2^{\Theta(d \log d)} 10 \log n = 2^{O\left(\frac{\log n}{\log \log n}\right)} 10 \log n = n^{o(1)}$$

DAGs. □

5.3 An upper bound for DAG covers with $\tilde{O}(m)$ additional edges

The goal of this section is to prove the following theorem.

Theorem 5.10. *Let G be an n -node weighted, directed graph with polynomially-bounded positive integer edge weights. Then there exists an $O(\log^3 n \cdot \log \log n)$ -approximate DAG cover with $O((m+n) \log^3 n)$ additional edges and $O(\log n)$ DAGs. Moreover, this DAG cover can be computed w.h.p. in $\tilde{O}(m)$ time.*

An essential tool we will need to prove Theorem 5.10 is the directed low-diameter decomposition introduced in [BNW22] and improved recently in [Bri+25]. The key properties of this low-diameter decomposition are summarized in Theorem 2.1 of [Bri+25], which we restate below.

Lemma 5.11 (Theorem 2.1 of [Bri+25]). *Let $G = (V, E, w)$ be an n -node weighted, directed graph with positive integer edge weights. Let d be a positive integer such that $d = O(\text{poly}(n))$. Then there exists a set of edges $E' \subseteq E$ with the following properties:*

1. *each SCC of $G \setminus E'$ has (weak) diameter at most d in G .*
2. *For every edge $e \in E$,*

$$\Pr[e \in E'] \leq \alpha \cdot \frac{w(e)}{d},$$

where $\alpha = O(\log n \cdot \log \log n)$.

3. *The set E' and the SCCs of $G \setminus E'$ can be computed in $\tilde{O}(m)$ time.*

For the remainder of this section, we use α to denote the coefficient α in property 2 of Theorem 5.11.

Construction of our DAG Cover $\mathcal{D}$

Let G be an n -node, weighted, directed graph with positive integer edge weights and maximum weight W . The key step of our construction will be a hierarchical decomposition of G obtained by repeatedly applying Lemma 5.11. Let $G_0 = G$, and let $\mathcal{F}_0$ denote the collection of all strongly connected components in G_0 . We will construct set family $\mathcal{F}_{i+1}$ and graph G_{i+1} from graph G_i , as follows.

1. Apply Lemma 5.11 to graph G_i with integer parameter $d_i = nW \cdot 2^{-i-1}$. Let $E_i \subseteq E(G_i)$ be the set of edges specified by Lemma 5.11.

2. Let $G_{i+1} = G_i - E_i$, and let $S_{i+1}^1, \dots, S_{i+1}^{k_{i+1}} \subseteq V(G)$ denote the SCCs of G_{i+1} . We let $\mathcal{F}_{i+1}$ denote the collection of SCCs of graph G_{i+1} .
3. Let $z = \lceil \lg(nW) \rceil$. We will terminate our recursion after computing graph G_z and collection $\mathcal{F}_z$.

Note that since $d_z < 1$, the graph G_z will be a DAG. We will use D^* to denote graph G_z . The following claim summarizes the properties of the sets $\mathcal{F}_i$ inherited from Lemma 5.11.

Claim 5.12. *The following properties of the $\mathcal{F}_i$'s and E_i 's are inherited from Lemma 5.11:*

1. Set family $\mathcal{F}_i$ is the collection of SCCs of graph G_i .
2. Each set $S \in \mathcal{F}_i$ has weak diameter at most $2d_i$ in G_{i-1} (and consequently, in G as well).
3. For every edge $e \in E(G_i)$,

$$\Pr[e \in E_i] \leq \alpha \cdot \frac{w(e)}{d_i}$$

We will need several preliminary notations and claims before we can construct the DAGs associated with the set family of $\mathcal{F}_i$'s. Fix an $i \in [0, z]$. We define a total order $<_i$ on the SCCs $\mathcal{F}_i$ in G_i . Informally, this will correspond to a specific topological order of the condensation graph of G_i . Fix two SCCs $S, T \in \mathcal{F}_i$, where $S \neq T$. If there exists a node $s \in S$ and a node $t \in T$ such that s can reach t in graph G_i , then we let $S <_i T$. If there exists distinct sets $S', T' \in \mathcal{F}_{i-1}$ satisfying $S \subseteq S'$ and $T \subseteq T'$ such that $S' <_{i-1} T'$, then we let $S <_i T$, as well. We observe that if there exists distinct $S, T \in \mathcal{F}_i$ such that $S <_i T$ and $T <_i S$, then that implies a contradiction of Property 1 of Claim 5.12. We conclude that $<_i$ is a partial order on the elements of $\mathcal{F}_i$. Finally, we convert partial order $<_i$ to a total order by completing the ordering arbitrarily.

Claim 5.13. *Relation $<_i$ is a total ordering of set family $\mathcal{F}_i$.*

We will now define a total order $<_D$ on the vertex set $V(G)$ that respects every total order $<_i$ for all $i \in [0, z]$. For each node $v \in V(G)$ and each $i \in [0, z]$, we let $S_i(v) \in \mathcal{F}_i$ denote the unique set in $\mathcal{F}_i$ containing node v . Given $s, t \in V(G)$, we will let $s <_D t$ if

$$s \neq t \quad \text{and} \quad S_i(s) <_i S_i(t),$$

for some $i \in [0, z]$. We will let $<_D$ be an arbitrary total order that satisfies this property for all $s \neq t \in V(G)$. We will use $s \leq_D t$ to denote that either $s <_D t$ or $s = t$.

Claim 5.14. *Relation $<_D$ is a total ordering of vertex set $V(G)$.*

Proof. Suppose towards contradiction that there exist nodes $s, t \in V(G)$, with $s \neq t$, such that $s <_D t$ and $t <_D s$. Then without loss of generality, $S_i(s) <_i S_i(t)$ and $S_j(t) <_j S_j(s)$ for some $i \leq j$. Since $S_j(s) \subseteq S_i(s)$ and $S_j(t) \subseteq S_i(t)$ and $S_i(s) <_i S_i(t)$, we conclude that $S_j(s) <_j S_j(t)$ by the definition of $<_j$. This contradicts Claim 5.13, so we conclude that $<_D$ is a total ordering of $V(G)$. $\square$

For each set $S \in \mathcal{F}_i$, we associate two *representative* nodes $r_1(S), r_2(S) \in S$. We choose $r_1(S)$ to be the first node in S under total order $<_D$, and we choose $r_2(S)$ to be the last node in S under total order $<_D$. For each node $v \in V(G)$ and each $i \in [0, z]$, we define the level i representative nodes of v , denoted as $r_i^1(v)$ and $r_i^2(v)$, to be

$$r_i^1(v) = r_1(S_i(v)) \quad \text{and} \quad r_i^2(v) = r_2(S_i(v)).$$

In particular, when $i = z$, we have that $r_z^1(v) = r_z^2(v) = v$, since D^* is a DAG. For each edge $(u, v) \in E(G)$, we define the *level* of edge (u, v) , denoted as $\ell(u, v)$, to be the smallest integer $i \in [0, z]$ such that $S_i(u) \neq S_i(v)$.

In addition to the above notation, we will need the following claim, which roughly states that we can approximately preserve distances between nodes s, t in an SCC $S \in \mathcal{F}_i$ when $s <_D t$, using a small collection of edges H that respect total order $<_D$.

Claim 5.15. *Let $S \in \mathcal{F}_i$ be a set in family $\mathcal{F}_i$. Then there exists a set of directed, weighted edges $H \subseteq S \times S$ of size $|H| = O(|S| \log |S|)$ with the following properties:*

1. *Every edge (u, v) in H respects total order $<_D$.*
2. *Every edge (u, v) in H has weight $w(u, v) = 2d_i$.*
3. *For all $s, t \in S$ with $s <_D t$,*

$$\text{dist}_G(s, t) \leq \text{dist}_H(s, t) \leq 4d_i.$$

4. *Set H can be computed in $O(|H|)$ time.*

Proof. Let $S = \{s_1, \dots, s_k\}$ be the nodes in S , ordered with respect to $<_D$. We define a path π that respects $<_D$ as follows:

$$\pi = (s_1, \dots, s_k).$$

We assign each edge in π weight d_i . Initially, we let $H = E(\pi)$. By [KP22c], for a directed path π on k nodes, there exists a collection of directed edges H' such that

- $|H'| = O(k \log k)$ and H' can be computed in $O(|H'|)$ time,
- H' is contained in the transitive closure of path π , and
- For every pair of nodes s, t in path π such that node s occurs on π before node t , there exists an $s \rightsquigarrow t$ path π' in $\pi \cup H'$ with $|\pi'| \leq 2$ edges.

Then we let $H = E(\pi) \cup H'$, with every edge in H assigned weight $2d_i$. Properties 1, 2, and 4 of our claim are clearly satisfied. We now verify Property 3 for all $s, t \in S$ with $s <_D t$:

- By Property 2, $\text{dist}_H(s, t) \geq 2d_i$. Moreover, $2d_i \geq \text{dist}_G(s, t)$ by Claim 5.12, so $\text{dist}_G(s, t) \leq \text{dist}_H(s, t)$.
- For each pair of nodes s, t in path π such that $s <_D t$, there exists an $s \rightsquigarrow t$ path π' in H with $|\pi'| \leq 2$ edges. Then $\text{dist}_H(s, t) \leq w(\pi') \leq 4d_i$, by Property 2. $\square$

We construct two DAGs D_1 and D_2 associated with our collection of $\mathcal{F}_i$'s. DAG D_1 will respect total order $<_D$, while DAG D_2 will respect the reverse of $<_D$. We construct DAG D_1 as follows.

Construction of DAG D_1 .

1. The vertex set of D_1 will be the same as G , i.e., $V(D_1) = V(G)$.
2. For each edge $(u, v) \in E(D^*)$, each $i, j \in [0, z]$ such that $\min(i, j) \geq \ell(u, v)$, and each $k, k' \in \{1, 2\}$, add the edge $(r_i^k(u), r_j^{k'}(v))$ to D_1 . Assign edge $(r_i^k(u), r_j^{k'}(v))$ weight $w_G(u, v) + 2d_i + 2d_j$. Since $r_z^1(v) = r_z^2(v) = v$ for all $v \in V(G)$, this procedure also adds edges of the form $(u, r_i^j(v))$ and $(r_i^j(u), v)$, for all $i \geq \ell(u, v)$ and $j \in \{1, 2\}$. Additionally, we add the edges in $E(D^*)$ with their original weights in G to DAG D_1 .
3. Fix an index $i \in [0, z]$, and fix a set $S_i^j \in \mathcal{F}_i$, where $j \in [1, k_i]$. Let $H_i^j \subseteq S_i^j \times S_i^j$ be the set of directed, weighted edges specified in Claim 5.15 with respect to set S_i^j . We will add the set of edges

$$H = \bigcup_{i \in [0, z], j \in [1, k_i]} H_i^j$$

to D_1 .

This completes the construction of D_1 . If there are parallel edges in D_1 from node u to node v , then we keep only the lowest weight edge from u to v . We will quickly verify that D_1 is a DAG.

Claim 5.16. *Graph D_1 is a DAG.*

Proof. Note that every edge in H respects total order $<_D$ by Claim 5.15. We will now show that every edge we add to D_1 in Step 2 respects $<_D$. Fix an edge $(r_i^x(u), r_j^y(v))$ added to D_1 in Step 2, for some $i \leq j \in [0, z]$, $x, y \in \{1, 2\}$, and $(u, v) \in E(D^*)$. Since $i \geq \ell(u, v)$, it follows that $S_i(u) \neq S_i(v)$. Additionally, since edge $(u, v) \in E(D^*) \subseteq E(G_i)$, it follows that $S_i(u) <_i S_i(v)$. Then since $r_i^x(u) \in S_i(u)$ and $r_j^y(v) \in S_j(v) \subseteq S_i(v)$, we conclude that

$$r_i^x(u) <_D r_j^y(v),$$

as desired. □

Construction of DAG D_2 .

1. The vertex set of D_2 will be the same as G , so $V(D_2) = V(G)$.
2. Fix an index $i \in [0, z]$, and fix a set $S_i^j \in \mathcal{F}_i$, where $j \in [1, k_i]$. Let $H_i^j \subseteq S_i^j \times S_i^j$ be the set of directed, weighted edges specified in Claim 5.15 with respect to set S_i^j . Let $H \subseteq V(G) \times V(G)$ be the directed, weighted set of edges

$$H = \bigcup_{i \in [0, z], j \in [1, k_i]} H_i^j.$$

Let H^R denote the set of weighted edges obtained by reversing the orientation of every edge in H . We add the edges in H^R to D_2 .

This completes the construction of graph D_2 . Note that for every edge (u, v) in graph D_2 , we have that $(v, u) \in E(D_1)$. Then by Claim 5.16, graph D_2 is also a DAG.

Construction of DAG Cover $\mathcal{D}$. We will construct our DAG cover $\mathcal{D}$ by repeating the following (random) procedure $10 \log n$ times:

1. Construct a collection of sets $\{\mathcal{F}_i\}_{i \in [0, z]}$. (Note that each set $\mathcal{F}_i$ is constructed randomly and inherits the probabilistic guarantees of Lemma 5.11.)
2. Construct DAGs D_1 and D_2 using the collection of sets $\{\mathcal{F}_i\}_{i \in [0, z]}$, and add D_1 and D_2 to $\mathcal{D}$.

Size and Time Analysis of our DAG Cover $\mathcal{D}$

In this section, we prove that our DAG cover $\mathcal{D}$ has the size claimed in Theorem 5.10, and we prove that our DAG cover can be constructed in the time claimed in Theorem 5.10.

Size Analysis. In Step 2 of the construction of DAG D_1 , we add at most $O(z^2)$ edges to D_1 for every edge $(u, v) \in E(D^*) \subseteq E(G)$. Then Step 2 of the construction of DAG D_1 contributes $O(mz^2) = O(m \log^2(nW))$ edges to D_1 . By Claim 5.15, the third step of the construction of DAG D_1 contributes at most

$$|H| \leq \sum_{i \in [0, z], j \in [1, k_i]} |H_i^j| \leq O(\log n) \cdot \sum_{i \in [0, z]} \sum_{j \in [1, k_i]} |S_i^j| = O(\log n) \cdot \sum_{i \in [0, z]} n = O(n \log(nW) \log n)$$

edges. Then we add at most

$$|E(D_1)| + |E(D_2)| = O(m \log^2 n + n \log^2 n)$$

edges to D_1 and D_2 , when $W = O(\text{poly}(n))$. Since we repeatedly construct random DAGs D_1 and D_2 exactly $10 \log n$ times, the number of additional edges and DAGs claimed in Theorem 5.10 is established.

Running Time Analysis. To construct our hierarchical decomposition $\{\mathcal{F}\}_i$, we make z calls to Lemma 5.11. This takes $\tilde{O}(\log(nW) \cdot m)$ time. The second step of the construction of DAG D_1 takes $O(mz^2) = O(m \log^2(nW))$ time. To construct all sets of edges H_i^j , where $i \in [0, z]$ and $j \in [1, k_i]$, it takes time $O(\sum_{i \in [0, z], j \in [1, k_i]} |H_i^j|) = O(n \log(nW) \log n)$ by Claim 5.15 and our earlier bound on $\cup_{i,j} |H_i^j|$ in our size analysis. We conclude that the constructions of DAGs D_1 and D_2 take $\tilde{O}(m)$ time when $W = O(\text{poly}(n))$. Since we repeatedly construct random DAGs D_1 and D_2 exactly $10 \log n$ times, the running time claimed in Theorem 5.10 is established.

Initial Distortion Analysis

In this section, we make some initial progress towards proving that our DAG cover $\mathcal{D}$ approximately preserves distances in G . First, we verify that for every DAG $D \in \mathcal{D}$, distances in D are at least distances in G .

Claim 5.17. *For each DAG $D \in \mathcal{D}$ and nodes $s, t \in V(G)$,*

$$\text{dist}_G(s, t) \leq \text{dist}_D(s, t).$$

Proof. We will prove that for every edge $(u, v) \in E(D)$,

$$\text{dist}_G(u, v) \leq w_D(u, v),$$

where w_D is the weight function associated with DAG D . This will immediately imply the stated claim. We split our proof into cases based on which step of the construction we added edge (u, v) to DAG D .

- Edge (u, v) was added in Step 2 of the construction of DAG D_1 . In this case, edge (u, v) is of the form

$$(u, v) = (r_i^x(u^*), r_j^y(v^*)),$$

for some $i, j \in [0, z]$ such that $\min(i, j) \geq \ell(u^*, v^*)$, $x, y \in \{1, 2\}$ and edge $(u^*, v^*) \in E(D^*)$. Recall that $w_D(u, v) = w_G(u^*, v^*) + 2d_i + 2d_j$. Then

$$\begin{aligned} \text{dist}_G(u, v) &= \text{dist}_G(r_i^x(u^*), r_j^y(v^*)) \\ &\leq \text{dist}_G(r_i^x(u^*), u^*) + \text{dist}_G(u^*, v^*) + \text{dist}_G(v^*, r_j^y(v^*)) \quad \text{by the triangle inequality} \\ &\leq \text{dist}_G(r_i^x(u^*), u^*) + w_G(u^*, v^*) + \text{dist}_G(v^*, r_j^y(v^*)) \\ &\leq 2d_i + w_G(u^*, v^*) + 2d_j \quad \text{by Theorem 5.12} \\ &= w_D(u, v), \end{aligned}$$

as desired.

- Edge (u, v) was added in Step 3 of the construction of DAG D_1 or in Step 2 of the construction of DAG D_2 . We will need the following observations:
 - Since edge $(u, v) \in H \cup H^R$, there exists a set $S_i^j \in \mathcal{F}_i$, where $i \in [0, z]$ and $j \in [1, k_i]$, such that $u, v \in S_i^j$.
 - By Property 2 of Claim 5.15, $w(u, v) = 2d_i$.
 - By Claim 5.12, set S_i^j has weak diameter at most $2d_i$ in G .

Then

$$\text{dist}_G(u, v) \leq 2d_i = w_D(u, v). \quad \square$$

We have shown that distances in our DAG cover are at least distances in G . We now establish the distortion upper bound guarantee of our DAG cover. Let D_1 and D_2 be the random DAGs

constructed in subsection 5.3, with associated set family $\{\mathcal{F}_i\}_i$ and graph family $\{G_i\}_i$. The key step in establishing our distortion upper bound will be to prove the following lemma.

Lemma 5.18. *For all $s, t \in V(G)$,*

$$\mathbb{E}[\min(\text{dist}_{D_1}(s, t), \text{dist}_{D_2}(s, t))] = O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Once we prove this lemma, our claimed distortion upper bound in Theorem 5.10 will follow from a simple application of Markov's inequality. The remainder of this section and section 5.3 will be devoted to developing the necessary claims and lemmas to prove Lemma 5.18; we will finally prove Lemma 5.18 in section 5.3.

First, we will show that paths in graph G_i have a natural structure with respect to the SCCs $\mathcal{F}_i$ of G_i .

Claim 5.19. *Let π be an $s \rightsquigarrow t$ path in G_i for some $s, t \in V(G)$ and $i \in [0, z]$. There exists a sequence of sets $S_1, \dots, S_k \subseteq \mathcal{F}_i$ in set family $\mathcal{F}_i$ with the following properties:*

1. *Path π contains exactly one edge $e_j = (u_j, v_j)$ of the form $e_j \in S_j \times S_{j+1}$ for all $j \in [1, k-1]$.*

Moreover,

- $\pi[s, u_1] \subseteq S_1$,
- $\pi[v_j, u_{j+1}] \subseteq S_{j+1}$ for all $j \in [1, k-2]$, and
- $\pi[v_{k-1}, t] \subseteq S_k$.

2. $S_{j_1} \neq S_{j_2}$ for all $j_1, j_2 \in [1, k]$ such that $j_1 \neq j_2$.

Proof. Recall that $\mathcal{F}_i$ is the set of all strongly connected components in G_i . Let $u, v \in \pi$ be nodes such that u comes before v in path π . Suppose that $u, v \in \pi \cap S$ for some $S \in \mathcal{F}_i$. This implies that $\pi[u, v] \subseteq S$ because S is an SCC in G_i . Then for every SCC $S \in \mathcal{F}_i$, set S intersects path π in a (possibly empty) contiguous subpath of π . Let sequence $S_1, \dots, S_k \subseteq \mathcal{F}_i$ be the SCCs in $\mathcal{F}_i$ that have nonempty intersection with π , written so that if $j_1 < j_2 \in [1, k]$ then subpath $\pi \cap S_{j_1}$ comes before subpath $\pi \cap S_{j_2}$ in path π . $\square$

We will now show that if nodes s and t are contained in the same set S in set family $\mathcal{F}_i$, then we can use the edges in H and H^R to upper bound the distances between s and t in D_1 and D_2 .

Claim 5.20. *Fix an index $i \in [0, z]$ and nodes $s, t \in S$ for some set $S \in \mathcal{F}_i$ in set family $\mathcal{F}_i$.*

- If $s <_D t$, then

$$\text{dist}_{D_1}(s, t) \leq 4d_i.$$

- if $s >_D t$, then

$$\text{dist}_{D_2}(s, t) \leq 4d_i.$$

Proof. This claim will follow directly from Claim 5.15 and our construction of DAGs D_1 and D_2 . Let $H_S \subseteq S \times S$ be the directed, weighted edges specified in Claim 5.15 with respect to set $S \in \mathcal{F}_i$. By construction, $H_S \subseteq E(D_1)$.

- If $s <_D t$, then

$$\text{dist}_{D_1}(s, t) \leq \text{dist}_{H_S}(s, t) \leq 4d_i,$$

by Property 3 of Claim 5.15.

- Otherwise, $s >_D t$. Let H_S^R be the set of weighted, directed edges obtained by reversing the orientations of the edges in H_S . By construction, $H_S^R \subseteq H^R \subseteq E(D_2)$. Then since $s >_D t$, we can again argue by Property 3 of Claim 5.15 that

$$\text{dist}_{D_2}(s, t) \leq \text{dist}_{H_S^R}(s, t) = \text{dist}_{H_S}(t, s) \leq 4d_i. \quad \square$$

For an event A , let $\mathbb{1}[A]$ denote the random variable that takes value 1 when event A holds, and value 0 otherwise. The following lemma makes some progress towards proving Lemma 5.18, by achieving an upper bound on the expected distortion between s and t in DAG D_2 , multiplied by the binary random variable $\mathbb{1}[s >_D t]$. Notice that if $s <_D t$, then $\text{dist}_{D_2}(s, t) = \infty$, so we cannot expect to get a reasonable upper bound on $\mathbb{E}[\text{dist}_{D_2}(s, t)]$ in general; this is why we upper bound $\mathbb{E}[\text{dist}_{D_2}(s, t) \cdot \mathbb{1}[s >_D t]]$ instead.

Lemma 5.21. *For all $s, t \in V(G)$,*

$$\mathbb{E}[\text{dist}_{D_2}(s, t) \cdot \mathbb{1}[s >_D t]] = O(\log^2 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Proof. Fix a shortest $s \rightsquigarrow t$ path π in G . Let X be the random variable $X = \text{dist}_{D_2}(s, t)$. Let A be the event that $s >_D t$. Let B_i be the event that path π is contained in graph G_i , for all $i \in [0, z]$.

Consider the scenario where event $A \cap B_i$ occurs, i.e., $s >_D t$, and path π is contained in G_i . Then there must exist an SCC $S \in \mathcal{F}_i$ of G_i such that $\pi \subseteq G_i[S]$. If this is not the case, then path π intersects with two or more SCCs in G_i , implying that $s <_D t$, a contradiction. Then we can apply

Claim 5.20 to argue that if event $A \cap B_i$ occurs, then $\text{dist}_{D_2}(s, t) \leq 4d_i$. In particular, this implies that

$$\mathbb{E}[X \mid A \cap B_i \cap \bar{B}_{i+1}] \leq 4d_i.$$

Observe that $\Pr[B_0] = 1$ since $\pi \subseteq G = G_0$. Then we can obtain the following inequality

$$\mathbb{E}[X \mid A] \leq \sum_{i=0}^z \mathbb{E}[X \mid A \cap B_i \cap \bar{B}_{i+1}] \Pr[B_i \cap \bar{B}_{i+1} \mid A].$$

Additionally, we observe that by Claim 5.12 and the union bound, for all $i \in [0, z]$,

$$\Pr[\bar{B}_{i+1} \mid B_i] \leq \sum_{e \in E(\pi)} \alpha \cdot \frac{w(e)}{d_i} = \frac{\alpha}{d_i} \cdot \text{dist}_G(s, t),$$

since π is an $s \rightsquigarrow t$ shortest path in G . Putting our observations together,

$$\begin{aligned} \mathbb{E}[X \cdot \mathbb{1}[s >_D t]] &= \mathbb{E}[X \mid A] \Pr[A] \\ &\leq \left(\sum_{i=0}^z \mathbb{E}[X \mid A \cap B_i \cap \bar{B}_{i+1}] \Pr[B_i \cap \bar{B}_{i+1} \mid A] \right) \Pr[A] \\ &= \sum_{i=0}^z \mathbb{E}[X \mid A \cap B_i \cap \bar{B}_{i+1}] \Pr[B_i \cap \bar{B}_{i+1} \cap A] \\ &\leq \sum_{i=0}^z \mathbb{E}[X \mid A \cap B_i \cap \bar{B}_{i+1}] \Pr[\bar{B}_{i+1} \mid B_i] \\ &\leq \sum_{i=0}^z 2d_i \cdot \Pr[\bar{B}_{i+1} \mid B_i] \\ &\leq \sum_{i=0}^z 2d_i \cdot \frac{\alpha}{d_i} \cdot \text{dist}_G(s, t) \\ &= (z+1) \cdot 2\alpha \cdot \text{dist}_G(s, t) \\ &= O(\log^2 n \cdot \log \log n) \cdot \text{dist}_G(s, t). \quad \square \end{aligned}$$

We will need the following technical lemma, which we prove using a similar argument as in Lemma 5.21. Recall that for any index $X \in [0, z]$, $d_X = nW2^{-X-1}$ is the diameter of the directed low-diameter decomposition that we apply to graph G_X . This lemma roughly states that an (s, t) -shortest path π survives to diameter $\tilde{O}(1) \cdot \text{dist}_G(s, t)$ in expectation.

Lemma 5.22. *Let $s, t \in V(G)$ be nodes such that s can reach t in G . Let π be an $s \rightsquigarrow t$ shortest*

path in G . Let $H \subseteq G$ be a subgraph of G such that $\pi \subseteq H$. For each $i \in [0, z]$ and subgraph $J \subseteq G$, let C_i^J be the event that $G_i = J$. Let $X \in [0, z]$ be the (random-valued) index such that $\pi \subseteq G_X$ and $\pi \not\subseteq G_{X+1}$. For each $i \in [0, z]$,

$$\mathbb{E}[d_X | C_i^H] \leq (z - i + 1) \cdot \alpha \cdot \text{dist}_G(s, t),$$

where $\alpha = O(\log n \cdot \log \log n)$ is the coefficient from the directed LDD in Theorem 5.11.

Proof. Fix a pair of nodes $s, t \in V(G)$ such that s can reach t in G . Let π be an $s \rightsquigarrow t$ shortest path in G , and let $H \subseteq G$ be a subgraph of G such that $\pi \subseteq H$. We will prove Lemma 5.22 for a fixed index $i \in [0, z]$.

For each $j \in [0, z]$, let B_j be the event that $\pi \subseteq G_j$. We can obtain the following equality:

$$\mathbb{E}[d_X | C_i^H] = \sum_{j=i}^z d_j \cdot \Pr[B_j \cap \bar{B}_{j+1} | C_i^H].$$

Additionally, we observe that by Claim 5.12 and the union bound, for all $j \in [i, z]$,

$$\Pr[\bar{B}_{j+1} | B_j \cap C_i^H] \leq \sum_{e \in E(\pi)} \frac{w(e) \cdot \alpha}{d_j} = \frac{\alpha}{d_j} \cdot \text{dist}_G(s, t),$$

since π is an $s \rightsquigarrow t$ shortest path in G . Putting our observations together,

$$\begin{aligned} \mathbb{E}[d_X | C_i^H] &= \sum_{j=i}^z d_j \cdot \Pr[B_j \cap \bar{B}_{j+1} | C_i^H] \\ &= \sum_{j=i}^z d_j \cdot \Pr[\bar{B}_{j+1} | B_j \cap C_i^H] \Pr[B_j | C_i^H] \\ &\leq \sum_{j=i}^z d_j \cdot \Pr[\bar{B}_{j+1} | B_j \cap C_i^H] \\ &\leq \sum_{j=i}^z d_j \cdot \frac{\alpha}{d_j} \cdot \text{dist}_G(s, t) \\ &\leq (z - i + 1) \cdot \alpha \cdot \text{dist}_G(s, t). \end{aligned}$$

□

In Lemma 5.21, we gave an upper bound on the expectation of random variable $\text{dist}_{D_2}(s, t) \cdot \mathbb{1}[s >_D t]$, for a pair of nodes $s, t \in V(G)$. This proof made heavy use of the fact that this random

variable took value 0 when $s <_D t$. Our next goal will be to give an upper bound on the expectation of the random variable $\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]$. We will need to introduce some additional notation first.

Additional Notation. For all nodes $s, t \in V(G)$ such that s can reach t in G (i.e., $(s, t) \in TC(G)$), we fix an $s \rightsquigarrow t$ shortest path $\pi_{s,t}$ in G . We may assume without loss of generality that our collection of paths $\{\pi_{s,t}\}_{(s,t) \in TC(G)}$ is *consistent*. Formally, this means that for any two paths $\pi_{s,t}$ and $\pi_{s',t'}$ in our collection, and for any two vertices $u, v \in V(G)$, if $u, v \in \pi_{s,t} \cap \pi_{s',t'}$ and node u precedes node v in both $\pi_{s,t}$ and $\pi_{s',t'}$, then we have that $\pi_{s,t}[u, v] = \pi_{s',t'}[u, v]$.

For all nodes $s, t \in V(G)$ such that s can reach t in G , and for each $i \in [0, z]$, let $B_i^{s,t}$ be the event that path $\pi_{s,t}$ is contained in G_i . For every subgraph $H \subseteq G$, let C_i^H be the event that $G_i = H$.

The following lemma formally states what we will prove about the expected size of the random variable $\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]$.

Lemma 5.23. *Let $s, t \in V(G)$, and let $i \in [0, z]$. Let $H \subseteq G$ be a subgraph of G such that $\pi_{s,t} \subseteq H$. Then*

$$\mathbb{E} [\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot (z - i + 1)^2 \cdot \alpha \cdot \text{dist}_G(s, t),$$

for a sufficiently large constant $c > 1$.

We will prove Lemma 5.23 by an induction argument in section 5.3. After proving Lemma 5.23, we finish the proof of Theorem 5.10 in section 5.3.

Induction proof of Lemma 5.23

In this section we will prove Lemma 5.23 using an induction argument. Recall that for all nodes $s, t \in V(G)$ such that s can reach t in G , and for each $i \in [0, z]$, $B_i^{s,t}$ denotes the event that path $\pi_{s,t}$ is contained in G_i , where $\pi_{s,t}$ is a $s \rightsquigarrow t$ shortest path in G specified in the previously. Likewise, for every subgraph $H \subseteq G$, C_i^H denotes the event that $G_i = H$.

Lemma 5.24. *Let $s, t \in V(G)$, and let $i \in [0, z]$. Let $H \subseteq G$ be a subgraph of G such that $\pi_{s,t} \subseteq H$. Then*

$$\mathbb{E} [\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot (z - i + 1)^2 \cdot \alpha \cdot \text{dist}_G(s, t),$$

for a sufficiently large constant $c > 1$.

We will prove Lemma 5.23 by induction on $k = z - i + 1$, starting with $k = 1$ and incrementing to $k = z + 1$. Let $c > 1$ denote the induction constant claimed in Lemma 5.23. We will specify

the precise value of c when we finish the induction step of our proof in Lemma 5.26. For each $k \in [1, z + 1]$, we define an induction statement S_k as follows.

Induction Statement S_k . *Let $i = z - k + 1$. Let $s, t \in V(G)$. Let $H \subseteq G$ be a subgraph of G such that $\pi_{s,t} \subseteq H$. Then*

$$\mathbb{E} [\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t).$$

We can prove statement S_1 with relative ease.

Claim 5.25 (Base Case: S_1). *Induction Statement S_1 is true.*

Proof. When $k = 1$, we have that $i = z - k + 1 = z$. If event C_i^H holds, then path $\pi_{s,t}$ is contained in graph $H = G_z = D^*$. Since we added the edges in $E(D^*)$ to DAG D_1 with their original edge weights, it follows that

$$\text{dist}_{D_1}(s, t) \leq \text{dist}_{D^*}(s, t) = \text{dist}_G(s, t).$$

We conclude that

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_z^H] \leq \mathbb{E}[\text{dist}_{D_1}(s, t) \mid C_z^H] \leq \text{dist}_G(s, t) \leq c \cdot \alpha \cdot \text{dist}_G(s, t),$$

as desired. □

For the remainder of this section, we will let k and i be fixed indices such that $k \in [2, z + 1]$ and $i = z - k + 1 \in [0, z - 1]$. Additionally, for the remainder of this section we will assume that Induction Statement S_{k-1} holds. By the end of this section, we will prove Lemma 5.26, which states that if S_{k-1} is true, then S_k is true as well.

Lemma 5.26. *If statement S_{k-1} holds, then statement S_k holds as well.*

Proving Lemma 5.26 will complete the inductive step of our proof of Lemma 5.23. The remainder of this section will be devoted towards eventually proving Lemma 5.26.

Proving Lemma 5.26. Our goal is to prove the upper bound

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t)$$

claimed in statement S_k , completing the proof of Lemma 5.26. As a warm up, we will prove that statement S_k holds in the special case where s and t lie in the same SCC in H .

Lemma 5.27. *Let $s, t \in V(G)$, and let $H \subseteq G$ be a subgraph of G such that $\pi_{s,t} \subseteq H$ and nodes s and t lie in the same SCC in H . If statement S_{k-1} holds, then*

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq (c \cdot (z - i)^2 + c_1) \cdot \alpha \cdot \text{dist}_G(s, t),$$

where $c_1 \geq 1$ is a sufficiently large universal constant that does not depend on the induction constant c . In particular, if $c \geq c_1$, then it follows that

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t).$$

Proof. We define a random variable X to be

$$X = \text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t].$$

We will make use of the following equality in our proof.

$$\begin{aligned} X &= \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] = \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] (\mathbb{1}[B_{i+1}^{s,t}] + \mathbb{1}[\bar{B}_{i+1}^{s,t}]) \\ &= \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] \mathbb{1}[B_{i+1}^{s,t}] + \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] \mathbb{1}[\bar{B}_{i+1}^{s,t}] \quad (5.1) \end{aligned}$$

Let $Y_1 = \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] \mathbb{1}[B_{i+1}^{s,t}]$, and let $Y_2 = \text{dist}_{D_1}(s, t) \mathbb{1}[s <_D t] \mathbb{1}[\bar{B}_{i+1}^{s,t}]$. Since $X = Y_1 + Y_2$, we can upper bound $\mathbb{E}[X \mid C_i^H]$ by upper bounding $\mathbb{E}[Y_1 \mid C_i^H]$ and $\mathbb{E}[Y_2 \mid C_i^H]$ separately.

We begin by upper bounding $\mathbb{E}[Y_1 \mid C_i^H]$. We will achieve this by applying the induction statement S_{k-1} . We observe the following sequence of inequalities:

$$\begin{aligned} &\mathbb{E}[Y_1 \mid C_i^H] \\ &= \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \cdot \mathbb{1}[B_{i+1}^{s,t}] \mid C_i^H] \\ &\leq \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid B_{i+1}^{s,t} \cap C_i^H]. \end{aligned}$$

We are almost ready to apply the induction hypothesis. Let $\mathcal{H}$ denote the family of all subgraphs $H' \subseteq H$ of H such that $\pi_{s,t} \subseteq H'$. Let $H^* \in \mathcal{H}$ be the graph in $\mathcal{H}$ that maximizes $\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_{i+1}^{H^*}]$, i.e., let

$$H^* = \operatorname{argmax}_{H' \in \mathcal{H}} \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_{i+1}^{H'}].$$

Notice that if event $B_{i+1}^{s,t} \cap C_i^H$ occurs, then there exists a subgraph $H' \in \mathcal{H}$ such that event

$C_{i+1}^{H'}$ occurs as well. Then by our choice of graph H^* , we obtain the following inequality.

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid B_{i+1}^{s,t} \cap C_i^H] \leq \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_{i+1}^{H^*}] \quad (5.2)$$

Continuing with our previous sequence of inequalities,

$$\begin{aligned} \mathbb{E}[Y_1 \mid C_i^H] &\leq \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid B_{i+1}^{s,t} \cap C_i^H] \\ &\leq \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_{i+1}^{H^*}] && \text{by inequality 5.2} \\ &\leq c \cdot (z - i)^2 \cdot \alpha \cdot \text{dist}_G(s, t). && \text{by statement } S_{k-1} \end{aligned}$$

We conclude from the above sequence of inequalities that

$$\mathbb{E}[Y_1 \mid C_i^H] \leq c \cdot (z - i)^2 \cdot \alpha \cdot \text{dist}_G(s, t). \quad (5.3)$$

Next we upper bound $\mathbb{E}[Y_2 \mid C_i^H]$. We will achieve this using Claim 5.20. We will also use of the following inequality which follows from the fact that $\pi_{s,t} \subseteq H$, combined with Claim 5.12 and the union bound.

$$\Pr[\bar{B}_{i+1}^{s,t} \mid C_i^H] \leq \sum_{e \in E(\pi_{s,t})} \frac{w(e) \cdot \alpha}{d_i} = \frac{\alpha}{d_i} \cdot \text{dist}_G(s, t) \quad (5.4)$$

Let A be the event that $s <_D t$. We observe the following sequence of inequalities:

$$\begin{aligned} \mathbb{E}[Y_2 \mid C_i^H] &= \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[A] \cdot \mathbb{1}[\bar{B}_{i+1}^{s,t}] \mid C_i^H] \\ &= \mathbb{E}[\text{dist}_{D_1}(s, t) \mid A \cap \bar{B}_{i+1}^{s,t} \cap C_i^H] \cdot \Pr[\bar{B}_{i+1}^{s,t} \mid A \cap C_i^H] \cdot \Pr[A \mid C_i^H] \\ &= \mathbb{E}[\text{dist}_{D_1}(s, t) \mid A \cap \bar{B}_{i+1}^{s,t} \cap C_i^H] \cdot \Pr[\bar{B}_{i+1}^{s,t} \cap A \mid C_i^H] \\ &\leq \mathbb{E}[\text{dist}_{D_1}(s, t) \mid A \cap \bar{B}_{i+1}^{s,t} \cap C_i^H] \cdot \Pr[\bar{B}_{i+1}^{s,t} \mid C_i^H] \\ &\leq 4d_i \cdot \Pr[\bar{B}_{i+1}^{s,t} \mid C_i^H] && \text{by Claim 5.20} \\ &\leq 4d_i \cdot \frac{\alpha}{d_i} \cdot \text{dist}_G(s, t) && \text{by inequality 5.4} \\ &= 4\alpha \cdot \text{dist}_G(s, t). \end{aligned}$$

We conclude from the above sequence of inequalities that

$$\mathbb{E}[Y_2 \mid C_i^H] \leq c_1 \cdot \alpha \cdot \text{dist}_G(s, t), \quad (5.5)$$

for $c_1 = 4$. Putting everything together, we get

$$\begin{aligned}\mathbb{E}[X \mid C_i^H] &= \mathbb{E}[Y_1 \mid C_i^H] + \mathbb{E}[Y_2 \mid C_i^H] && \text{by equation 5.1} \\ &\leq (c \cdot (z - i)^2 + c_1) \cdot \alpha \cdot \text{dist}_G(s, t). && \text{by inequalities 5.3 and 5.5}\end{aligned}$$

In particular, if $c \geq c_1$, then it follows that

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot (z - i + 1)^2 \cdot \alpha \cdot \text{dist}_G(s, t) = c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t).$$

□

By Lemma 5.27, if nodes s and t lie in the same SCC in subgraph $H \subseteq G$, then statement S_k holds. Therefore, when proving Lemma 5.26 we may assume without loss of generality that s and t do not lie in the same SCC in graph H . For the remainder of this section, fix a pair of nodes $s, t \in V(G)$, and let $H \subseteq G$ be a fixed subgraph of G such that $\pi_{s,t} \subseteq H$ and nodes s and t do not lie in the same SCC in H . We define a random variable X to be

$$X = \text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t].$$

We recall that our goal in proving Lemma 5.26 is to show that

$$\mathbb{E}[X \mid C_i^H] \leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t).$$

Suppose that event C_i^H holds. Let $S_1, \dots, S_q \in \mathcal{F}_i$, $q > 1$, be the sequence of SCCs of H described in Claim 5.19 with respect to path $\pi_{s,t}$ in graph H . Let $\{(u_j, v_j)\}_{j \in [1, q-1]}$ be the associated set of edges, where edge (u_j, v_j) is the unique edge in the intersection $E(\pi_{s,t}) \cap (S_j \times S_{j+1})$. We observe that $(u_j, v_j) \in E(D^*) \subseteq E(D_1)$ for all $j \in [1, q-1]$.

We define vertices $x_j, y_j \in V(G)$ for all $j \in [0, q-2]$. Let $x_0 = s$ and $y_{q-2} = t$. For each $j \in [1, q-2]$, let $y_{j-1} = v_j$ and let $x_j = v_j$, so that $x_j = y_{j-1}$. See Figure 5.2 for a visualization of path $\pi_{s,t}$, the SCCs $\{S_i\}_i$, the edges $\{(u_i, v_i)\}_i$, and the vertices $\{x_i, y_i\}_i$.

For each $j \in [0, q-2]$, let X_j be the random variable

$$X_j = \text{dist}_{D_1}(x_j, y_j).$$

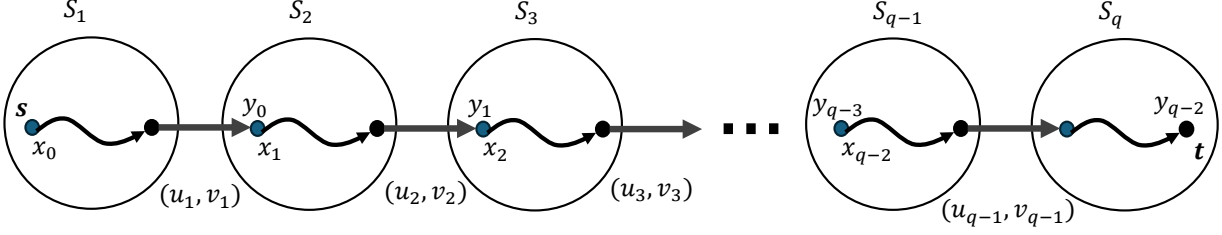


Figure 5.2: A visualization of path $\pi_{s,t}$ in subgraph G_i of G . We upper bound the distance from s to t in DAG D_1 by upper bounding the distance from x_i to y_i in DAG D_1 for all $i \in [1, q-2]$.

We will use the following upper bound on $\mathbb{E}[X \mid C_i^H]$ in our proof of Lemma 5.26.

$$\mathbb{E}[X \mid C_i^H] \leq \mathbb{E} \left[\sum_{j=0}^{q-2} \text{dist}_{D_1}(x_j, y_j) \cdot \mathbb{1}[s <_D t] \mid C_i^H \right] \leq \sum_{j=0}^{q-2} \mathbb{E}[X_j \mid C_i^H] \quad (5.6)$$

Inequality 5.6 tells us that to upper bound $\mathbb{E}[X \mid C_i^H]$, it is sufficient to upper bound $\mathbb{E}[X_j \mid C_i^H]$ for each $j \in [0, q-2]$. This will be our next goal.

Lemma 5.28. *If statement S_{k-1} holds for a sufficiently large induction constant c , then for each $j \in [0, q-2]$,*

$$\mathbb{E}[X_j \mid C_i^H] \leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(x_j, y_j)$$

Proof. Fix an index $j \in [0, q-2]$. To upper bound $\mathbb{E}[X_j \mid C_i^H]$, we essentially need to handle four different cases based on the ordering of vertices x_j, u_{j+1}, v_{j+1} , and y_j under $<_D$. We briefly outline these four cases and the different inequalities we will use to handle them, before proceeding with the formal upper bound.

Let $\pi_1 := \pi_{x_j, u_{j+1}} = \pi_{s,t}[x_j, u_{j+1}]$, and let $\pi_2 := \pi_{v_{j+1}, y_j} = \pi_{s,t}[v_{j+1}, y_j]$ (these equalities follow from the consistency of our collection of paths $\{\pi_{s,t}\}_{(s,t) \in TC(G)}$). Let $Z_1 \in [0, z]$ denote the (random-valued) index such that $\pi_1 \subseteq G_{Z_1}$ and $\pi_1 \not\subseteq G_{Z_1+1}$. Let $Z_2 \in [0, z]$ denote the (random-valued) index such that $\pi_2 \subseteq G_{Z_2}$ and $\pi_2 \not\subseteq G_{Z_2+1}$. Recall that we use $x \leq_D y$ to denote that either $x <_D y$ or $x = y$.

- **Case 1:** $x_j \leq_D u_{j+1}$ and $v_{j+1} \leq_D y_j$. In this case we will apply the following triangle inequality

$$\text{dist}_{D_1}(x_j, y_j) \leq \text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_{D_1}(u_{j+1}, v_{j+1}) + \text{dist}_{D_1}(v_{j+1}, y_j).$$

Since $(u_{j+1}, v_{j+1}) \in E(D_1)$ and $(u_{j+1}, v_{j+1}) \in \pi_{s,t}$, it follows that

$$\text{dist}_{D_1}(u_{j+1}, v_{j+1}) = w_G((u_{j+1}, v_{j+1})) = \text{dist}_G(u_{j+1}, v_{j+1}).$$

We conclude that

$$X_j \leq \text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) + \text{dist}_{D_1}(v_{j+1}, y_j). \quad (5.7)$$

We will use inequality 5.7 to handle the case where $x_j \leq_D u_{j+1}$ and $v_{j+1} \leq_D y_j$.

- **Case 2:** $x_j \leq_D u_{j+1}$ **and** $v_{j+1} >_D y_j$. In this case we will apply the following triangle inequality

$$\text{dist}_{D_1}(x_j, y_j) \leq \text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_{D_1}(u_{j+1}, r_{Z_2}^1(v_{j+1})) + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j).$$

We will make use of the fact that

$$\text{dist}_{D_1}(u_{j+1}, r_{Z_2}^1(v_{j+1})) \leq w_{D_1}((u_{j+1}, r_{Z_2}^1(v_{j+1}))) = \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_2},$$

since $(u_{j+1}, r_{Z_2}^1(v_{j+1})) \in E(D_1)$, to obtain the following inequality:

$$X_j \leq \text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_2} + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j). \quad (5.8)$$

We will use inequality 5.8 to handle the case where $x_j \leq_D u_{j+1}$ and $v_{j+1} >_D y_j$.

- **Case 3:** $x_j >_D u_{j+1}$ **and** $v_{j+1} \leq_D y_j$. In this case we will apply the following triangle inequality

$$\text{dist}_{D_1}(x_j, y_j) \leq \text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + \text{dist}_{D_1}(r_{Z_1}^2(u_{j+1}), v_{j+1}) + \text{dist}_{D_1}(v_{j+1}, y_j).$$

We will make use of the fact that $\text{dist}_{D_1}(r_{Z_1}^2(u_{j+1}), v_{j+1}) \leq \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_1}$ since $(r_{Z_1}^2(u_{j+1}), v_{j+1}) \in E(D_1)$, to obtain the following inequality

$$X_j \leq \text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1} + \text{dist}_G(u_{j+1}, v_{j+1}) + \text{dist}_{D_1}(v_{j+1}, y_j). \quad (5.9)$$

We will use inequality 5.9 to handle the case where $x_j >_D u_{j+1}$ and $v_{j+1} \leq_D y_j$.

- **Case 4:** $x_j >_D u_{j+1}$ **and** $v_{j+1} >_D y_j$. In this case we will apply the following triangle

inequality

$$\text{dist}_{D_1}(x_j, y_j) \leq \text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + \text{dist}_{D_1}(r_{Z_1}^2(u_{j+1}), r_{Z_2}^1(v_{j+1})) + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j).$$

We will make use of the fact that $\text{dist}_{D_1}(r_{Z_1}^2(u_{j+1}), r_{Z_2}^1(v_{j+1})) \leq \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_1} + d_{Z_2}$ since $(r_{Z_1}^2(u_{j+1}), r_{Z_2}^1(v_{j+1})) \in E(D_1)$, to obtain the following inequality

$$X_j \leq \text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1} + \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_2} + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j). \quad (5.10)$$

We will use inequality 5.10 to handle the case where $x_j >_D u_{j+1}$ and $v_{j+1} >_D y_j$.

In order to apply our four case analysis formally, we will use the following equality, which can be proved using the fact that $\mathbb{1}[x \leq_D y] + \mathbb{1}[x >_D y] = 1$ for any $x, y \in V(G)$.

$$\begin{aligned} X_j &= X_j \mathbb{1}[x_j \leq_D u_{j+1}] \mathbb{1}[v_{j+1} \leq_D y_j] + X_j \mathbb{1}[x_j \leq_D u_{j+1}] \mathbb{1}[v_{j+1} >_D y_j] \\ &\quad + X_j \mathbb{1}[x_j >_D u_{j+1}] \mathbb{1}[v_{j+1} \leq_D y_j] + X_j \mathbb{1}[x_j >_D u_{j+1}] \mathbb{1}[v_{j+1} >_D y_j] \end{aligned} \quad (5.11)$$

Inserting inequalities 5.7, 5.8, 5.9, and 5.10 into equation 5.11, we obtain

$$\begin{aligned} X_j &\leq (\text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) + \text{dist}_{D_1}(v_{j+1}, y_j)) \mathbb{1}[x_j \leq_D u_{j+1}] \mathbb{1}[v_{j+1} \leq_D y_j] \\ &+ \left(\text{dist}_{D_1}(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_2} + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) \mathbb{1}[x_j \leq_D u_{j+1}] \right) \mathbb{1}[v_{j+1} >_D y_j] \\ &+ \left(\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_1} + \text{dist}_{D_1}(v_{j+1}, y_j) \right) \mathbb{1}[x_j >_D u_{j+1}] \mathbb{1}[v_{j+1} \leq_D y_j] \\ &\quad + \left(\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + \text{dist}_G(u_{j+1}, v_{j+1}) + d_{Z_1} + d_{Z_2} + \text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) \right) \\ &\quad \cdot \mathbb{1}[x_j >_D u_{j+1}] \mathbb{1}[v_{j+1} >_D y_j]. \end{aligned} \quad (5.12)$$

We can rearrange terms on the right side of inequality 5.12 to get

$$\begin{aligned} X_j &\leq \text{dist}_{D_1}(x_j, u_{j+1}) \mathbb{1}[x_j \leq_D u_{j+1}] \cdot (\mathbb{1}[v_{j+1} \leq_D y_j] + \mathbb{1}[v_{j+1} >_D y_j]) \\ &\quad + \text{dist}_{D_1}(v_{j+1}, y_j) \mathbb{1}[v_{j+1} \leq_D y_j] \cdot (\mathbb{1}[x_j \leq_D u_{j+1}] + \mathbb{1}[x_j >_D u_{j+1}]) \\ &+ \text{dist}_G(u_{j+1}, v_{j+1}) \cdot (\mathbb{1}[x_j \leq_D u_{j+1}] + \mathbb{1}[x_j >_D u_{j+1}]) \cdot (\mathbb{1}[v_{j+1} \leq_D y_j] + \mathbb{1}[v_{j+1} >_D y_j]) \\ &\quad + \left((\text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) + d_{Z_2}) \mathbb{1}[v_{j+1} >_D y_j] \right) \cdot (\mathbb{1}[x_j \leq_D u_{j+1}] + \mathbb{1}[x_j >_D u_{j+1}]) \\ &+ \left((\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1}) \mathbb{1}[x_j >_D u_{j+1}] \right) \cdot (\mathbb{1}[v_{j+1} \leq_D y_j] + \mathbb{1}[v_{j+1} >_D y_j]). \end{aligned} \quad (5.13)$$

We can simplify inequality 5.13 using the fact that for any $x, y \in V(G)$, $\mathbb{1}[x \leq_D y] + \mathbb{1}[x >_D y] = 1$.

$$\begin{aligned}
X_j &\leq \text{dist}_{D_1}(x_j, u_{j+1}) \mathbb{1}[x_j <_D u_{j+1}] \\
&\quad + \text{dist}_{D_1}(v_{j+1}, y_j) \mathbb{1}[v_{j+1} <_D y_j] \\
&\quad + \text{dist}_G(u_{j+1}, v_{j+1}) \\
&\quad + (\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1}) \mathbb{1}[x_j >_D u_{j+1}] \\
&\quad + (\text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) + d_{Z_2}) \mathbb{1}[v_{j+1} >_D y_j]. \quad (5.14)
\end{aligned}$$

Let W_1, W_2, W_3, W_4, W_5 be the five terms on the right side of inequality 5.14, written in the order they appear. To finish the proof of Lemma 5.28, it will be sufficient to upper bound $\mathbb{E}[W_p \mid C_i^H]$ for each $p \in [1, 5]$.

We will upper bound the conditional expectations of W_1 and W_2 by appealing to Lemma 5.27. We will focus on W_1 first. Notice that $\pi_1 = \pi_{x_j, u_{j+1}} \subseteq \pi_{s,t} \subseteq H$ and nodes x_j and u_{j+1} lie in the same SCC in H . Then we can directly apply Lemma 5.27 to obtain the inequality

$$\mathbb{E}[W_1 \mid C_i^H] = \mathbb{E}[\text{dist}_{D_1}(x_j, u_{j+1}) \cdot \mathbb{1}[x_j <_D u_{j+1}] \mid C_i^H] \leq (c \cdot (z - i)^2 + c_1) \cdot \alpha \cdot \text{dist}_G(x_j, u_{j+1}), \quad (5.15)$$

where $c_1 \geq 1$ is a universal constant that does not depend on induction constant c . Likewise, since $\pi_2 = \pi_{v_{j+1}, y_j} \subseteq \pi_{s,t} \subseteq H$ and nodes v_{j+1} and y_j lie in the same SCC in H , we directly apply Lemma 5.27 to prove the following inequality.

$$\mathbb{E}[W_2 \mid C_i^H] = \mathbb{E}[\text{dist}_{D_1}(v_{j+1}, y_j) \cdot \mathbb{1}[v_{j+1} <_D y_j] \mid C_i^H] \leq (c \cdot (z - i)^2 + c_1) \cdot \alpha \cdot \text{dist}_G(v_{j+1}, y_j) \quad (5.16)$$

Now we will upper bound the conditional expectations of W_4 and W_5 by appealing to Lemma 5.22. We will focus on W_4 first. Let A be the event that $x_j >_D u_{j+1}$. Suppose that event $A \cap C_i^H$ occurs. Recall that Z_1 is the random-valued index such that $\pi_1 \subseteq G_{Z_1}$ and $\pi_1 \not\subseteq G_{Z_1+1}$. Since $x_j >_D u_{j+1}$, there must exist an SCC $S \in \mathcal{F}_{Z_1}$ of G_{Z_1} such that $x_j, u_{j+1} \in S$; if this were not true, then since $x_j \rightsquigarrow u_{j+1}$ path π_1 satisfies $\pi_1 \subseteq G_{Z_1}$ it would imply that $x_j <_D u_{j+1}$, a contradiction. In particular, this means that $r_{Z_1}^2(u_{j+1}) \in S$. By the above discussion and Claim 5.20, we obtain the inequality

$$\mathbb{E}[\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) \mid A \cap C_i^H] \leq \mathbb{E}[4d_{Z_1} \mid A \cap C_i^H]. \quad (5.17)$$

We observe the following sequence of inequalities:

$$\begin{aligned}
& \mathbb{E}[W_4 \mid C_i^H] \\
&= \mathbb{E}[(\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1}) \mathbb{1}[x_j >_D u_{j+1}] \mid C_i^H] \\
&= \mathbb{E}[\text{dist}_{D_1}(x_j, r_{Z_1}^2(u_{j+1})) + d_{Z_1} \mid A \cap C_i^H] \cdot \Pr[A \mid C_i^H] \\
&\leq \mathbb{E}[5d_{Z_1} \mid A \cap C_i^H] \cdot \Pr[A \mid C_i^H] && \text{by inequality 5.17} \\
&\leq 5\mathbb{E}[d_{Z_1} \mid C_i^H] \\
&\leq c_2 \cdot (z - i + 1) \cdot \alpha \cdot \text{dist}_G(x_j, u_{j+1}), && \text{by Lemma 5.22}
\end{aligned}$$

where $c_2 \geq 1$ is a sufficiently large universal constant that does not depend on the induction constant c . We have shown

$$\mathbb{E}[W_4 \mid C_i^H] \leq c_2 \cdot (z - i + 1) \cdot \alpha \cdot \text{dist}_G(x_j, u_{j+1}). \quad (5.18)$$

An identical argument will work for bounding $\mathbb{E}[W_5 \mid C_i^H]$ as well; we present this argument for completeness. Let A be the event that $v_{j+1} >_D y_j$. Suppose that event $A \cap C_i^H$ occurs. Recall that Z_2 is the random-valued index such that $\pi_2 \subseteq G_{Z_2}$ and $\pi_2 \not\subseteq G_{Z_2+1}$. Since $v_{j+1} >_D y_j$, there must exist an SCC $S \in \mathcal{F}_{Z_2}$ of G_{Z_2} such that $v_{j+1}, y_j \in S$; if this were not true, then since $v_{j+1} \rightsquigarrow y_j$ path π_2 satisfies $\pi_2 \subseteq G_{Z_2}$ it would imply that $v_{j+1} <_D y_j$, a contradiction. In particular, this means that $r_{Z_2}^1(v_{j+1}) \in S$. By the above discussion and Claim 5.20, we obtain the inequality

$$\mathbb{E}[\text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) \mid A \cap C_i^H] \leq \mathbb{E}[4d_{Z_2} \mid A \cap C_i^H]. \quad (5.19)$$

We observe the following sequence of inequalities:

$$\begin{aligned}
& \mathbb{E}[W_5 \mid C_i^H] \\
&= \mathbb{E}[(\text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) + d_{Z_2}) \mathbb{1}[v_{j+1} >_D y_j] \mid C_i^H] \\
&= \mathbb{E}[\text{dist}_{D_1}(r_{Z_2}^1(v_{j+1}), y_j) + d_{Z_2} \mid A \cap C_i^H] \cdot \Pr[A \mid C_i^H] \\
&\leq \mathbb{E}[5d_{Z_2} \mid A \cap C_i^H] \cdot \Pr[A \mid C_i^H] && \text{by inequality 5.19} \\
&\leq 5\mathbb{E}[d_{Z_2} \mid C_i^H] \\
&\leq c_2 \cdot (z - i + 1) \cdot \alpha \cdot \text{dist}_G(v_{j+1}, y_j), && \text{by Lemma 5.22}
\end{aligned}$$

where $c_2 \geq 1$ is a sufficiently large universal constant that does not depend on the induction constant c . We have shown

$$\mathbb{E}[W_5 \mid C_i^H] \leq c_2 \cdot (z - i + 1) \cdot \alpha \cdot \text{dist}_G(v_{j+1}, y_j). \quad (5.20)$$

We are ready to finish the proof of Lemma 5.28. We will make use of the fact that

$$\text{dist}_G(x_j, y_j) = \text{dist}_G(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) + \text{dist}_G(v_{j+1}, y_j), \quad (5.21)$$

since path $\pi_{s,t}$ is a $s \rightsquigarrow t$ shortest path in G . We complete the proof of Lemma 5.28 with the following sequence of inequalities.

$$\begin{aligned} \mathbb{E}[X_j \mid C_i^H] &\leq \sum_{p=1}^5 \mathbb{E}[W_p \mid C_i^H] && \text{by inequality 5.14} \\ &\leq \sum_{p \in \{1,2,4,5\}} \mathbb{E}[W_p \mid C_i^H] + \text{dist}_G(u_{j+1}, v_{j+1}) && \text{since } W_3 = \text{dist}_G(u_{j+1}, v_{j+1}) \end{aligned}$$

Applying inequalities 5.15, 5.16, 5.18, and 5.20, we continue

$$\begin{aligned} \mathbb{E}[X_j \mid C_i^H] &\leq (c \cdot (z-i)^2 + c_1 + c_2(z-i+1)) \cdot \alpha \cdot \text{dist}_G(x_j, u_{j+1}) + \text{dist}_G(u_{j+1}, v_{j+1}) \\ &\quad + (c \cdot (z-i)^2 + c_1 + c_2(z-i+1)) \cdot \alpha \cdot \text{dist}_G(v_{j+1}, y_j). \end{aligned}$$

Using equation 5.21, this yields

$$\begin{aligned} \mathbb{E}[X_j \mid C_i^H] &\leq (c \cdot (z-i)^2 + c_1 + c_2(z-i+1)) \cdot \alpha \cdot \text{dist}_G(x_j, y_j) && \text{by equation 5.21} \\ &\leq (c \cdot (z-i)^2 + (c_1 + c_2)(z-i+1)) \cdot \alpha \cdot \text{dist}_G(x_j, y_j) \\ &\leq c \cdot (z-i+1)^2 \cdot \alpha \cdot \text{dist}_G(x_j, y_j) && \text{by letting } c \geq c_1 + c_2 \\ &\leq c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(x_j, y_j). \end{aligned}$$

Notice that we require induction constant c to be at least $c \geq c_1 + c_2$, where c_1 is the constant in the statement of Lemma 5.27, and c_2 is the constant in inequality 5.18. $\square$

With Lemma 5.28 proved, we can now complete the inductive step of our induction proof of Lemma 5.23.

Lemma 5.29. *If statement S_{k-1} holds, then statement S_k holds as well.*

Proof. We are ready to finish our proof of the inductive step by combining all of our proven

inequalities.

$$\begin{aligned}
\mathbb{E}[X \mid C_i^H] &\leq \sum_{j=0}^{q-2} \mathbb{E}[X_j \mid C_i^H] && \text{by inequality 5.6} \\
&\leq \left(\sum_{j=0}^{q-2} c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(x_j, y_j) \right) && \text{by Lemma 5.28} \\
&= c \cdot k^2 \cdot \alpha \cdot \text{dist}_G(s, t),
\end{aligned}$$

where the last equality follows from the fact that $\text{dist}_G(s, t) = \sum_{j=0}^{q-2} \text{dist}_G(x_j, y_j)$, since $\pi_{s,t}$ is a shortest path in G . Note that Lemma 5.28 requires that the induction constant c in statements S_{k-1} and S_k has size at least $c \geq c_1 + c_2$, where c_1 is the constant in the statement of Lemma 5.27, and c_2 is the constant in inequality 5.18. We have completed the inductive step of the proof of Lemma 5.23. $\square$

By Claim 5.25 and Lemma 5.26 we have proven that Inductive Statement S_k holds for all $k \in [1, z + 1]$. This immediately implies Lemma 5.23.

Lemma 5.30. *Let $s, t \in V(G)$, and let $i \in [0, z]$. Let $H \subseteq G$ be a subgraph of G such that $\pi_{s,t} \subseteq H$. Then*

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_i^H] \leq c \cdot (z - i + 1)^2 \cdot \alpha \cdot \text{dist}_G(s, t),$$

for a sufficiently large constant $c > 1$.

Finishing the proof of Theorem 5.10

With Lemma 5.23 in hand, we can now easily finish our proof of Theorem 5.10.

Claim 5.31. *For all $s, t \in V(G)$,*

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]] = O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Proof. Recall that C_0^G is the event that $G_0 = G$. Notice that by our construction, $G_0 = G$ and shortest path $\pi_{s,t}$ satisfies $\pi_{s,t} \subseteq G_0$, unconditionally. Then by Lemma 5.23,

$$\mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]] = \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t] \mid C_0^G] \leq O(z^2 \alpha) \cdot \text{dist}_G(s, t).$$

Since $z = O(\log n)$ and $\alpha = O(\log n \cdot \log \log n)$, the claim follows. $\square$

We are ready to prove Lemma 5.18, which we restate below.

Lemma 5.32. *For all $s, t \in V(G)$,*

$$\mathbb{E}[\min(\text{dist}_{D_1}(s, t), \text{dist}_{D_2}(s, t))] = O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Proof.

$$\begin{aligned} \mathbb{E}[\min(\text{dist}_{D_1}(s, t), \text{dist}_{D_2}(s, t))] &\leq \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]] + \mathbb{E}[\text{dist}_{D_2}(s, t) \cdot \mathbb{1}[s >_D t]] \\ &\leq O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t), \end{aligned}$$

where the final inequality follows from Lemma 5.21 and Claim 5.31. $\square$

We can now finish the distortion analysis and the proof of Theorem 5.10.

Proof of Theorem 5.10. We have finished the running time analysis and size analysis for Theorem 5.10. What remains is to prove that with high probability, for all $s, t \in V(G)$, there exists $D \in \mathcal{D}$ such that

$$\text{dist}_D(s, t) \leq O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Fix a pair of nodes $s, t \in V(G)$. Since $\mathbb{E}[\min(\text{dist}_{D_1}(s, t), \text{dist}_{D_2}(s, t))] = O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t)$ by Lemma 5.18, by Markov's inequality we conclude that with probability at least $1/2$,

$$\min(\text{dist}_{D_1}(s, t), \text{dist}_{D_2}(s, t)) \leq O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t).$$

Then with probability at least

$$1 - \left(\frac{1}{2}\right)^{10 \log n} \geq 1 - n^{-10},$$

there exists a DAG $D \in \mathcal{D}$ such that $\text{dist}_D(s, t) \leq O(\log^3 n \cdot \log \log n) \cdot \text{dist}_G(s, t)$. Then by applying the union bound over all pairs of nodes $s, t \in V(G)$, we conclude that $\mathcal{D}$ is an $O(\log^3 n \cdot \log \log n)$ -distance-preserving DAG cover of G with high probability. $\square$

We will also quickly prove Theorem 5.33, which we restate below.

Theorem 5.33. *Let G be an n -node weighted, directed graph with positive integer edge weights and polynomial aspect ratio. Then there exists a distribution $\mathcal{D}$ of DAGs over vertex set $V(G)$ satisfying the following properties:*

1. The support of distribution $\mathcal{D}$ contains at most $O((m+n)\log^3 n)$ additional edges.
2. Let $D \sim \mathcal{D}$ be a DAG sampled from distribution $\mathcal{D}$. For all pairs of nodes $(s, t) \in TC(G)$, s can reach t in D with probability $1/2$. Moreover,

$$\mathbb{E}[\text{dist}_D(s, t) \mid s \rightsquigarrow_D t] = O(\log^3 n \cdot \log \log n \cdot \text{dist}_G(s, t)),$$

where $s \rightsquigarrow_D t$ denotes the event that s can reach t in D .

3. We can sample a DAG from distribution $\mathcal{D}$ in $\tilde{O}(m)$ time.

Proof. We define a random graph D from our distribution $\mathcal{D}$ as follows. Construct random graphs D_1 and D_2 as in section 5.3, and choose graph D from set $\{D_1, D_2\}$ uniformly at random. Notice that the first property and the third property of Theorem 5.33 holds immediately from Theorem 5.10.

Let $(s, t) \in TC(G)$. Notice that by our construction of DAGs D_1 and D_2 , s can reach t in DAG D_1 if and only if $s <_D t$, where $<_D$ is the total order on $V(G)$ defined in subsection 5.3. Likewise, s can reach t in DAG D_2 if and only if $s >_D t$. Then it immediately follows that s can reach t in sampled DAG D with probability $1/2$ if $s \neq t$.

To bound the (conditional) expected distortion, we now consider two cases:

- **Case 1:** DAG $D = D_1$. Then

$$\mathbb{E}[\text{dist}_D(s, t) \mid s \rightsquigarrow_D t] = 2 \cdot \mathbb{E}[\text{dist}_{D_1}(s, t) \cdot \mathbb{1}[s <_D t]] = O(\log^3 \cdot \log \log n \cdot \text{dist}_G(s, t)),$$

by Claim 5.31.

- **Case 2:** DAG $D = D_2$. Then

$$\mathbb{E}[\text{dist}_D(s, t) \mid s \rightsquigarrow_D t] = 2 \cdot \mathbb{E}[\text{dist}_{D_2}(s, t) \cdot \mathbb{1}[s >_D t]] = O(\log^2 n \cdot \log \log n \cdot \text{dist}_G(s, t)),$$

by Lemma 5.21.

□

Chapter 6

Conclusion

This thesis studied graph compression and graph preconditioning, with the goal of understanding when large graphs can be compressed, augmented, or decomposed while preserving the structural information needed for computation. The results give new upper and lower bounds for several fundamental objects: distance preservers, strong connectivity preservers, hopsets, shortcut sets, and DAG covers.

On the compression side, Chapter 2 proved new bounds for distance preservers. For directed unweighted graphs with p demand pairs, we gave an exact distance preserver with $\tilde{O}(n^{5/6}p^{2/3} + n)$ edges, beating the classical consistent-tiebreaking bound in a natural parameter regime. We also proved an $\Omega(n^{2/3}p^{2/3})$ lower bound for finite-stretch directed preservers, separating finite-stretch preservation from pure reachability preservation. In addition, we showed that progress on undirected exact distance preservers would automatically imply progress in the directed setting, via a reduction from directed graphs to DAGs and then to undirected graphs.

Chapter 3 studied compression for directed strong connectivity. We proved that every n -vertex directed graph admits a k -fault-tolerant strong connectivity preserver with $O(k 4^k n \log n)$ edges, which is near-linear for constant k . We also gave k -connectivity preservers with $O(n\sqrt{nk})$ edges. These results show that strong connectivity is much more compressible than arbitrary directed reachability, while also revealing directed phenomena that do not occur in the undirected setting. In particular, fault tolerance and higher-order connectivity are essentially equivalent in undirected graphs up to constant factors, but they lead to different sparsity behavior in directed graphs.

On the preconditioning side, Chapter 4 proved lower bounds for diameter-reducing augmentations. For exact hopsets, we showed that the folklore sampling construction is essentially optimal: even $O(n)$ -size exact hopsets cannot reduce the hopbound below $\tilde{\Omega}(n^{1/2})$ in general. For shortcut sets, we improved the lower bound for $O(n)$ -size shortcut sets to $\tilde{\Omega}(n^{1/4})$. Together, these results clarify the difference between exact distance preservation, approximate distance preservation, and

reachability preservation.

Chapter 5 introduced and studied DAG covers as a directed analogue of tree covers and probabilistic tree embeddings. We proved that, if the number of additional edges is measured only as a function of n , then even $n^{2-o(1)}$ added edges may still require $n^{1-o(1)}$ DAGs. In contrast, when the edge budget is nearly linear in the input sparsity, every weighted directed graph admits an $\tilde{O}(1)$ -distance-preserving DAG cover with $O((m+n)\log^3 n)$ additional edges and $O(\log n)$ DAGs. This gives a directed preconditioning primitive analogous in spirit to tree covers in undirected graphs.

A recurring theme throughout the thesis is that directed and undirected graphs are neither entirely analogous nor entirely separate. For some problems, such as exact distance preservers, directed preservers can be related to undirected ones through an intermediate DAG structure. For others, such as fault-tolerant strong connectivity preservers and DAG covers, classical undirected ideas admit meaningful directed generalizations. Even then, however, the directed setting introduces important new subtleties and separations that did not exist in undirected graphs. These distinctions reveal a richer directed landscape than one would obtain by simply translating undirected results to directed graphs.

Another unifying theme is the usefulness of the path-system perspective. Many of the problems studied here can be reduced to understanding collections of paths: how they intersect, how they overlap, and how it affects the worst-case size of these systems. In particular, this viewpoint appears in upper bounds and reductions for distance preservers in Chapter 2, and in the low-overlap path systems underlying the hopset and shortcut-set lower bounds in Chapter 4. By isolating the combinatorial structure of the relevant paths in the graph, path systems provide a flexible framework for approaching graph compression problems. I believe that further developing this perspective could lead to sharper bounds for a variety of problems in graph compression and preconditioning.

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